UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023
or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to
Commission File No. 001-7784

Lumen Logo Blue_Black.jpg

Lumen Technologies, Inc.

(Exact name of registrant as specified in its charter)

Louisiana	**72-0651161**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)
100 CenturyLink Drive,	
Monroe, Louisiana	**71203**
(Address of principal executive offices)	(Zip Code)

(318) 388-9000
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, par value $1.00 per share	LUMN	New York Stock Exchange
Preferred Stock Purchase Rights	N/A	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports) and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T during the preceding 12 months (or for such shorter period that the registrant was required to submit and post such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, smaller reporting company or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and “emerging growth company” in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer	☒	Accelerated Filer	☐	Non-accelerated Filer	☐	Smaller Reporting Company	☐
						Emerging Growth Company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management’s assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☒

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

On February 20, 2024, 1,009,755,821 shares of common stock were outstanding. The aggregate market value of the voting stock held by non-affiliates as of June 30, 2023 was $2.3 billion.

DOCUMENTS INCORPORATED BY REFERENCE:

Portions of the Registrant's Proxy Statement to be furnished in connection with the 2024 annual meeting of shareholders are incorporated by reference in Part III of this report.

Auditor Name: KPMG LLP Auditor Location: Denver, Colorado Auditor Firm ID: 185

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Unless the context requires otherwise, (i) references in this report on Form 10-K, for all periods presented, to "Lumen Technologies, Inc.," "Lumen Technologies" or "Lumen,"" "we," "us," the "Company" and "our" refer to Lumen Technologies, Inc. and its consolidated subsidiaries and (ii) references in this report to "Level 3" refer to Level 3 Parent, LLC and its predecessor, Level 3 Communications, Inc., which we acquired on November 1, 2017.

PART I

Special Note Regarding Forward-Looking Statements

This report and other documents filed by us under the federal securities law include, and future oral or written statements or press releases by us and our management may include, forward-looking statements about our business, financial condition, operating results or prospects. These "forward-looking" statements are defined by, and are subject to the "safe harbor" protections under the federal securities laws. These statements include, among others:

- forecasts of our anticipated future results of operations, cash flows or financial position;
- statements concerning the anticipated impact of our completed, pending or proposed transactions, investments, product development, participation in government programs, Quantum Fiber buildout plans, and other initiatives, including synergies or costs associated with these initiatives;
- statements about our liquidity, profitability, profit margins, tax position, tax assets, tax rates, asset values, contingent liabilities, growth opportunities, growth rates, acquisition and divestiture opportunities, business prospects, regulatory and competitive outlook, market share, product capabilities, investment and expenditure plans, business strategies, securities repurchase plans, leverage, capital allocation plans, financing or refinancing alternatives and sources, and pricing plans; and
- other similar statements of our expectations, beliefs, future plans and strategies, anticipated developments and other matters that are not historical facts, many of which are highlighted by words such as “may,” “will,” “would,” “could,” “should,” “plans,” “believes,” “expects,” “anticipates,” “estimates,” "forecasts," “projects,” "proposes," "targets," “intends,” “likely,” “seeks,” “hopes,” or variations or similar expressions with respect to the future.

These forward-looking statements are based upon our judgment and assumptions as of the date such statements are made concerning future developments and events, many of which are beyond our control. These forward-looking statements, and the assumptions upon which they are based, (i) are not guarantees of future results, (ii) are inherently speculative and (iii) are subject to a number of risks and uncertainties. Actual events and results may differ materially from those anticipated, estimated, projected or implied by us in those statements if one or more of these risks or uncertainties materialize, or if our underlying assumptions prove incorrect. All of our forward-looking statements are qualified in their entirety by reference below to factors that could cause our actual results to differ materially from those anticipated, estimated, projected or implied by us in those forward-looking statements. These factors include but are not limited to:

- the effects of intense competition from a wide variety of competitive providers, including decreased demand for our more mature service offerings and increased pricing pressures;
- the effects of new, emerging or competing technologies, including those that could make our products less desirable or obsolete;
- our ability to successfully and timely attain our key operating imperatives, including simplifying and consolidating our network, simplifying and automating our service support systems, attaining our Quantum Fiber buildout schedule, replacing aging or obsolete plant and equipment, strengthening our relationships with customers and attaining projected cost savings;
- our ability to safeguard our network, and to avoid the adverse impact of cyber-attacks, security breaches, service outages, system failures, or similar events impacting our network or the availability and quality of our services;

- the effects of ongoing changes in the regulation of the communications industry, including the outcome of legislative, regulatory or judicial proceedings relating to content liability standards, intercarrier compensation, universal service, service standards, broadband deployment, data protection, privacy and net neutrality;

- our ability to generate cash flows sufficient to fund our financial commitments and objectives, including our capital expenditures, operating costs, debt repayments, taxes, pension contributions and other benefits payments;

- our ability to effectively retain and hire key personnel and to successfully negotiate collective bargaining agreements on reasonable terms without work stoppages;

- our ability to successfully adjust to changes in customer demand for our products and services, including increased demand for high-speed data transmission services and artificial intelligence services;

- our ability to successfully maintain the quality and profitability of our existing product and service offerings, to introduce profitable new offerings on a timely and cost-effective basis and to transition customers from our legacy products to our newer offerings;

- our ability to successfully and timely implement our corporate strategies, including our deleveraging and buildout strategies;

- our ability to successfully and timely realize the anticipated benefits from our 2022 and 2023 divestitures, and to successfully operate and transform our remaining business;

- changes in our operating plans, corporate strategies, or capital allocation plans, whether based upon changes in our cash flows, cash requirements, financial performance, financial position, market or regulatory conditions or otherwise;

- the impact of any future material acquisitions or divestitures that we may transact;

- the negative impact of increases in the costs of our pension, healthcare, post-employment or other benefits, including those caused by changes in markets, interest rates, mortality rates, demographics or regulations;

- the potential negative impact of customer and shareholder complaints, government investigations, security breaches or service outages impacting us or our industry;

- adverse changes in our access to credit markets on favorable terms, whether caused by changes in our financial position, lower credit ratings, unstable markets, debt covenant restrictions or otherwise;

- our ability to meet the terms and conditions of our debt obligations and covenants, including our ability to make transfers of cash in compliance therewith;

- the impact of any purported notice of default or notice of acceleration arising from alleged breach of covenants under our credit documents;

- our ability to consummate the transactions contemplated by our amended and restated transaction support agreement entered into on January 22, 2024 (the "TSA") on the currently anticipated timeline or at all, including the ability of the parties to successfully negotiate definitive agreements with respect to all matters covered by the term sheet included therein and the occurrence of events that may give rise to failure to satisfy any of the conditions to consummating such transactions or a right of any of the parties to terminate the TSA;

- our ability to maintain favorable relations with our security holders, key business partners, suppliers, vendors, landlords and lenders;

- our ability to timely obtain necessary hardware, software, equipment, services, governmental permits and other items on favorable terms;

- our ability to meet evolving environmental, social and governance ("ESG") expectations and benchmarks, and effectively communicate and implement our ESG strategies;
- the potential adverse effects arising out of allegations regarding the release of hazardous materials into the environment from network assets owned or operated by us or our predecessors, including any resulting governmental actions, removal costs, litigation, compliance costs, or penalties;
- our ability to collect our receivables from, or continue to do business with, financially-troubled customers;
- our ability to continue to use or renew intellectual property used to conduct our operations;
- any adverse developments in legal or regulatory proceedings involving us;
- changes in tax, pension, healthcare or other laws or regulations, in governmental support programs, or in general government funding levels, including those arising from governmental programs promoting broadband development;
- our ability to use our net operating loss carryforwards in the amounts projected;
- the effects of changes in accounting policies, practices or assumptions, including changes that could potentially require additional future impairment charges;
- the effects of adverse weather, terrorism, epidemics, pandemics, rioting, vandalism, societal unrest, or other natural or man-made disasters or disturbances;
- the potential adverse effects if our internal controls over financial reporting have weaknesses or deficiencies, or otherwise fail to operate as intended;
- the effects of changes in interest rates or inflation;
- the effects of more general factors such as changes in exchange rates, in operating costs, in public policy, in the views of financial analysts, or in general market, labor, economic, public health or geopolitical conditions; and
- other risks referenced in the "Risk Factors" section or other portions of this report or other of our filings with the U.S. Securities and Exchange Commission (the "SEC").

Additional factors or risks that we currently deem immaterial, that are not presently known to us or that arise in the future could also cause our actual results to differ materially from our expected results. Given these uncertainties, investors are cautioned not to unduly rely upon our forward-looking statements, which speak only as of the date made. We undertake no obligation to publicly update or revise any forward-looking statements for any reason, whether as a result of new information, future events or developments, changed circumstances, or otherwise. Furthermore, any information about our intentions contained in any of our forward-looking statements reflects our intentions as of the date of such forward-looking statement, and is based upon, among other things, existing regulatory, technological, industry, competitive, economic and market conditions, and our assumptions as of such date. We may change our intentions, strategies or plans (including our capital allocation plans) at any time and without notice, based upon any changes in such factors, in our assumptions or otherwise.

ITEM 1. BUSINESS

Business Overview and Purpose

We are a facilities-based technology and communications company that provides a broad array of integrated products and services to our domestic and global business customers and our domestic mass markets customers. We operate one of the world’s most interconnected networks. Our platform empowers our customers to swiftly adjust digital programs to meet immediate demands, create efficiencies, accelerate market access and reduce costs, which allows our customers to rapidly evolve their IT programs to address dynamic changes. Our specific products and services are detailed below under the heading “Segments and Products & Services.”

We conduct our operations under the following three brands:

- "Lumen," which is our flagship brand for serving the enterprise and wholesale markets
- "Quantum Fiber," which is our brand for providing fiber-based services to residential and small business customers
- "CenturyLink," which is our long-standing brand for providing mass-marketed copper-based services, managed for cash flow and optimal efficiency.

With approximately 170,000 on-net buildings and 350,000 route miles of fiber optic cable globally, we are among the largest providers of communications services to domestic and global enterprise customers. Our terrestrial fiber optic long-haul network throughout North America and Asia Pacific connects to metropolitan fiber networks that we operate.

As further discussed immediately below under the heading “Acquisitions and Divestitures,” we sold (i) our Latin American business and a portion of our incumbent local exchange business ("ILEC") during 2022 and (ii) our business conducted in Europe, the Middle East and Africa ("EMEA") during 2023.

For a discussion of certain risks applicable to our business, see “Risk Factors” in Item 1A of Part I of this report.

Acquisitions and Divestitures

General

Since being incorporated in 1968, we have grown principally through acquisitions. By 2008, we had become one of the largest providers of rural telephone services in the United States. Since then, we acquired Embarq Corporation in mid-2009, Qwest Communications International Inc. in early 2011 and Level 3 Communications, Inc. in late 2017. These acquisitions substantially changed our customer base, geographic footprint, business strategies and mix of products and services.

We continue to evaluate the possibility of acquiring additional assets or divesting assets in exchange for cash, securities or other properties, and at any given time may be engaged in discussions or negotiations regarding additional acquisitions or divestitures. We generally do not announce our acquisitions or divestitures until we have entered into a preliminary or definitive agreement.

Divestitures of the Latin American, ILEC and EMEA Businesses

On August 1, 2022, affiliates of Level 3 Parent, LLC, an indirect wholly-owned subsidiary of Lumen Technologies, Inc., sold Lumen’s Latin American business. On October 3, 2022, we and certain of our affiliates sold the portion of our facilities-based ILEC business primarily conducted within 20 Midwestern and Southeastern states. On November 1, 2023, affiliates of Level 3 Parent, LLC sold Lumen's operations in Europe, the Middle East and Africa (the "EMEA business").

See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses to our consolidated financial statements in Item 8 of Part II of this report for additional information on these transactions.

Financial Highlights

The following table summarizes the results of our consolidated operations:

	Years Ended December 31,		
	2023[1]	2022[1]	2021
	(Dollars in millions)		
Operating revenue	$ 14,557	17,478	19,687
Operating expenses	24,141	17,383	15,402
Operating (loss) income	$ (9,584)	95	4,285
Net (loss) income	$ (10,298)	(1,548)	2,033

(1) During 2023 and 2022, we recorded non-cash, non-tax-deductible goodwill impairment charges of $10.7 billion and $3.3 billion, respectively. For additional information, see Note 3—Goodwill, Customer Relationships and Other Intangible Assets to our consolidated financial statements in Item 8 of Part II of this report.

During 2023, 2022 and 2021, approximately 6.5%, 8.6% and 9.4%, respectively, of our consolidated revenue was derived outside the U.S.

The following table summarizes certain selected financial information from our consolidated balance sheets:

	As of December 31,	
	2023	2022
	(Dollars in millions)	
Total assets	$ 34,018	45,612
Total long-term debt[1]	19,988	20,572
Total stockholders' equity	417	10,374

(1) For additional information on our total long-term debt, see Note 7—Long-Term Debt and Credit Facilities to our consolidated financial statements in Item 8 of Part II of this report. For information on our total obligations, see "Management's Discussion and Analysis of Financial Condition and Results of Operations—Liquidity and Capital Resources—Future Contractual Obligations" in Item 7 of Part II of this report.

The summary financial information appearing above should be read in conjunction with, and is qualified by reference to, our consolidated financial statements and notes thereto in Item 8 of Part II of this report and “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in Item 7 of Part II of this report.

Strategy

Our over-arching strategic goal is to digitally connect people, data, and applications quickly, securely, and effortlessly. To attain this goal, we strive to, among other things:

- strengthen our digital self-service product ordering platforms;
- expand our offering of secure edge computing services;
- create a more adaptive network;
- expand our network capacity through our Quantum Fiber buildout plan and other initiatives;
- monetize our non-core assets and deliver cost-effective operations;
- manage our non-fiber business for cash flow; and
- strengthen our financial position and performance through debt and cost reduction efforts.

Our Stakeholders

We believe that regular communications with our stakeholders is a vital component of Lumen's success. Our "North Star" guides us to operate with transparency and infuses clarity into the communications we have with all of our stakeholders including our investors, employees, customers, vendors, partners and our global community.

Employees and Human Capital Resources

To position Lumen for growth and success, we have made changes to our executive leadership team that have played a critical role in modernizing our business, attracting new talent and invigorating our culture. Lumen’s highly competitive business requires attracting, developing and retaining a motivated team inspired by leadership, engaged in meaningful work, motivated by career growth opportunities and thriving in a culture that embraces diversity, inclusion and belonging. Understanding and anticipating the priorities of our current and future employees is important to our future success. We aim to bring together the best mix of diverse talent to develop the brightest ideas to transform industries across the globe. At December 31, 2023, we had approximately 28,000 employees world-wide, including approximately 3,700 outside the U.S.

Attracting, Developing and Retaining Talent

Our recruiting, development and retention objectives focus on treating talent as a differentiator and a leading indicator of business performance. We strive to hire and retain the best talent available to provide outstanding opportunities for career advancement and to champion fair selections and best hiring practices. We have implemented diverse interview panels, which include at least one woman or person of color, to minimize the potential for unconscious bias in our recruitment process. We have also developed a non-biased pre-hire assessment process. We have established a framework of competency-based success profiles and regular career development and training programs, which we believe empower our employees to pursue their professional goals and improve employee engagement and retention. We invest in broad-based development for our employees in various ways such as skills-building programs, on-demand learning options, tuition reimbursement and tailored intern and mentoring programs, along with a suite of leadership development courses. In an effort to create more development opportunities for all employees, we are expanding our mentoring and leadership development programs, with added focus on development for diverse employees.

We believe we have made significant strides in attracting, engaging, and hiring a diverse group of early career employees through our internship program, our numerous sales and operations academies, and our "pathways in technology" program. We have also increased our focus on fostering internal mobility and providing more visibility and career advancement opportunities to our workforce through our internal communications platforms. Developing strong leaders who can move our company forward is a priority for Lumen.

We gauge the efficacy of our programs, identify opportunities for improvement, and pursue solutions through tracking and analyzing data in a variety of ways, including conducting annual talent reviews and measuring our progress toward goals specified in our development, diversity and inclusion plans.

Diversity, Inclusion & Belonging

We believe that diversity stimulates creativity, spurs innovation and helps drive profitability, which is why we strive to create inclusive, welcoming workplaces where everyone can feel at home, be their authentic selves and thrive. Realizing greater diversity across all levels of an organization is, and will continue to be, an ongoing journey. Our Diversity & Inclusion Steering Committee, comprised of a cross-functional team of senior executives, including our Chief Diversity & Inclusion Officer, oversees and champions our diversity, inclusion and belonging strategy. We aim for the highest standards of fairness and equal opportunity in recruitment, hiring, promotions, job assignments and compensation (including undertaking periodic gender and race/ethnicity pay equity studies of our U.S., non-represented employees and making pay adjustments when warranted). Inclusive recruiting and outreach programs for diverse candidates, supportive and engaging employee resource groups, and management-led listening circles are among some of Lumen's initiatives to create greater diversity and belonging among our employees.

Positive Corporate Culture

Our employees are critical to Lumen's success and we believe creating a positive, inclusive culture is essential to attracting and retaining engaged employees. We want our employees to be proud to work with us and fully engaged to share in our purpose maximize the world's digital potential. Lumen is transforming from the bottom up by building a culture of teamwork, trust, and transparency. Lumen's cultural transformation strategy incorporates a wide variety of communication and training activities encouraging collaboration among our colleagues around the world. We measure the program's efficacy and identify opportunities for improvements through an engagement survey distributed approximately every six months.

Health & Wellness

We are committed to promoting the health, safety and well-being of our employees, business partners and global communities. Lumen strives for an above-average safety performance as we continue our investments in programs and training to support health and safety. We want all of our employees to thrive, and we regularly re-evaluate how to best support our employees' well-being through benefits and resources. We design our current benefit and wellness programs to drive engagement that positively impacts our culture, job satisfaction, recruiting and retention programs. We offer progressive employee benefits and enhancements that recognize the diverse needs of our people and their families.

Labor Relations

At December 31, 2023, approximately 21% of our U.S. workforce was represented by a union, either the Communications Workers of America or the International Brotherhood of Electrical Workers. A small number of our overseas employees are represented by unions or another representative body. We recognize the critical role that our supervisors and managers play in fostering a productive and respectful work environment, and we encourage employees to work directly with their supervisors, where possible, to efficiently and effectively resolve workplace concerns. We also respect our employees' rights to voluntarily establish and join unions and similar associations without unlawful interference. We strive to work collaboratively with the unions, councils and associations that represent our workers.

Customer Success

Our customers range from individual households to global enterprises. Whether our network supports remote education to under-served communities or a multi-national work-from-home enterprise, all customers are impacted by the quality and reliability of our products and services. Understanding how each customer accesses and uses our products and services informs the type of customer engagement to best meet their expectations. Lumen's Customer Experience ("CX") team takes the lead in driving customer obsession and guiding the company toward our North Star by listening and learning from our customers, then acting to meet their needs. This assists us in accomplishing our mission of igniting business growth by connecting people, data and applications - quickly, securely, and effortlessly. We believe a strong experience leads to satisfied customers and engaged employees who are encouraged to recommend creative solutions.

We highly value both customer and employee suggestions. We offer our customers several channels for communicating with us, including voice, text, email, chat and social media, among others. We are driving a digital-first culture that allows our customers to configure, order and rapidly deploy our services through an all-digital, self-service set of tools. Since 2019, we have hosted an annual CX event, during which we invite customers to collaborate directly with us.

While careful listening to customers is the best source of customer experience feedback, we believe overlaying it with employee feedback is the most effective way to continuously improve. We regularly invite our front-line employees to provide feedback on opportunities to improve our capabilities.

Partners and Vendors

We seek to engage with those partners and vendors who best contribute to our customers' success. Lumen seeks to co-innovate with a comprehensive group of strategic partners to create solutions focused exclusively on our customers' business and IT requirements. Through our open and interoperable approach, we seek to identify the optimal platform for serving our customers – whether ours or a third party's. When necessary, Lumen incorporates market-leading technologies to optimize application performance and streamline integration throughout the IT stack to ensure seamless integration and interoperability. Lumen has collaborated with a host of technology partners, in an effort to integrate different technologies to improve our products and services. We believe this collaboration has strengthened our capability to tailor and manage scalable solutions that customers control.

Given these efforts to better serve our customers, we are materially reliant on a wide range of vendors to support our organization and partners to support our strategy. We work with, and rely on, other communications companies that lease us transmission capacity or sell us various services necessary for our current operations, as well as a wide range of software, hardware and equipment suppliers. We believe that co-innovating with other companies enables us to more rapidly improve our customer offerings.

In addition, we provide services to our customers in Latin America and EMEA through contractual relationships with third-party carriers. Under these arrangements, the third-party carriers invoice us for their services, and we pass along those charges to our customers through our invoices.

Environmental Stewardship and Sustainability

We believe our commitment to environmental sustainability promotes the financial health of our business and strengthens our relations with our employees, communities, customers and investors.

In early 2022, we formed the Sustainability Management Committee ("SMC") comprised of employees from across the business. The SMC designs and oversees our company-wide sustainability program, including the monitoring of climate-related issues, and is responsible for driving our sustainability agenda with the Board and senior leadership. Additionally, our Environment, Health and Safety ("EHS") team is responsible for overseeing and implementing our EHS and environmental sustainability initiatives.

The EHS program framework focuses on the following key areas:

- Environmental compliance and management: The Lumen EHS team assesses and reviews our company programs, operational facilities and waste management vendors. We monitor environmental legislative activity and collaborate with other internal groups to develop documented practices and procedures that support compliance with applicable laws and regulations.

- Energy and emissions: In an effort to reduce our carbon footprint, we continue to identify and implement energy efficiency and greenhouse gas ("GHG") emissions reduction initiatives. In November 2023, we announced early achievement of our 2018-2025 science-based GHG emissions-reduction targets. We remain committed to exploring ways to reduce GHG emissions through our operational, customer and employee initiatives.

- Water: Lumen uses the World Resource Institute's Aqueduct Water Risk Atlas to assess susceptibility to future water stress across our areas of operation. We strive to reduce our water consumption, especially in the water-stressed communities where we operate. We track our usage and closely monitor abnormalities to improve water efficiencies and reduce site discharge.

- Waste: We are committed to reusing and recycling products, minimizing material use and carefully managing our waste. Each year, we divert millions of pounds of electronic and communications equipment from landfills. We recycle telecommunications equipment, and our modem/router takeback program allows customers to return their equipment, which are then either reused or sent to an R2-certified recycler.

- Supplier environmental assessment: We expect our suppliers to embrace and share our commitment to compliance and sustainability efforts. As reflected in our Supplier Code of Conduct, we expect our suppliers to use reasonable efforts to employ environmentally preferred and energy-efficient services, and to work with their own suppliers to assess and address environmental and sustainability issues within their supply chains.

- Climate preparedness: We evaluate various climate change risks to our ongoing operations when we consider expanding our network or facilities. Our comprehensive business continuity program focuses on prevention, collaboration, communication, response and recovery to assist us in quickly resolving disruptive events. Weather events such as severe flooding and hurricanes can impact our ability to deliver services, so business resiliency and adaptability is key to the long-term viability of our business.

- Occupational Health and Safety: The EHS team conducts risk assessments, reviews safety incident data and monitors health and safety legislation to develop policies and procedures designed to minimize safety hazards and support compliance with applicable laws and regulations. We continuously monitor safety performance to identify trends and evaluate opportunities to eliminate or reduce the risks of workplace hazards.

Our Network

Our network, through which we provide most of our products and services, consists of fiber-optic and copper cables, high-speed transport equipment, electronics, voice switches, data switches, routers, and various other equipment. We operate part of our network with leased assets, and a substantial portion of our equipment with licensed software.

At December 31, 2023, our network (owned and leased) included (i) approximately 350,000 route miles of fiber optic plant and (ii) multiple gateway and transmission facilities used in connection with operating our network throughout North America.

At December 31, 2023, our domestic network connected to (i) approximately 170,000 buildings, which we refer to as "Fiber On-net" buildings, serving our enterprise customer base and (ii) approximately 21.8 million broadband-enabled units capable of serving our Mass Markets customer base. At December 31, 2023, approximately 3.7 million of our Mass Markets broadband-enabled units were capable of receiving services from our fiber-based infrastructure, with the remainder connected with copper-based infrastructure. Our domestic network also included at such date central office and other equipment that enables us to provide telephone service as an ILEC.

As discussed in "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7 of Part II of this report and Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses to our consolidated financial statements in Item 8 of Part II of this report, we sold portions of our network during 2022 and 2023.

As noted elsewhere in this report, we view our network as one of our most critical assets. We have devoted, and plan to continue to devote, substantial resources to (i) simplify and modernize our network and legacy systems (ii) retire aging or obsolete systems and plant and (iii) expand our network to address demand for enhanced or new products. A key element of our network expansion plan is our Quantum Fiber buildout project. Under this project, we propose over the next several years to construct additional fiber optic infrastructure to enable us to provide Quantum Fiber broadband services to several million additional urban and suburban locations in our remaining ILEC markets.

Although we own most of our network, we lease a substantial portion of our core fiber network from several other communication companies under arrangements that will periodically need to be renewed or replaced to support our current network operations.

As a critical infrastructure provider, we are a constant target of cyber-attacks from a wide range of intruders, including advanced persistent threat actors. From time to time in the ordinary course of our business, we experience security incidents and disruption in our services. We develop and maintain systems and programs designed to protect against cyber-attacks and network outages. The development, maintenance and operation of these systems and programs is costly and requires ongoing monitoring and updating as technologies change and efforts to bypass security measures become more sophisticated and evolve rapidly.

For additional information regarding our systems, network assets, network risks, capital expenditure requirements and reliance upon third parties, see “Risk Factors” in Item 1A of Part I of this report.

Competition

We compete in a dynamic and highly competitive market in which demand for high-speed, secure data services continues to grow. We expect continued intense competition from a wide variety of sources under these evolving market conditions. In addition to competition from large international communications providers, we are facing competition from a growing number of sources, including systems integrators, hyperscalers, cloud service providers, software networking companies, infrastructure companies, cable companies, wireless service providers, device providers, resellers and smaller niche providers.

Our ability to compete hinges upon effectively enhancing and better integrating our existing products, introducing new products on a timely and cost-effective basis, meeting changing customer needs, providing high-quality information security to build customer confidence and combat cyber-attacks, extending our core technology into new applications and anticipating emerging technological and industry changes. Depending on the applicable market and services, competition can be intense, especially if competitors in the market have network assets better suited to customer needs, faster transmission speeds or lower prices, or, in certain overseas markets, are national or regional incumbent communications providers that have a longer history of providing service in the market.

We compete to provide services to business customers based on a variety of factors, including the comprehensiveness and reliability of our network, our data transmission speeds, price, the latency of our available network services, the scope of our integrated offerings, the reach and peering capacity of our IP network, and customer service. Competition from large communications providers, systems integrators, hyperscalers and others have increased pricing pressures with respect to several key products and services that we offer to our enterprise and wholesale business customers. In particular, several hyperscalers have recently built their own data transmission facilities, which has reduced demand for our network services.

Competition to provide broadband services to our mass markets customers remains high. Market demand for our broadband services could be adversely affected by (i) advanced wireless data transmission technologies, including fixed wireless and low-earth-orbit satellite services, and (ii) continued enhancements to cable-based services, each of which generally provides faster average broadband transmission speeds than our copper-based infrastructure. In addition, several established or new communications companies, infrastructure companies or municipalities have built or are building new fiber-based networks to provide high-speed broadband services in existing or unserved markets, frequently with the support of governmental subsidies. Our network expansion and innovation strategy is focused largely on addressing these competitive pressures. To meet these demands and remain competitive, we are continuing to invest in network capacity, security, reliability, flexibility and design innovations, including through our Quantum Fiber buildout initiative.

For our traditional voice services, providers of wireless voice, social networking, videoconferencing and electronic messaging services are significant competitors as many customers are increasingly using these services to communicate, resulting in the long-term systemic decline in our traditional voice services. Other potential sources of competition include non-carrier systems that are capable of bypassing our local networks, either partially or completely, through various means. Developments in software have permitted new competitors to offer affordable networking products that historically required more expensive hardware investment. We anticipate that all these trends will continue to place downward pressures on the use of our voice network.

Additionally, the Telecommunications Act of 1996 obligates ILECs, including those operated by us, to permit competitors to interconnect their facilities to the ILEC's network and to take various other steps that are designed to promote competition, including obligations to (i) negotiate interconnection agreements in good faith, (ii) provide nondiscriminatory "unbundled" access to specific portions of the ILEC's network and (iii) permit competitors to physically or virtually collocate their plant on the ILEC's property. As a result of the above-described regulatory and technological developments, we also face competition from competitive local exchange carriers ("CLECs"), particularly in densely populated areas. CLECs provide competing services through (i) reselling an ILEC's local services, (ii) using an ILEC's unbundled network elements, (iii) operating their own facilities or (iv) a combination thereof.

Additional information about competitive pressures is located under the heading "Risk Factors—Business Risks" in Item 1A of Part I of this report.

Sales and Marketing

Sales Channels

Our enterprise sales and marketing approach focuses on solving complex customer problems with advanced technology and network solutions – striving to make core networks services compatible with digital tools. We also rely on our call center personnel and a variety of channel partners to promote sales of services that meet the needs of our customers. To meet the needs of different customers, our offerings include both stand-alone services and bundled services designed to provide a complete offering of integrated services.

Our business customers range from small business offices to the world's largest global enterprise customers. Our direct sales representatives generally market our business services to members of in-house IT departments or other highly-sophisticated customers with deep technological experience. These individuals typically satisfy their IT requirements by contracting with us or a rapidly evolving group of competitors, or by deploying in-house solutions. We also market our products and services through inbound call centers, telemarketing and third parties, including telecommunications agents, system integrators, value-added resellers and other telecommunications firms. We support our distribution through digital advertising, events, television advertising, website promotions and public relations. We maintain local offices in most major and secondary markets within the U.S. and many of the primary markets of the other countries in which we provide services.

Similarly, our sales and marketing approach to our mass market customers emphasizes customer-oriented sales, marketing and service with a local presence. Our approach includes marketing our products and services primarily through direct sales representatives, inbound call centers, telemarketing and third parties, including retailers, satellite television providers, door to door sales agents and digital marketing firms.

Segments and Products & Services

We structure our segments and customer-facing sales channels to align with how we support our customers. We believe this reporting structure provides greater transparency into how we are performing against our strategy, including focusing on growth opportunities and managing declining legacy services.

Segments

We report our financial performance using two segments, as described below:

- *Business Segment:* Under our Business segment, we provide our products and services under four sales channels to meet the needs of our enterprise and commercial customers; and

- *Mass Markets Segment:* Under our Mass Markets segment, we provide products and services to residential and small business customers.

The following table shows the composition of our operating revenue by segment for the years ended December 31, 2023, 2022 and 2021:

	Years Ended December 31,			Percent Change	
	2023	2022	2021	2023 vs 2022	2022 vs 2021
Percentage of revenue:					
Business	79 %	75 %	72 %	4 %	3 %
Mass Markets	21 %	25 %	28 %	(4)%	(3)%
Total operating revenue	100 %	100 %	100 %		

For additional information on our segment data, including information on certain centrally-managed assets and expenses not reflected in our segment results, see Note 17—Segment Information to our consolidated financial statements in Item 8 of Part II of this report and "Management's Discussion and Analysis of Financial Condition and Results of Operations—Reporting Segments" in Item 7 of Part II of this report.

Products & Services

At December 31, 2023, we categorized our products and services revenue among the following product categories for the Business segment:

- *Grow, which includes products and services that we anticipate will grow, including:*
    - *Dark Fiber.* We control an extensive array of unlit optical fiber known as “dark fiber,” which has been laid but not yet been equipped with the equipment necessary for it to transmit data. We provide access to this unlit optical fiber to customers who are interested in building their networks with this high-bandwidth, highly secure optical technology. We also provide professional services to engineer these networks, and in some cases, manage them for customers;
    - *Edge Cloud Services.* We provide access to both public and private cloud solutions that allow our customers to optimize cost and performance by offloading workloads. Lumen’s cloud access products are designed to leverage our network edge to provide low-latency secure services for our customers. Additionally, we provide cloud orchestration tools that allow customers to shift work between cloud environments dynamically;
    - *Internet Protocol ("IP").* Our IP services provide global internet access over a high performance, diverse network. Our fiber network spans approximately 350,000 route miles globally with extensive off-net access solutions across North America and Asia Pacific;
    - *Managed Security Services.* We provide enterprise security solutions that help our customers secure networks, mitigate malicious attacks and identify potential security threats. These services include DDoS mitigation, remote and premise-based firewalls, professional consulting and management services, and threat intelligence services;
    - *Software-Defined Wide Area Networks ("SD WAN")* . We offer Lumen-managed and co-managed SD-WAN solutions to help reduce the complexity and business risk of network transformation on a single, automated platform that coordinates the full spectrum of connectivity types. Our tools, technology and hands-on expertise provide the ability to design, deploy and evolve with business needs while maintaining complete visibility, security and control;
    - *Secure Access Service Edge ("SASE").* We offer Lumen Secure Access Service Edge (SASE) as a comprehensive network and security solution using a cloud-first architecture, centered around zero-trust security principles. The service is delivered from a choice of multiple SASE software partners, offers flexible service management options, and is available on our IP backbone with several access options to connect and protect customer networks;

- *Unified Communications and Collaboration ("UC&C")*. We provide access to various unified communications platforms. This offering includes both individual, license-based service models and more robust options that transform a customer's inbound and outbound calling platform; and

- *Optical Services*. We deliver high bandwidth optical wavelength networks to customers requiring an end-to-end solution with ethernet technology for a scalable amount of bandwidth connecting sites or providing high-speed access to cloud computing resources.

- *Nurture, which includes our more mature offerings, such as:*

  - *Ethernet.* We deliver a robust array of networking services built on ethernet technology. Ethernet services include point-to-point and multi-point equipment configurations that facilitate data transmissions across metropolitan areas and larger enterprise-class wide area networks. Our ethernet technology is also used by wireless service providers for data transmission via our fiber-optic cables connected to their towers; and

  - *VPN Data Networks.* Leveraging our extensive fiber-optic network, we create private networks tailored to our customers' needs. These technologies enable service providers, enterprises and government entities to streamline multiple networks into a single, cost-effective solution that simplifies the transmission of voice, video, and data over a single secure network.

- *Harvest, which includes our legacy services managed for cash flow, including:*

  - *Voice Services.* We offer our customers a complete portfolio of traditional Time Division Multiplexing ("TDM") voice services including primary rate interface service, local inbound service, switched one-plus, toll free, long distance and international services; and

  - *Private Line.* We deliver private line services, a direct circuit or channel specifically dedicated for connecting two or more organizational sites. Private line service offers a high-speed, secure solution for frequent transmission of large amounts of data between sites, including wireless backhaul transmissions.

- *Other, which includes:*

  - *Equipment.* We sell and install certain communications equipment.

  - *IT Solutions.* We craft technology solutions for our customers and often manage these solutions on an ongoing basis. These services frequently enhance equipment or networks owned, acquired, or controlled by the customer and often include our consulting or software development services; and

  - *Other Legacy Services.* We continue to provide certain services based on older platforms to support our customers as they transition to newer technology. These services include Synchronous Optical Network ("SONET") based ethernet, legacy data hosting services, and conferencing services.

At December 31, 2023, we reported our products and services revenue among the following categories for the Mass Markets segment:

- *Fiber Broadband*, under which we provide high speed broadband services to residential and small business customers utilizing our fiber-based network infrastructure;

- *Other Broadband*, under which we provide primarily lower speed broadband services to residential and small business customers utilizing our copper-based network infrastructure; and

- *Voice and Other,* under which we derive revenues from (i) providing local and long-distance voice services, professional services, and other ancillary services, and (ii) federal broadband and state support programs.

Research, Development & Intellectual Property

As of December 31, 2023, we held approximately 3,100 patents and patent applications in the U.S. and other countries. We have also received licenses to use patents held by others, which give us the freedom to operate our business without the risk of interruption from the holder of the patented technology. We plan to continue to file new patent applications as we enhance and develop products and services, and we plan to continue to seek opportunities to expand our patent portfolio through strategic acquisitions and licensing.

In addition to our patent rights, we have rights in various trade names, trademarks, copyrights and other intellectual property that we use to conduct our business. Our services often use the intellectual property of others, including licensed software. We also occasionally license our intellectual property to others as we deem appropriate.

For information on various litigation risks associated with owning and using intellectual property rights, see “Risk Factors—Business Risks” in Item 1A of Part I of this report, and Note 18—Commitments, Contingencies and Other Items to our consolidated financial statements in Item 8 of Part II of this report.

Regulation of Our Business

Our domestic operations are regulated by the Federal Communications Commission (the "FCC"), by various state regulatory commissions and occasionally by local agencies. Our non-domestic operations are regulated by supranational groups (such as the European Union, or EU), national agencies and frequently state, provincial or local bodies. Generally, we must obtain and maintain operating licenses from these bodies in most areas where we offer regulated services.

Changes in the composition and leadership of the FCC, state regulatory commissions and other agencies that regulate our business could have significant impacts on our revenue, expenses, competitive position and prospects. Changes in the composition and leadership of these agencies are often difficult to predict, which makes future planning more difficult.

The following description discusses some of the major regulations affecting our operations, but others could have a substantial impact on us as well. For additional information, see “Risk Factors” in Item 1A of Part I of this report.

Federal Regulation of Domestic Operations

General

The FCC regulates the interstate services we provide, including the business data service charges we bill for wholesale network transmission and intercarrier compensation, including the interstate access charges that we bill other communications companies in connection with the origination and termination of interstate phone calls. Additionally, the FCC regulates several aspects of our business related to international communications services, privacy, public safety and network infrastructure, including (i) our access to and use of local telephone numbers, (ii) our provision of emergency 911 services and (iii) our use or removal (potentially on a reimbursable basis) of equipment produced by certain vendors deemed to cause potential national security risks. We could incur substantial penalties if we fail to comply with the FCC's applicable regulations.

Many of the FCC's regulations adopted in recent years remain subject to judicial review and additional rule-makings, thus increasing the difficulty of determining the ultimate impact of these changes on us and our competitors.

Universal Service

Between 2015 and 2021, we received approximately $500 million annually through Phase II of the FCC's Connect America Fund ("CAF II"), a program that ended on December 31, 2021. In connection with the CAF II funding, we were required to meet certain specified infrastructure buildout requirements in 33 states by the end of 2021, which required substantial capital expenditures. In the first quarter of 2022, we recognized $59 million of previously deferred revenue related to the conclusion of the CAF II program based upon our final buildout and filing submissions. The government has the right to audit our compliance with the CAF II program. The ultimate outcome of any remaining examinations is unknown, but could result in a liability to us in excess of our reserve accruals established for these matters.

In January 2020, the FCC created the Rural Digital Opportunity Fund ("RDOF") program, a federal support program designed to fund broadband deployment in rural America. For the first phase of this program, RDOF Phase I, the FCC ultimately awarded $6.4 billion in support payments to be paid in equal monthly installments over 10 years. We were awarded RDOF funding in several of the states in which we operate and began receiving monthly support payments during the second quarter of 2022. We received approximately $17 million in annual RDOF Phase I support payments for each of the years ended December 31, 2023 and 2022 and expect to receive this same amount each year thereafter during the program period.

Federal officials have proposed changes to current programs and laws that could impact us, including proposals designed to increase broadband access, increase competition among broadband providers, lower broadband costs and re-adopt "net neutrality" rules similar to those adopted under a prior administration. In late 2021, the U.S. Congress enacted legislation that appropriated $65 billion to improve broadband affordability and access, primarily through federally funded state grants. As of the date of this report, various state and federal agencies are continuing to take steps to make this funding available to eligible applicants, including us. Although it remains premature to speculate on the ultimate impact of this legislation on us, we anticipate that the release of this funding would increase competition for broadband customers in newly-served areas.

For additional information about these programs, see (i) Note 4—Revenue Recognition to our consolidated financial statements in Item 8 of Part II of this report and (ii) "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7 of Part II of this report.

Broadband Regulation

In February 2015, the FCC adopted an order regulating broadband internet access services ("BIAS") as a Title II utility service under the Communications Act of 1934. In December 2017, the FCC voted to repeal the classification of BIAS as a Title II utility service and to preempt states from imposing substantial regulations on broadband services. Opponents of this change appealed this action in federal court. Several states have also opposed the change and have proposed, implemented or enacted laws or orders focused on state-specific Internet service regulation. In October 2019, the federal court upheld the FCC's classification decision but vacated a part of its preemption ruling. Various courts are considering or have ruled upon the issue of the enforceability of state broadband regulation, and additional litigation and appeals are expected with respect to this issue. In addition, members of the current administration and various consumer interest groups have advocated in favor of reclassifying BIAS as a Title II utility service. The ultimate impact of these pending judicial matters and calls for additional regulation are currently unknown to us, although the imposition of heightened regulation of our Internet operations could potentially hamper our ability to operate our data networks efficiently, restrict our ability to implement network management practices necessary to ensure quality service, increase the cost of operating, maintaining and upgrading our network, and otherwise negatively impact our current operations.

State Regulation of Domestic Operations

Historically ILECs, including ours, have been regulated as "common carriers," and state regulatory commissions have generally exercised jurisdiction over intrastate voice telecommunications services and their associated facilities. In recent years, most states have reduced their regulation of ILECs. State regulatory commissions generally continue to (i) set the rates that telecommunications companies charge each other for exchanging traffic, (ii) administer support programs designed to subsidize the provision of services to high-cost rural areas, (iii) regulate the purchase and sale of ILECs, (iv) require ILECs to provide service under publicly-filed tariffs setting forth the terms, conditions and prices of regulated services, (v) limit ILECs' ability to borrow and pledge their assets, (vi) regulate transactions between ILECs and their affiliates and (vii) impose various other service standards.

In most states, switched and business data services and interconnection services are subject to price regulation, although the extent of regulation varies by type of service and geographic region. In addition, Voice-Over-Internet Protocol services are regulated by state regulators, but more lightly than ILEC services. State agencies also regulate certain aspects of non-ILEC communications businesses, including administering the payment of federal subsidies to support broadband infrastructure construction.

Data Privacy Laws and Regulations

Various foreign, federal and state laws govern our storage, maintenance and use of customer data, including a wide range of consumer protection, data protection, privacy, intellectual property and similar laws. Data privacy regulations are complex and vary across jurisdictions. As a company providing global services, we must comply with various jurisdictional data privacy regulations, including the General Data Protection Regulation ("GDPR") in the EU and similar laws adopted by various other jurisdictions in certain of our domestic and overseas markets. Domestically, the number of state privacy laws continues to increase. The application, interpretation and enforcement of these laws are often uncertain, and may be interpreted and applied inconsistently from jurisdiction to jurisdiction. These regulations require careful handling of personal and customer data and could have a significant impact on our business, especially if we violate any of those regulations.

Anti-Bribery and Corruption Regulations

As a provider of global services, we must comply with complex foreign and U.S. laws and regulations governing business ethics and practices, such as the U.S. Foreign Corrupt Practices Act, the U.K. Bribery Act and other local laws prohibiting corrupt payments to governmental officials and anti-competition regulations. We have compliance policies, programs and training designed to prevent non-compliance with such anti-corruption regulations in the U.S. and other jurisdictions.

Regulation of International Operations

Our subsidiaries operating outside of the U.S. are subject to various regulations in the markets where service is provided. The scope of regulation varies from country to country. The communications regulatory regimes in certain of our non-domestic markets are in the process of development. Many issues, including the pricing of services, have not been addressed fully, or even at all.

Our overseas operations are also subject to various other domestic or non-domestic laws or regulations, including various laws or regulations governing exports and imports of various goods or technologies and certain sanctioned business activities.

In 2020, the United Kingdom ("UK") terminated its membership in the EU ("Brexit") and has entered into related separation agreements with the EU regarding data sharing, financial services and other matters. Following the sale of our EMEA operations on November 1, 2023, we conduct only limited operations within the UK and EU. Consequently, we do not anticipate Brexit will have a substantial impact on our business.

Other Regulations

Our networks and properties are subject to numerous federal, state and local laws and regulations, including laws and regulations governing the use, storage and disposal of hazardous materials, the release of pollutants into the environment and the remediation of contamination. Our contingent liabilities under these laws are further described in Note 18—Commitments, Contingencies and Other Items. Certain federal and state agencies, including attorneys general, monitor and exercise oversight related to consumer protection issues. We are also subject to codes that regulate our trenching and construction operations or that require us to obtain permits, licenses or franchises to operate. Such regulations are enacted by municipalities, counties, state, federal or other regional governmental bodies, and can vary widely from jurisdiction to jurisdiction as a result. Such regulations may also require us to pay substantial fees.

Seasonality

Overall, our business is not materially impacted by seasonality. Our network-related operating expenses are, however, generally higher in the second and third quarters of the year. From time to time, weather related problems have resulted in increased costs to repair our network and respond to service calls in some of our markets. The amount and timing of these costs are subject to the weather patterns of any given year but have generally been highest during the third quarter and have been related to damage from severe storms, including hurricanes, tropical storms and tornadoes in our markets along the Atlantic and Gulf of Mexico coastlines.

Additional Information

From time to time, we may make investments in other communications or technology companies. For further information on regulatory, technological and competitive factors that could impact our revenue, see "Regulation" and "Competition" above under this Item 1 and "Risk Factors" below under Item 1A. For more information on the financial contributions of our various services, see "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7 of Part II of this report.

Website Access and Important Investor Information

We were incorporated in Louisiana in 1968. Our website is *www.lumen.com*. We routinely post important investor information in the "Investor Relations" section of our website at *ir.lumen.com*. The information contained on, or that may be accessed through, our website is not part of this report or any other periodic reports that we file with the SEC. Any references to our website in this report or any other periodic reports that we file with the SEC are provided for convenience only, and are not intended to make any of our website information a part of this or such other reports. You may obtain free electronic copies of annual reports on Form 10-K, quarterly reports on Form 10-Q, and current reports on Form 8-K of us and two of our principal subsidiaries, and amendments to those reports, in the "Investor Relations" section of our website (*ir.lumen.com*) under the heading "FINANCIALS" and subheading "SEC Filings." These reports are also available on the SEC's website at *www.sec.gov*. From time to time, we also use our website to webcast our earnings calls and certain of our meetings with investors or other members of the investment community.

We have adopted a written code of conduct that serves as the code of ethics applicable to our directors, officers and employees, in accordance with applicable laws and rules promulgated by the SEC and the New York Stock Exchange. In the event that we make any changes (other than by a technical, administrative or non-substantive amendment) to, or provide any waivers from, the provisions of our code of conduct applicable to our directors or executive officers, we intend to disclose these events on our website or in a report on Form 8-K filed with the SEC. The code of conduct, as well as copies of our guidelines on significant governance issues and the charters of our key board committees, are also available in the "Governance" section of our website at *www.lumen.com/en-us/about/governance* or in print to any shareholder who requests them by sending a written request to our Corporate Secretary at Lumen Technologies, Inc., 100 CenturyLink Drive, Monroe, Louisiana, 71203.

In connection with filing this report, our chief executive officer and chief financial officer made the certifications regarding our financial disclosures required under the Sarbanes-Oxley Act of 2002, and its related regulations. In addition, during 2023, our chief executive officer certified to the New York Stock Exchange that she was unaware of any violations by us of the New York Stock Exchange's corporate governance listing standards.

As a large complex organization, we are from time to time subject to litigation, disputes, governmental or internal investigations, consent decrees, service outages, security breaches or other adverse events. We typically publicly disclose these occurrences (and their ultimate outcomes) only when we determine these disclosures to be material to investors or otherwise required by applicable law.

We typically disclose material non-public information by disseminating press releases, making public filings with the SEC, or disclosing information during publicly accessible meetings or conference calls. Nonetheless, from time to time we have used, and intend to continue to use, our website and social media accounts to augment our disclosures.

Although at various times we answer questions raised by securities analysts, it is against our policy to disclose to them selectively any material non-public information or other confidential information. Investors should not assume that we agree with any statement or report issued by an analyst with respect to our past or projected performance. To the extent that reports issued by securities analysts contain any projections, forecasts or opinions, such reports are not our responsibility.

Unless otherwise indicated, information contained in this report and other documents filed by us under the federal securities laws concerning our views and expectations regarding the technology or communications industries are based on estimates made by us using data from industry sources and making assumptions based on our industry knowledge and experience. You should be aware that we have not independently verified data from industry or other third-party sources and cannot guarantee its accuracy or completeness.

We have developed methodologies for calculating certain of our statistical data, including route miles, broadband subscribers, broadband-enabled units, on-net buildings and similar metrics. We may calculate these amounts differently from other industry participants.

Our principal executive offices and telephone number are listed on the cover page of this report.

ITEM 1A. RISK FACTORS

The following discussion identifies material factors that could (i) materially and adversely affect our business, financial condition, results of operations or prospects or (ii) cause our actual results to differ materially from our anticipated results, projections or other expectations. The following information should be read in conjunction with the other portions of this annual report, including “Special Note Regarding Forward-Looking Statements”, “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in Item 7 and our consolidated financial statements and related notes in Item 8. All references to "Notes" in this Item 1A of Part I refer to the Notes to Consolidated Financial Statements included in Item 8 of Part II of this report. Please note the following discussion is not intended to comprehensively list all risks or uncertainties faced by us. Our operations or actual results could also be similarly impacted by additional risks and uncertainties that are not currently known to us, that we currently deem to be immaterial, that arise in the future or that are not specific to us. In addition, certain of the risks described below apply only to a part or segment of our business.

Business Risks

Challenges with integrating or modernizing our existing applications and systems could harm our performance.

To succeed, we need to integrate, update and upgrade our existing applications and systems, including many legacy systems from past acquisitions. We cannot assure you we will be able to integrate our legacy IT systems, modernize our infrastructure, timely retire aging or obsolete systems or deploy a master data management platform. These modernization efforts will require efficient allocation of resources, development capacity, greater use of artificial intelligence (“AI”) and other emerging technologies, access to subject-matter experts, development of a sustainable operating model and successful collaboration between legal, privacy and security personnel. Any failure to timely accomplish these initiatives may negatively affect our (i) customer and employee experiences, (ii) ability to meet regulatory, legal or contractual obligations, (iii) network stability, (iv) ability to realize anticipated efficiencies, (v) ability to timely repair infrastructure and respond to service outages or (vi) ability to deliver services to our customers at required speed and scale.

We may not be able to create the global digital experience expected by customers.

Our customers expect us to create and maintain a global digital experience, including (i) automation and simplification of our offerings and (ii) digital self-service access to our products, services and customer support. To do so, we must timely and successfully complete the digital transformation of our operations that is currently underway. Effective digital transformation is a complex, dynamic process requiring efficient allocation and prioritization of resources, simplification of our product portfolio, faster product deployments, retirement of obsolete systems, migration of data and corresponding workforce and system development. We cannot assure you we will be able to timely effect the successful digital transformation necessary to develop or deliver a global digital experience expected by our customers. If we are unable to do so, we could lose existing customers or fail to attract new ones, either of which could prevent us from attaining our financial goals.

We operate in an intensely competitive industry and existing and future competitive pressures could harm our performance.

Each of our business and mass market offerings faces increasingly intense competition, with increased pressure to timely offer digitally integrated services, from a wide range of sources under evolving market conditions that have increased the number and variety of companies that compete with us. Some of our current and potential competitors: (i) offer products or services that are substitutes for our traditional wireline services, including wireless broadband, wireless voice and non-voice communication services, (ii) offer a more comprehensive range of communications products and services, (iii) operate systems that enable them to provision services easier and faster, (iv) have greater financial, provisioning, technical, engineering, research, development, marketing, customer relations or other resources, (v) conduct operations or raise capital at a lower cost, (vi) are subject to less regulation, (vii) have stronger brand names, (viii) have deeper or more long-standing relationships with key customers, or (ix) have larger operations than ours, any of which may enable them to compete more successfully for customers, strategic partners and acquisitions. In recent years, competitive pressures have commoditized pricing for some of our products and services and lowered market prices for many of our other products and services. Continued competitive pressures will likely place further downward pressure on market pricing.

Our ability to successfully compete could be hampered if we fail to timely develop and market innovative technology solutions that address changing customer demands.

The technology and communications industry has been and continues to be impacted by significant technological changes, which are enabling an increasing variety of companies to compete with us. Many of these technological changes are (i) displacing or reducing demand for certain of our services, (ii) enabling the development of competitive products or services, (iii) enabling customers to reduce or bypass use of our networks or (iv) reducing our profit margins. For example, as service providers continue to invest in 5G and low earth orbit satellite networks and services, their services could reduce demand for our network services. Increasingly, customers are demanding more technologically advanced products that suit their evolving needs, including traditional and generative AI services. As we note below, several of our competitors have dedicated substantially more resources to their development. If we fail to develop competitive AI services, our business and financial performance could be adversely impacted.

To remain competitive, we will need to accurately predict and respond to changes in technology, to continue developing products and services attractive to our customers, to timely provision our products and services, to maintain and expand our network to enable it to support customer demands for greater transmission capacity and speeds, and to discontinue outdated products and services on a cost-effective basis. Our ability to do so could be restricted by various factors, including limitations of our existing network, technology, capital or personnel. If we fail at that, we could lose customers or fail to attract new ones.

We may be unable to attract, develop and retain leaders and employees with the right skillsets and technical expertise.

We may be unable to attract and retain skilled and motivated leaders and employees who possess the right skillsets and technical, managerial and development expertise to execute our plans for transformation, innovation and strategic growth. We operate in a highly competitive and expanding industry, where competition for highly skilled employees has grown increasingly intense and competitors have targeted hiring our employees. We have experienced, and may continue to experience, higher than anticipated levels of employee attrition. Further, the increased availability of remote working arrangements, largely driven by the COVID-19 pandemic, has expanded the pool of companies that can compete for our employees and employee candidates. We believe some of our competitors with greater resources and fewer cost constraints than us have from time to time been able to offer compensation, benefits or accommodations in excess of what we are able to offer. These risks to attracting and retaining key personnel may have been exacerbated by the impacts of the low trading price of our common stock, which, as discussed below, restricted our ability in 2023 to offer competitive equity incentive compensation to our key employees. Our failure to successfully attract and retain key personnel could materially adversely impact our business or financial performance.

Under our current work guidelines implemented in 2022, nearly half of our employees work fully from home, and a substantial portion of the remainder work partly from home under "hybrid" work schedules. These work arrangements may impair our ability to maintain our collaborative and innovative culture, and may cause disruptions among our employees, including decreases in productivity, challenges in collaboration between on-site and off-site employees and, potentially, employee dissatisfaction and attrition. If our attempts to operate under a hybrid working environment are not successful, our business could be adversely impacted.

The pandemic, inflation and other events over the past couple years have increased employees' expectations regarding compensation, workplace flexibility and work-home balance. These developments have intensified certain of our above-described challenges and made it relatively more difficult for us to attract and retain top talent.

Uncertainty regarding our future prospects could adversely impact our ability to maintain satisfactory relations with our employees, customers, vendors and others.

Developments related to our negotiations with creditors, coupled with concerns regarding continued declines in our revenues and increased leverage, have (i) created uncertainties about our future ability to improve our financial performance and refinance or extend our upcoming debt maturities and (ii) placed downward pressure on the per share trading price of our common stock.

These uncertainties coupled with a low stock trading price could adversely impact our ability to attract, retain and motivate our employees. We grant equity-based incentive awards to key personnel, the value of which is tied to our stock price, our financial performance or both. During 2023, the low trading price of our stock limited our ability under our 2018 equity incentive plan to grant equity incentive awards in aggregate amounts consistent with our prior practices. Our ability to attract, retain and motivate our employees could be weakened if (i) the anticipated value of such equity-based incentive awards does not materialize, (ii) our equity-based compensation otherwise ceases to be viewed as a valuable benefit, (iii) our total compensation package is not viewed as being competitive, or (iv) we do not obtain the shareholder approval needed to continue granting equity-based incentive awards in the amounts we believe are necessary.

Similarly, customers, vendors, landlords, banks or other third parties may be less willing to transact business with us if they believe our future is uncertain, any of which could adversely impact our business, financial performance, financial position or future prospects.

Under certain specified circumstances, a low stock price could also cause the New York Stock Exchange to initiate proceedings to delist our securities from trading on the New York Stock Exchange. If our securities were ultimately delisted for any reason, we believe the liquidity and market price of our shares would decrease and fewer investors would be willing to own our shares. In addition, a low stock price could limit our ability to raise capital through the issuance of capital stock and could limit the number of financial analysts willing to publish reports about us.

We could be harmed if our reputation is damaged.

We believe our Lumen and other brand names and our reputation are important corporate assets that help us attract and retain customers and talented employees. However, our corporate reputation is susceptible to material damage by events such as disputes with customers or competitors, cyber-attacks or service outages, internal control deficiencies, delivery failures, compliance violations, government investigations or legal proceedings. Similar events impacting one of our competitors could result in negative publicity for our entire industry that indirectly harms our business. We may also experience reputational damage if customers, vendors, employees, advocacy groups, regulators, investors, the media, social media influencers or others criticize our services, operations or public positions. For instance, we could be harmed if our customer experience scores, as measured by "NPS" (Net Promoter Score) and "CHS" (Customer Health Score), for our products and services are low or declining relative to our competitors. In addition, the reputational risk of unauthorized disclosure of confidential company or customer data could increase to the extent our employees inappropriately use social networking sites or other emerging technologies, such as generative AI tools.

There is a risk that negative or inaccurate information about Lumen, even if based on rumor or misunderstanding, could adversely affect our business. Damage to our reputation could be difficult, expensive and time-consuming to repair. Damage to our reputation could also reduce the value and effectiveness of the Lumen brand name and could reduce investor confidence in us, having a material adverse impact on the value of our securities.

We could be harmed by cyber-attacks.

Our vulnerability to cyber-attacks is heightened by several features of our operations, including (i) our material reliance on our owned and leased networks to conduct our operations, (ii) our transmission of large amounts of data over our systems and (iii) our processing and storage of sensitive customer data.

As further described in Item 1C of this annual report, cyber-attacks on our systems may stem from a variety of sources and take many forms. Cyber-attacks can put at risk personally identifiable customer data or protected health information, thereby implicating stringent domestic and foreign data protection laws. These threats may also arise from failure or intrusions of systems owned, operated or controlled by other unaffiliated operators, upon whom we are materially reliant to operate our business. Various other factors could intensify these risks, including, (i) our maintenance of information in digital form stored on servers connected to the Internet, (ii) our use of open and software-defined networks, (iii) the challenges of operating and maintaining our complex multi-continent network composed of legacy and acquired properties, which is more difficult to safeguard than newer fully-integrated networks, (iv) growth in the size and sophistication of our customers and their service requirements, (v) increased use of our network due to greater demand for data services, (vi) our increased incidence of employees working from remote locations and (vii) the increased difficulty of defending against attacks that use AI-generated social engineering, increasingly malicious code and increasingly sophisticated phishing techniques.

As a critical infrastructure service provider, we and our customers are constant targets of cyber-attacks. The number of these attacks against us increased in 2023. Despite our efforts to prevent these events, some of these attacks have resulted in security incidents. On March 27, 2023, we filed with the U.S. Securities and Exchange Commission a Current Report on Form 8-K announcing two cybersecurity incidents, including one that involved a sophisticated threat actor that had accessed our internal information technology systems. Since filing that report, we have taken the measures described therein to assess, contain and remediate both incidents, including working with outside forensic firms. Based on information known to us at this time, we continue to believe that these incidents have neither had nor are likely to have a material adverse impact on our ability to serve our customers or our business, operations or financial results.

We believe the importance of our network to global internet data flows will continue to make it a target to a wide range of threat actors, including nation state actors and other advanced persistent threat actors. Moreover, the risk of incidents is likely to continue to increase due to several factors, including (i) the increasing sophistication of cyber-attacks, (ii) the wider accessibility of cyber-attack tools and (iii) growing threats from Chinese, Russian and other state actors due to heightened geopolitical tensions. It should also be noted that defenses against cyber-attacks currently available to us and others are unlikely to prevent intrusions by a highly-determined, highly-sophisticated threat actor. Consequently, you should assume that we will continue to experience cyber incidents in the future. Thus far, none of our past security incidents have had a material adverse effect on us, and we continue to take steps designed to limit our cyber risks. Nonetheless, we cannot assure you that future cyber incidents or events will not ultimately have a material adverse impact on our ability to serve our customers or our business, operations or financial results.

Although we maintain insurance coverage that may, subject to policy terms and conditions (including self-insured deductibles, coverage restrictions and monetary coverage caps), cover certain aspects of our cyber risks, such insurance coverage may be unavailable or insufficient to cover our losses.

Cyber-attacks could (i) disrupt the proper functioning of our networks and systems, which could in turn disrupt the operations of our customers, (ii) result in the destruction, loss, theft, misappropriation or release of proprietary, confidential, sensitive, classified or otherwise valuable information of ours, our employees, our customers or our customers' end users, (iii) require us to notify customers, regulatory agencies or the public of data incidents, (iv) damage our reputation or result in a loss of business, (v) require us to provide credits for future service to our customers or to offer expensive incentives to retain customers, (vi) subject us to claims by our customers or regulators for damages, fines, penalties, license or permit revocations or other remedies, (vii) result in the loss of industry certifications or (viii) require significant management attention or financial resources to remedy the resulting damages or to change our systems. Any or all of the foregoing developments could have a material adverse impact on us.

We could be harmed by outages in our network or various platforms, or other failures of our services.

From time to time in the ordinary course of our business, we experience outages in our network, hosting, cloud or IT platforms, or failures of our products or services (including basic and enhanced 911 emergency services) to perform in the manner anticipated. These disruptions expose us to several of the same risks listed above for cyber-attacks, including the loss of customers, the issuance of credits or refunds, and regulatory fines. We remain vulnerable to future disruptions due to several factors, including the challenges of maintaining and replacing aging or obsolete network elements, human error, continuous changes in our network, the introduction of new products or technologies, vulnerabilities in our vendors or supply chain, aberrant employees and hardware and software limitations. The process for remediating any interruptions, outages, delays or cessations of service could be more expensive, time-consuming, disruptive and resource intensive than planned. Delayed sales, lower margins, fines or lost customers resulting from future disruptions could have a material adverse impact on our business, reputation, results of operations, financial condition, cash flows and stock price.

Several of our services continue to experience declining revenue, and our efforts to offset these declines may not be successful.

Primarily as a result of the competitive and technological changes discussed above, we have experienced a prolonged systemic decline in our local voice, long-distance voice, network access and private line revenues. Consequently, we have experienced declining consolidated revenues (excluding acquisitions) for a prolonged period and have not been able to realize cost savings sufficient to fully offset the decline. More recently, we have experienced declines in revenue derived from a broader array of our products and services, including those marketed to our enterprise customers and customers with global locations. We have thus far been unable to reverse our annual revenue losses (excluding acquisitions). In addition, most of our more recent product and service offerings generate lower profit margins and may have shorter lifespans than our traditional communication services, and some can be expected to experience slowing or no growth in the future. Some of our new product offerings have reduced or displaced our sale of older product offerings. Accordingly, we may not be successful in attaining our goal of achieving future revenue growth.

Our operations, financial performance and liquidity are materially reliant on key suppliers, vendors and other third parties.

Our ability to conduct our operations could have a material adverse impact on us if certain of our arrangements with third parties were terminated, including those further described below.

Reliance on other communications providers. To offer certain services in certain of our markets, we must either purchase services or lease network capacity from, or interconnect our network with, the infrastructure of other communications carriers or cloud companies who typically compete against us in those markets. Our reliance on these supply or interconnection arrangements limits our control over the delivery and quality of our services. In addition, we are exposed to the risk that other companies may be unwilling or unable to continue or renew these arrangements in the future. Those risks are heightened when the other company is a competitor who may benefit from terminating the agreement or imposing price increases. Additionally, several companies rely on our network to transmit their data or voice traffic. Their reliance on our network exposes us to the risk that they may transfer all or a portion of this traffic from our network to alternative networks owned, constructed or leased by them, thereby reducing our revenue. Certain of our hyperscaler customers have built infrastructure that has reduced their reliance on us.

Reliance on key suppliers and vendors . We depend on a limited number of suppliers and vendors to provide us, directly or through other suppliers, with equipment and services relating to our network infrastructure, including fiber optic cable, software, optronics, transmission electronics, digital switches, routing equipment, customer premise equipment, and related components. We also rely on software and service vendors or other parties to assist us with operating, maintaining and administering our business, including billing, security, provisioning and general operations. Our operations could be adversely affected if any of these vendors experience business interruptions, security incidents, litigation or other issues that interfere with their ability to deliver their products or services on a timely basis.

Reliance on key licensors. We rely on key technologies licensed from third parties to deliver certain of our products and services. Our agreements with these licensors may expire or be terminated, and some of the licenses may not be available to us in the future on terms acceptable to us or at all. Moreover, if we incorporate licensed technology into our network, we may have limited flexibility to deploy different technologies from alternative licensors.

Reliance on key customer contracts. We have several complex high-value national and global customer contracts. These contracts are frequently impacted by a variety of factors that could reduce or eliminate the profitability of these contracts. Moreover, we would be adversely impacted if we fail to renew major contracts upon their expiration.

Reliance on landowners. We rely on rights-of-way, colocation agreements, franchises and other authorizations granted by governmental bodies, railway companies, utilities, carriers and other third parties to locate a portion of our network equipment over, on or under their respective properties. A significant number of these authorizations are scheduled to lapse over the next five to ten years, unless we are able to extend or renew them. Further, some of our operations are subject to licensing and franchising requirements imposed by municipalities or other governmental authorities. Our operations could be adversely affected if any of these authorizations are cancelled, or otherwise terminate or lapse, or if the landowner requests price increases. Similarly, our buildout plans can be delayed if we cannot receive necessary landowner authorizations or governmental permits. We cannot assure you we will be able to successfully extend these arrangements when their terms expire, or to enter into new arrangements that may be necessary to implement our network expansion opportunities.

Climate change could disrupt our operations, cause us to incur substantial additional capital and operating costs or negatively affect our business.

A substantial number of our domestic facilities are located in coastal states, which subjects them to the risks associated with severe tropical storms, hurricanes and tornadoes, and many other of our facilities are subject to the risk of earthquakes, floods, fires, tornadoes or other similar casualty events. From time to time these events (including Hurricane Ian in 2022 in Florida) have disrupted our operations, and similar future events could cause substantial damages, including downed transmission lines, flooded facilities, power outages, fuel shortages, network delays or failures, damaged or destroyed property and equipment, and work interruptions. Due to substantial deductibles, coverage limits and exclusions, and limited availability, we have typically recovered only a portion of our losses through insurance. Our system redundancy and other measures we take to protect our infrastructure and operations from the impacts of such events may be ineffective or inadequate to sustain our operations following such events. Any of these occurrences could result in lost revenues from business interruption, damage to our reputation and reduced profits.

Climate change may increase the frequency or severity of natural disasters and other extreme weather events in the future, which would increase our exposure to the above-cited risks and could disrupt our supply chain from our key suppliers and vendors.

Our environmental, social and governance (ESG) commitments, programs and disclosures may expose us to reputational, legal and business risks.

Our reputation and brands could be impacted by our public commitments to various corporate environmental, social and governance (ESG) initiatives, including our political contributions, our advocacy positions, and our goals for sustainability, inclusion and diversity. These initiatives, goals, or commitments could be difficult to achieve and costly to implement. To the extent that our required and voluntary disclosures about ESG matters increase, we could be criticized for their accuracy, adequacy, or completeness. We could fail to achieve, or be perceived to fail to achieve, our ESG-related initiatives, goals, or commitments. In addition, we could be criticized for the timing, scope or nature of these initiatives, goals, commitments, or for any revisions to them. Our actual or perceived failure to achieve our ESG-related initiatives, goals, commitments, or to meet evolving stakeholder expectations or standards could adversely impact us by resulting in legal or regulatory proceedings against us, customer or employee attrition, reputational damage, or other negative impacts on our business.

We face other business risks.

We face other business risks, including among others, (i) the difficulties of managing and administering an organization that offers a complex set of products to a diverse range of customers across several continents, (ii) the risks and uncertainties inherent in acquiring or disposing of businesses, or engaging in other strategic transactions, and (iii) the adverse effects of terrorism, rioting, vandalism or social unrest.

Legal and Regulatory Risks

We are subject to an extensive, evolving regulatory framework that could create operational or compliance costs.

As explained in greater detail elsewhere in this annual report, (i) our domestic operations are regulated by the FCC and other federal, state and local agencies and (ii) our international operations are regulated by a wide range of various foreign and international bodies. We cannot assure you we will be successful in obtaining or retaining all regulatory licenses necessary to carry out our business in our various markets. Even if we are, the prescribed service standards and conditions imposed on us under these licenses and related laws may increase our costs, limit our operational flexibility or result in third-party claims.

We are subject to numerous requirements and interpretations under various international, federal, state and local laws, rules and regulations, which are often quite detailed and occasionally in conflict with each other. Accordingly, we cannot ensure we will always be considered to be in compliance with all these requirements at any single point in time.

Various governmental agencies, including state attorneys general with jurisdiction over our operations, have routinely in the past investigated our business practices either in response to customer complaints or on their own initiative, and are expected to continue to do the same in the future. Certain of these investigations have resulted in substantial fines in the past. On occasion, we have resolved such matters by entering into consent decrees, which are court orders that frequently restrict our future conduct. If breached by us, these consent decrees expose us not only to contractual remedies, but also to judicial enforcement via contempt of court proceedings, any of which could have material adverse consequences. Additionally, future investigations can potentially result in enforcement actions, litigation, fines, settlements or reputational harm, or could cause us to change our sales practices or operations.

Our prior participation in the FCC's CAF II program and current participation in the FCC's RDOF program subjects us to certain financial risks. If we are not in compliance with FCC measures by the end of the CAF II and RDOF programs, we could incur substantial penalties or forfeitures, including but not limited to being suspended or disbarred from future governmental programs or contracts for a significant period of time, which could have a material adverse impact on our financial condition.

We provide products or services to various federal, state and local agencies. Our failure to comply with complex governmental regulations and laws applicable to these programs, or the terms of our governmental contracts, could result in us suffering substantial negative publicity or penalties, being suspended or debarred from future governmental programs or contracts for a significant period of time and in certain instances could lead to the revocation of our FCC licenses. Moreover, certain governmental agencies frequently reserve the right to terminate their contracts for convenience or if funding is unavailable. If our governmental contracts are terminated for any reason, or if we are suspended or debarred from governmental programs or contracts, it could have a material adverse impact on our results of operations and financial condition.

A variety of state, national, foreign and international laws and regulations apply to the collection, use, retention, protection, security, disclosure, transfer and other processing of personal and other data. The European Union and other international regulators, as well as some state governments, have recently enacted or enhanced data privacy legal requirements, and other governments are considering establishing similar or stronger protections. Many of these laws are complex and change frequently and often conflict with the laws in other jurisdictions. Some of our customers impose similar requirements on us that are equally or more demanding. If we fail to comply with any of these governmental or contractual requirements, we could incur potential substantial penalties and reputational damage.

Adapting and responding to changing regulatory requirements has historically materially impacted our operations. We believe evolving regulatory developments and regulatory uncertainty could continue to have a material impact on our business. In particular, our business could be materially impacted if the U.S. Congress amends or eliminates current federal law limitations on the liability of private network providers, such as us, against claims related to third party content stored or transmitted on private networks, as currently proposed by certain governmental officials, legislative leaders and consumer interest groups. We could also be materially affected if currently pending proposals to increase the regulation of internet service providers or to further strengthen data privacy laws are implemented. In addition, federal and state agencies that regulate the support program payments we receive or the fees that we charge for certain of our regulated services can, and from time to time do, reduce the amounts we receive or can charge. The variability of these laws could also hamper the ability of us and our customers to plan for the future or establish long-term strategies.

Third-party content stored or transmitted on our networks could result in liability or otherwise damage our reputation.

While we disclaim liability for third-party content in most of our service contracts, as a private network provider we potentially could be exposed to legal claims relating to third-party content stored or transmitted on our networks. Such claims could involve, among others, allegations of defamation, invasion of privacy, copyright infringement, or aiding and abetting restricted activities such as online gambling or pornography. Although we believe our liability for these types of claims is limited under current law, suits against other carriers have been successful and we cannot assure you that our defenses will prevail. Such third-party content could also result in adverse publicity and damage our reputation. Moreover, as noted above, pending proposals to change the law could materially heighten our legal exposure.

Our pending legal proceedings could have a material adverse impact on us.

There are several potentially material proceedings pending against us. Results of these legal proceedings cannot be predicted with certainty. As of any given date we could have exposure to losses under proceedings in excess of our accrued liability. For each of these reasons, any of the proceedings described in Note 18—Commitments, Contingencies and Other Items, as well as current litigation not described therein or future litigation, could have a material adverse effect on our business, reputation, financial position, operating results, the trading price of our securities and our ability to access the capital markets. We can give you no assurances as to the ultimate impact of these matters on us.

We may not be successful in protecting and enforcing our intellectual property rights.

We rely on various patents, copyrights, trade names, trademarks, service marks, trade secrets and other similar intellectual property rights, as well as confidentiality agreements and procedures, to establish and protect our proprietary rights. For a variety of reasons, however, these steps may not fully protect us, including due to inherent limitations on the ability to enforce these rights. If we are unsuccessful in protecting or enforcing our intellectual property rights, our business, competitive position, results of operations and financial condition could be adversely affected.

Issues related to the development and use of artificial intelligence (AI) could give rise to legal or regulatory actions, damage our reputation or otherwise materially harm our business.

We currently incorporate AI technology in certain of our products and services and in our business operations. AI is currently being developed in a highly competitive and rapidly evolving environment by a wide variety of technology companies, many of which are dedicating substantially more resources than we are to research and development initiatives. Due to the complexity of its design and algorithms, AI presents various risks and challenges, and its use could have unintended adverse consequences. While we aim to develop and use AI responsibly and attempt to identify and mitigate ethical and legal issues presented by its use, we may be unsuccessful in identifying or resolving issues before they arise. The Company's use of AI may give rise to risks related to harmful content, inaccurate output, bias, intellectual property infringement or misappropriation, defamation, privacy incidents, and cybersecurity vulnerabilities, among others. The United States, the European Union and other governmental bodies have taken initial steps to regulate AI, which could ultimately increase AI’s legal risks or decrease its usefulness. For all these reasons, we cannot assure you that our use of AI will not harm our business, operations or reputation.

We have been accused of infringing the intellectual property rights of others and will likely face similar accusations in the future.

We routinely receive notices from third parties or are named in lawsuits filed by third parties claiming we have infringed or are infringing their intellectual property rights. We are currently responding to several of these notices and claims and expect this industry-wide trend will continue. If these claims succeed, we could be required to pay significant monetary damages, to cease using the applicable technology or to make royalty payments to continue using the applicable technology. If we are required to take one or more of these actions, our revenues or profit margins may decline, our operations could be materially impaired or we may be required to stop selling or redesign one or more of our products or services, any of which could have a material adverse impact on our business. Similarly, from time to time, we may need to obtain the right to use certain patents or other intellectual property from third parties to be able to offer new products and services. If we cannot obtain rights to use any required technology from a third party on reasonable terms, our ability to offer new products and services may be prohibited, restricted, made more costly or delayed.

Failure to extend or renegotiate our collective bargaining agreements or work stoppages could have a material impact on us.

As of December 31, 2023, approximately 21% of our employees were members of various bargaining units represented by labor unions. Although we have agreements with these labor unions, we cannot predict the outcome of our future negotiations of these agreements. We may be unable to reach new agreements, and union employees may engage in strikes, work slowdowns or other labor actions, which could materially disrupt our ability to provide services and increase our costs. Even if we succeed in reaching new or replacement agreements, they may impose significant new costs on us that impair our competitive position.

Our international operations expose us to various regulatory, currency, tax, legal and other risks.

Our international operations are subject to U.S. and non-U.S. laws and regulations regarding operations in international jurisdictions in which we provide services, either directly or indirectly through our contractual arrangements with other carriers. These numerous and sometimes conflicting laws and regulations include anti-corruption laws, anti-competition laws, trade restrictions, economic sanctions, tax laws, immigration laws, environmental laws, privacy laws and accounting requirements. Many of these laws are complex and change frequently. There is a risk that these laws or regulations may materially restrict our ability to deliver services in various international jurisdictions or expose us to the risk of fines, penalties or license revocations if we are determined to have violated applicable laws or regulations. Additionally, these laws or regulations may potentially impact our customers and result in foregone business or penalties to us if we fail to comply with any applicable sanctions or restrictions on our activities.

Many non-U.S. laws and regulations relating to communications services are more restrictive than U.S. laws and regulations. We are subject to the GDPR of the European Union and the United Kingdom, as well as various other laws governing privacy rights, data protection and cybersecurity laws in other regions. These laws and regulations continue to proliferate and evolve, are becoming more complex and increasingly conflict among the various countries in which we operate, which has resulted in greater compliance risk and cost for us. Moreover, many countries are still in the early stages of providing for and adapting to a liberalized telecommunications market, which could make it more difficult for us to obtain licenses and conduct our operations.

In addition to these international regulatory risks, some of the other risks inherent in conducting business internationally include: economic, social and political instability, with the attendant risks of terrorism, kidnapping, extortion, civic unrest, potential seizure or nationalization of assets; currency and exchange controls, repatriation restrictions and fluctuations in currency exchange rates, problems collecting accounts receivable; the difficulty or inability in certain jurisdictions to enforce contract or intellectual property rights; reliance on certain third parties with whom we lack extensive experience; supply chain challenges; and challenges in securing and maintaining the necessary physical and telecommunications infrastructure.

Our operations and financial results could be impacted by changes in multilateral conventions, treaties, tariffs or other arrangements between or among sovereign nations, including most recently Brexit.

Media reports concerning our legacy infrastructure could expose us to governmental actions, removal costs, litigation, compliance costs, penalties or reputational damage.

Media reports issued in mid-2023 alleged that certain lead-sheathed cables that are part of our copper-based network infrastructure pose public health and environmental risks. Such allegations may subject us to legislative or regulatory actions, removal costs, litigation, compliance costs or penalties. Accordingly, we may incur substantial expenses, which could have a material adverse impact on our financial results or condition.

We may also experience reputational harm from negative assertions about the public health or environmental impact of our lead-sheathed cables, which could adversely affect our business, even if such allegations ultimately prove to be inaccurate. Such damage to our reputation could be difficult, expensive and time-consuming to repair. Damage to our reputation could reduce investor confidence in us and have a material adverse impact on the value of our securities.

Financial Risks

Our significant debt levels expose us to a broad range of risks.

As of December 31, 2023, we had approximately $11.4 billion of outstanding consolidated secured indebtedness, $8.5 billion of outstanding consolidated unsecured indebtedness (excluding (i) finance lease obligations, (ii) unamortized premiums, net and (iii) unamortized debt issuance costs) and approximately $1.8 billion of unused borrowing capacity under our revolving credit facility.

Our significant levels of debt and related debt service obligations could adversely affect us in several respects, including:

- requiring us to dedicate a substantial portion of our cash flow from operations to the payment of interest and principal on our debt, thereby reducing the funds available to us for other purposes, including acquisitions, capital expenditures and strategic initiatives;
- hindering our ability to capitalize on business opportunities and to plan for or react to changing market, industry, competitive or economic conditions;
- making us more vulnerable to economic or industry downturns, including interest rate increases (especially with respect to our variable rate debt);
- placing us at a competitive disadvantage compared to less leveraged companies;
- adversely impacting other parties' perception of Lumen, including but not limited to existing or potential customers, vendors, employees or creditors;
- making it more difficult or expensive for us to obtain any necessary future financing or refinancing, including the risk that this could force us to sell assets or take other less desirable actions to raise capital; and
- increasing the risk that we may not meet the financial or non-financial covenants contained in our debt agreements or timely make all required debt payments, either of which could result in the acceleration of some or all of our outstanding indebtedness.

The effects of each of these factors could be intensified if we increase our borrowings or experience any downgrade in our credit ratings or those of our affiliates. Subject to certain limitations and restrictions, the current terms of our debt instruments and our subsidiaries' debt instruments permit us or them to incur additional indebtedness.

We expect to periodically require financing, and we cannot assure you we will be able to obtain such financing on terms that are acceptable to us, or at all.

We expect to periodically require financing in the future to refinance existing indebtedness and potentially for other purposes. Our ability to arrange additional financing will depend on, among other factors, our financial position, performance, credit ratings, and debt covenants. Prior allegations that we have breached covenants in our credit documents could dissuade potential lenders from extending credit to us, unless and until we satisfactorily address these concerns through the execution of additional credit agreements, the receipt of waivers or other similar actions. Our ability to obtain additional financing could also depend on prevailing market conditions, which could be adversely affected by (i) general market conditions, such as disruptions in domestic or overseas sovereign or corporate debt markets, geo-political instabilities, trade restrictions, pandemics, contractions or limited growth in the economy or other similar adverse economic developments in the U.S. or abroad, and (ii) specific conditions in the communications industry. Instability in the domestic or global financial markets has from time to time resulted in periodic volatility and disruptions in capital markets that have partially or severely limited the ability of leveraged companies like us to obtain debt financing. For these and other reasons, we can give no assurance additional financing for any of these purposes will be available on terms acceptable to us, or at all.

If we are unable to make required debt payments or refinance our debt, we would likely have to consider other options, such as selling assets, issuing additional securities, cutting or delaying costs or otherwise reducing our cash requirements, or negotiating with our lenders to restructure our applicable debt. Our current and future debt instruments may restrict, or market or business conditions may limit, our ability to complete some of these actions on favorable terms, or at all. For these and other reasons, we cannot assure you we could implement these steps in a sufficient or timely manner, or at all. Nor can we assure you that these steps, even if successfully implemented, would not be detrimental to our operations, financial performance or future prospects.

We have a highly complex debt structure, which could impact the rights of our investors.

Lumen Technologies, Inc. and various of its subsidiaries owe substantial sums pursuant to various debt and financing arrangements, certain of which are guaranteed by other principal subsidiaries. Over half of the debt of Lumen Technologies, Inc. is guaranteed by certain of its principal domestic subsidiaries, some of which have pledged substantially all of their assets (including certain of their respective subsidiaries) to secure their guarantees. The remainder of the debt of Lumen Technologies, Inc. is neither guaranteed nor secured. Over half of the debt of Level 3 Financing, Inc. is (i) secured by a pledge of substantially all of its assets and (ii) guaranteed on a secured basis by certain of its affiliates. The remainder of the debt of Level 3 Financing, Inc. is not secured by any of its assets, but is guaranteed on an unsecured basis by certain of its affiliates. As of the date of this annual report, substantial amounts of debt are also owed by two direct or indirect subsidiaries of Qwest Communications International Inc. Most of the over 200 subsidiaries of Lumen Technologies, Inc. have neither borrowed money nor guaranteed any of the debt of Lumen Technologies, Inc. or its affiliates. As such, investors in our consolidated debt instruments should be aware that (i) determining the priority of their rights as creditors is a complex matter which is substantially dependent upon the assets and earning power of the entities that issued or guaranteed (if any) the applicable debt and (ii) a substantial portion of such debt is structurally subordinated to all liabilities of the non-guarantor subsidiaries of Lumen Technologies, Inc. to the extent of the value of those subsidiaries that are obligors.

Our various debt agreements include restrictions and covenants that could (i) limit our ability to conduct operations or borrow additional funds, (ii) restrict our ability to engage in inter-company transactions, and (iii) lead to the acceleration of our repayment obligations in certain instances.

Under our consolidated debt and financing arrangements, the issuer of the debt is subject to various covenants and restrictions, the most restrictive of which pertain to the debt of Lumen Technologies, Inc. and Level 3 Financing, Inc.

Lumen Technologies, Inc.'s senior secured credit facilities and secured notes contain several significant limitations restricting our ability to, among other things, borrow additional money or issue guarantees; pay dividends or other distributions to shareholders; make loans; create liens on assets; sell assets; transact with its affiliates and engage in mergers, consolidations or other similar transactions. These restrictive covenants could have a material adverse impact on our ability to operate or reconfigure our business, to issue additional priority debt, to pursue acquisitions, divestitures or strategic transactions, or to otherwise pursue our plans and strategies.

The debt and financing arrangements of Level 3 Financing, Inc. contain substantially similar limitations that restrict their operations on a standalone basis as a separate restricted group. Consequently, certain of these covenants may significantly restrict our ability to engage in transactions with Level 3, including receiving cash from Level 3, or distributing cash from Level 3 to other of our affiliated entities.

Lumen Technologies, Inc.'s senior secured credit facilities, as well as the term loan debt of Qwest Corporation, also contain financial maintenance covenants which are described further in Note 7—Long-Term Debt and Credit Facilities.

The failure of Lumen Technologies, Inc. or any of its subsidiaries to comply with the above-described restrictive or financial covenants could result in an event of default, which, if not cured or waived, could accelerate our debt repayment obligations. Certain of our debt instruments have cross-default or cross-acceleration provisions. When present, these provisions could have a wider impact on liquidity than might otherwise arise from a default or acceleration of a single debt instrument.

Certain debtholders of Level 3 may seek to claim that Level 3's use of proceeds following the sale of its Latin American business resulted in potential defaults under its credit documents.

On July 25, 2023, the Company received a letter from representatives purporting to act on behalf of holders of approximately 37% of Lumen Technologies, Inc.'s funded debt and approximately 56% of Level 3's funded debt requesting a meeting to discuss the Company's upcoming debt maturities as well as what the letter referred to as an apparent event of default by Level 3 relating to Level 3's use of proceeds from the divestiture of its Latin American business.

If the transactions contemplated by the TSA are consummated, the participating creditors would waive and release us from any claims or remedies arising out of any such breaches to the extent permitted under the Company's debt agreements and applicable law. However, there can be no assurance that these transactions will be consummated, or that other creditors will not seek to assert claims against us. If the transactions contemplated by the TSA are not consummated, there can be no assurance that participating creditors would not attempt to deliver purported notices of default, or seek to declare the principal amount of their debt holdings due and payable, together with accrued interest. Any such acceleration also could allow lenders under our senior secured credit facilities to declare all funds borrowed to be due and payable, to terminate their commitments thereunder, and to cease making further loans. Secured debtholders could also institute foreclosure proceedings against their collateral. Although the Company would vigorously dispute any and all such actions, any such actions may result in an outcome that could have a material adverse impact on our business, operations and financial condition, and any such actions could force us to seek bankruptcy protection. In addition, responding to or defending against any claims of default, including through litigation, may require us to expend significant funds and management time and attention, and could adversely impact our ability to obtain financing in the future or to refinance our existing indebtedness.

The transactions contemplated by the TSA may not be consummated as contemplated, on the currently expected timeline or at all, and even if such transactions are consummated, we may not achieve their anticipated benefits.

We expect that the completion of the transactions contemplated by the TSA will enhance our liquidity and extend our debt maturities. However, completion of these transactions is subject to the satisfaction of certain conditions and the TSA permits certain specified lender groups and the Company to terminate the agreement under various specified circumstances.

As a result, any or all of the transactions may not be consummated as originally contemplated, on the currently expected timeline, or at all. Accordingly, we may not be able to realize the expected benefits from these transactions on a timely basis or at all. Even if we are successful in completing the transactions contemplated by the TSA, we may not realize some or all of the expected benefits from such transactions. We have incurred, and will continue to incur, significant costs, expenses and fees for professional services and other transaction costs in connection with the transactions contemplated by the TSA, and these fees and costs are payable by us regardless of whether such transactions are consummated.

If we are successful in completing the transactions contemplated by the TSA, the Company will be subject to higher levels of interest, which could have important consequences, including, (i) limiting our ability to obtain additional financing to fund future working capital, capital expenditures, acquisitions or other general corporate requirements, and increasing our cost of borrowing; (ii) requiring a substantial portion of our cash flows to be dedicated to payments on our obligations instead of for other purposes; and (iii) each of the other factors specified above under the heading ""–Our significant debt levels expose us to a broad range of risks."

In addition, the agreements that will govern the Company's indebtedness to be executed in connection with the consummation of the transactions contemplated by the TSA will contain significant additional restrictions that could limit the Company's ability to engage in activities that may be in our long-term best interest, including certain restrictions on our ability to incur indebtedness, incur liens, enter into mergers or consolidations, dispose of assets, enter into affiliate transactions, pay dividends, make acquisitions and make investments, loans and advances. These restrictions may affect our ability to execute our business strategies, limit our ability to raise additional debt or equity financing needed to operate our business, including during economic or business downturns, and limit our ability to compete effectively or take advantage of new business opportunities. Our failure to comply with those covenants could result in an event of default which, if not cured or waived, could result in the acceleration of all our debt.

Our cash flows may not adequately fund all of our cash requirements.

Each segment of our business is very capital intensive. We expect to continue to require significant capital to pursue our Quantum Fiber buildout plans and to otherwise maintain, upgrade and expand our network infrastructure and product offerings, based on several factors, including (i) changes in customers' service requirements; (ii) our need to replace aging or obsolete infrastructure; (iii) our continuing need to expand and improve our network to remain competitive and meet customer demand; and (iv) our regulatory commitments. Any failure to make appropriate capital expenditures could adversely impact our financial performance or prospects. We will also continue to need substantial amounts of cash to meet our fixed commitments and other business objectives, including without limitation funding our debt repayments, operating costs, maintenance expenses, tax obligations, periodic pension contributions and other benefits payments. As discussed elsewhere in this annual report, our revenues have decreased for several years, which, coupled with asset divestitures and other factors, has placed downward pressure on our cash flows. For all these reasons, we cannot assure you our future cash flows from operating activities will be sufficient to fund all of our cash requirements in the manner currently contemplated.

As a holding company, we rely on payments from our operating companies to meet our obligations.

As a holding company, substantially all of our income and operating cash flow is dependent upon the earnings of our subsidiaries and their distribution of those earnings to us in the form of dividends, loans or other payments. As a result, we rely upon our subsidiaries to generate cash flows in amounts sufficient to fund our obligations, including the payment of our long-term debt. Our subsidiaries are separate and distinct legal entities and have no obligation to pay any amounts owed by us, except to the extent they have guaranteed such payments. Similarly, subject to limited exceptions for tax-sharing or cash management purposes, our non-guarantor subsidiaries have no obligation to make any funds available to us to repay our obligations, whether by dividends, loans or other payments. As discussed in greater detail elsewhere herein, restrictions imposed by credit instruments or other agreements applicable to Level 3 and certain of our other subsidiaries limit the amount of funds our subsidiaries are permitted to transfer to us, including the amount of dividends that may be paid to us. Moreover, our rights to receive assets of any subsidiary upon its liquidation or reorganization would be effectively subordinated to the claims of creditors of that subsidiary, including trade creditors. In addition, the laws under which our subsidiaries were organized typically restrict the amount of dividends they may pay. The ability of our subsidiaries to transfer funds could be further restricted under applicable state or federal tax laws, regulatory orders or regulations. For all these reasons, you should not assume our subsidiaries will be able in the future to generate and distribute to us cash in amounts sufficient to fund our cash requirements.

We may not be able to fully utilize our NOLs.

As of December 31, 2023, we had approximately $800 million of federal net operating loss carryforwards ("NOLs"), which remain subject to limitations under Section 382 of the Internal Revenue Code and related regulations ("Section 382"). These limitations could restrict our ability to use these NOLs in the amounts we project. In an effort to safeguard our NOLs, we have maintained an NOL rights agreement which is scheduled to lapse in late 2026, assuming it is ratified by our shareholders at our 2024 annual shareholder meeting. At December 31, 2023, we also had substantial state NOLs which we believe are subject to legal and practical limitations on our ability to realize their full benefit. We cannot assure you we will be able to utilize these NOLs as projected or at all.

Increases in costs for pension and healthcare benefits for our active and retired employees may have a material impact on us.

As of December 31, 2023, our company-sponsored benefit plans that cover our current and former U.S.-based employees had approximately 24,000 active employee participants, approximately 55,000 active and retired employees and surviving spouses eligible for post-retirement healthcare benefits, approximately 21,000 pension retirees and approximately 7,000 former employees with vested pension benefits. As of such date, our domestic pension plans and our other domestic post-retirement benefit plans were substantially underfunded from an accounting standpoint. We also maintain benefit plans for a much smaller base of our non-U.S. employees. The cost to fund the pension and healthcare benefit plans for our active and retired employees has a significant impact on our profitability. Our costs of maintaining our pension and healthcare plans, and the future funding requirements for these plans, are affected by several factors, including investment returns on funds held by our applicable plan trusts; changes in prevailing interest rates and discount rates or other factors used to calculate the funding status of our plans; increases in healthcare costs generally or claims submitted under our healthcare plans specifically; the longevity and payment elections of our plan participants; changes in plan benefits; and the impact of the continuing implementation, modification or potential repeal of current federal healthcare and pension funding laws and regulations promulgated thereunder. Increased costs under these plans could reduce our profitability and increase our funding commitments to our pension plans.

See Note 11—Employee Benefits for additional information regarding the funded status of our pension plans and our other post-retirement benefit plans.

Lapses in our disclosure controls and procedures or internal control over financial reporting could materially and adversely affect us.

We maintain (i) disclosure controls and procedures designed to provide reasonable assurances regarding the accuracy and completeness of our SEC reports and (ii) internal control over financial reporting designed to provide reasonable assurance regarding the reliability and compliance with U.S. generally accepted accounting principles (“GAAP”) of our financial statements. We cannot assure you these measures will be effective. Our management previously identified two material weaknesses that, while successfully remediated during 2019, were costly to remediate and delayed the filing of our annual report on Form 10-K for the year ended December 31, 2018.

If we are required to record additional intangible asset impairments, we will be required to record a significant charge to earnings and reduce our stockholders' equity.

As of December 31, 2023, approximately 22% of our total consolidated assets reflected on the consolidated balance sheet included in this annual report consisted of goodwill, customer relationships and other intangible assets. From time to time, including most recently in the fourth and second quarter of 2023 and in the fourth quarter of 2022, we have recorded large non-cash charges to earnings in connection with required reductions of the value of our intangible assets. If our intangible assets are determined to be impaired in the future, we may be required to record additional significant, non-cash charges to earnings during the period in which the impairment is determined to have occurred. Any such charges could, in turn, have a material adverse effect on our results of operation or financial condition.

High inflation could continue to adversely impact us.

Although inflation has recently been declining, during the past three years our operations were impacted by the highest domestic inflation rates in decades. If inflation rates remain elevated, our operations will likely continue to be impacted. Potential impacts of high inflation include (i) lower revenue if inflationary pressures cause customers to defer, decrease or cancel their expenditures on our products and services, (ii) lower margins if we cannot offset the higher cost of our labor and supplies by raising our prices or reducing our other expenses, (iii) higher interest costs to the extent inflation places upwards pressure on prevailing interest rates and (iv) as noted above, potential difficulties retaining personnel if we do not match the salary increase expectations of our workforce.

We face other financial risks.

We face other financial risks, including among others the risk that:

- downgrades in our credit ratings or unfavorable financial analyst reports regarding us or our industry could adversely impact the liquidity or market prices of our outstanding debt or equity securities;

- a change of control of us or certain of our affiliates could accelerate a substantial portion of our outstanding indebtedness in an amount that we might not be able to repay; and

- ongoing attempts of the United States, various foreign countries and supranational or international organizations to reform taxes or identify new tax sources could materially impact our taxes, or that one or more of our ongoing tax audits or examinations could result in tax liabilities that differ materially from those we have recognized in our consolidated financial statements.

Divestiture Risks

We may be unable to realize the anticipated benefits of our recently completed divestitures.

In connection with divesting our Latin American and EMEA businesses and a portion of our ILEC business in 2022 and 2023, we completed internal restructurings and entered into multi-year agreements with the purchasers to provide certain transitional services and to provide or receive certain commercial services.

We anticipate that it will be challenging and time-consuming to continue providing transition services to the purchasers of our divested operations. We may incur or experience (i) greater tax or other costs or realize fewer benefits than anticipated under our post-closing agreements with the purchasers, (ii) operational or commercial difficulties segregating the divested assets from our retained assets, (iii) disputes with the purchasers regarding the nature and sufficiency of the transition services we provide or the terms and conditions of our commercial agreements with the purchasers, (iv) potential disputes with creditors concerning the transactions or use of the proceeds therefrom, (v) higher vendor costs due to reduced economies of scale or other similar dis-synergies, (vi) weaker performance to the extent segregation and support of the divested businesses distracts or diverts personnel and resources from the operation, digitization, and transformation of our retained business, (vii) losses or increased inefficiencies from stranded or underutilized assets, (viii) the loss of any customers dissatisfied with our services post-closing, (ix) challenges in retaining and attracting personnel or (x) the loss of vendors or customers due to our inability to assign contracts with their consent.

The divestitures will reduce our future cash flows. If our remaining business fails to perform as expected, the divestitures could exacerbate certain of the other financial risks specified in this Item 1A, including our ability to fund all of our current cash requirements.

General Risk Factors

Unfavorable general economic, societal, health or environmental conditions could negatively impact us.

Unfavorable general economic, societal, health or environmental conditions, including unstable economic and credit markets, or depressed economic activity caused by trade wars, epidemics, pandemics, wars, societal unrest, rioting, civic disturbances, natural disasters, terrorist attacks, environmental disasters, political instability or other factors, could negatively affect our business or operations in a variety of ways.

We currently do not pay dividends to our common shareholders and any decision to adopt or continue a stock repurchase plan is entirely discretionary.

We discontinued paying dividends to our holders of common stock in the fourth quarter of 2022, and have no current plans to pay dividends in respect of our common stock for the foreseeable future.

From time to time we adopt share repurchase plans. Holders of our common stock should be aware that repurchases of our common stock under any such plans are completely discretionary and may be suspended or discontinued at any time and for any reason without prior notice.

Shareholder or debtholder activism efforts could cause a material disruption to our business.

While we always welcome constructive input from our shareholders and regularly engage in dialogue with our shareholders to that end, activist shareholders may from time to time engage in proxy solicitations, advance shareholder proposals or otherwise attempt to effect changes or acquire control over us. Responding to these actions can be costly and time-consuming and may disrupt our operations and divert the attention of our board and management. These adverse impacts could be intensified if activist shareholders advocate actions that are not supported by other shareholders, our board or management. The recent increase in the activism of debtholders could increase the risk of claims being made under our debt agreements.

Our agreements and organizational documents and applicable law could similarly limit another party's ability to acquire us.

A number of provisions in our organizational documents and various provisions of applicable law or our Section 382 rights agreement may delay, defer or prevent a future takeover of us unless the takeover is approved by our board. These provisions (which are described further in our Registration Statement on Form 8-A/A filed with the SEC on March 2, 2015) could deprive our shareholders of any related takeover premium.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Risk management and strategy

As a technology and communications company that globally transmits large amounts of information over our networks, we recognize the critical importance of maintaining the security and integrity of information and systems under our control. We view cybersecurity risk as one of our principal enterprise-wide risks, subject to control and monitoring at various levels of management throughout the Company. We dedicate significant resources towards programs designed to identify, assess, manage, mitigate and respond to cybersecurity threats.

As described in Item 1A “Risk Factors,” several features of our operations heighten our susceptibility to cyber-attacks, including (i) our material reliance on our owned and leased networks to conduct our operations, (ii) our transmission of large amounts of data over our systems and (iii) our processing and storage of sensitive customer data. Cyber-attacks on our systems may stem from a variety of sources, including fraud, malice or sabotage on the part of foreign nations, third parties, vendors, or employees and attempts by outside parties to gain access to sensitive data that is stored in or transmitted across our network. Cyber-attacks can take many forms, including computer hackings, computer viruses, ransomware, worms or other destructive or disruptive software, denial of service attacks, or other malicious activities.

To identify, assess and mitigate cybersecurity risk, we have implemented a global information security management program that includes administrative, technical, and physical safeguards. We leverage a defense-in-depth model to identify, detect, protect and respond to threats to our information systems. Our security operations center provides advanced threat detection and response capabilities. We maintain an insider threat program to detect, investigate and mitigate insider threat risks to Lumen assets, data, services and personnel globally.
Our privacy and cybersecurity policies encompass information security, incident response procedures, and vendor management. Our risk management team works closely with our Information Technology, Privacy, Product, and Operations departments to continuously evaluate emerging cyber risk. We monitor existing or proposed privacy and cybersecurity laws, regulations and guidance that are or may be applicable to us in the regions where we operate, including in the European Union and the United Kingdom where we are subject to GDPR, as well as various other laws governing privacy rights, data protection and cybersecurity in other regions. As a U.S. government contractor we are required to comply with extensive governmental regulations and standards regarding cyber security.

We periodically engage both internal and external auditors and consultants to assess and enhance our program. These independent external auditors and consultants are accredited under various information security standards, including those administered by the International Organization for Standardization and the PCI Security Council. These engagements typically include penetration testing, third-party certifications, compliance assessments, audits, and assessments of vulnerabilities and emerging threats. We also periodically deploy our Internal Audit processes to conduct additional reviews and assessments. We also share and receive threat intelligence with government agencies, cyber analysis centers and cybersecurity associations.

As noted elsewhere in this annual report, we are materially reliant on a variety of third-party service providers to operate our business, which exposes us to the risk of cyber incidents impacting those providers’ systems. We have a vendor risk management program that assesses, manages and oversees risks associated with third-party service providers who have access to our data and systems. We maintain ongoing monitoring to ensure their compliance with our cybersecurity standards.

Despite our efforts to prevent security incidents, (i) some of these attacks have resulted in security incidents (although thus far we do not believe that any of these incidents has resulted in a material adverse effect on our operating results or financial condition) and (ii) future security incidents are likely (some of which could have a material adverse effect on our operating results or financial condition). See Item 1A “Risk Factors” for a further discussion of cybersecurity risks.

We maintain an Incident Response Playbook that provides a set of guidelines for our stakeholders to follow when handling any data incident. This Playbook describes how we assess incidents and how our security team shares information about such incidents with others at Lumen, including senior leadership and, if warranted, with some or all members of the Board of Directors. These escalation provisions, together with our Disclosure Controls and Procedures, are designed to ensure that appropriate representatives throughout the Company are available to assess how to respond to such incidents and make any necessary public notifications.

Our Incident Response Team (“CIRT”) is notified of all cybersecurity incidents, and is responsible for detecting and coordinating responses to security incidents. This team regularly assesses its communication plan to confirm that its members can be alerted quickly in the event of an actual crisis and meet as a team to discuss response options. The CIRT also addresses each incident, unless it determines that an incident is sufficiently serious. In those instances, it will notify our Cyber Security Watch Team, which is responsible for addressing cybersecurity incidents that raise more significant risks.

Our Cyber Security Watch Team (“CSWAT”) is comprised of senior IT, operations, risk, legal and compliance leaders across business segments. In addition to addressing our more significant cyber incidents, CSWAT manages risks from matters related to business continuity, including risks posed by cybersecurity threats, and implements controls to mitigate such operational risks. Among other processes, this team reviews our programs and processes related to information security, third party risk, vendor management, facilities, unplanned downtime, business disruption, business continuity and disaster recovery.

Governance

As part of our overall risk management approach, we prioritize the identification and management of cybersecurity risk at several levels, including Board oversight, executive commitment and employee training. Our Risk and Security Committee, comprised of independent directors from our Board, assists the Board in overseeing our cybersecurity and data privacy risk. Specifically, our Risk and Security Committee, which meets quarterly, (i) receives periodic reports from our Chief Security Officer (“CSO”) on security programs, including incident reports, (ii) reviews risk assessments from information security, privacy, and internal audit management teams with respect to cybersecurity, including the adequacy and effectiveness of the Company’s internal controls regarding cybersecurity; (iii) reviews emerging cybersecurity developments and threats; (iv) reviews compliance with applicable laws and industry standards; and (v) periodically reviews our strategy to mitigate cybersecurity risks, such as our cyber insurance coverage and contingency plans in the event of security incidents or other system disruptions. At least quarterly, our Risk and Security Committee provides reports to the full Board regarding matters recently discussed by the Committee, which enables the full Board to provide additional oversight of our cyber risks and cyber processes. The full Board also reviews our cybersecurity risks in connection with its annual review of our enterprise risk mitigation programs.

Our CSO has worked in the public and private sectors in information security since 1997 and has been a chief security officer since 2017. His technical and process certifications include CISSP, ITIL Foundation, Six Sigma Certified, CISCO CCNP, and CCNA. He oversees the implementation and compliance of our information security standards and mitigation of information security related risks.

We also have management level committees and response teams who support our processes to assess and manage cybersecurity risk as follows:

- The Risk Oversight Committee (“ROC”), whose core members include the CFO, Chief Technology Officer, Chief Product Officer, and General Counsel, is responsible for making risk management decisions to ensure consideration of all relevant factors and alignment with our overall risk mitigation strategy. The ROC also oversees key risk management activity to help ensure accountability, adequacy of resourcing, implementation of Company directives, and alignment of oversight provided by the Board and senior management.

- The Technology Security and Privacy Council, co-chaired by the CSO, Chief Information Officer, and Chief Privacy Officer, brings together IT, legal and internal audit personnel, and other function leads. The Security and Privacy Council provides a forum for these cross-functional members of management to consider emerging technologies, such as artificial intelligence and emerging cybersecurity risks; review cybersecurity and privacy regulations; approve, review and update policies and standards as appropriate; and promote cross-functional collaboration to manage cybersecurity and privacy risks across the enterprise.

At the day-to-day operational level, we maintain an experienced information security team who are tasked with implementing our privacy and cybersecurity program and support the CSO in implementing our detection, reporting, security and mitigation functions. This team and the CSO work to develop and implement tools and processes designed to assist in identifying, containing and remediating cybersecurity incidents, and periodically retain consultants to assist with these activities. We also periodically hold employee trainings on our privacy, cybersecurity and information management policies, conduct phishing tests and generally seek to promote a company-wide awareness of cybersecurity risk through broad-based communications and educational initiatives.

ITEM 2. PROPERTIES

Our property, plant and equipment consists principally of fiber-optic and metallic cables, high-speed transport equipment, electronics, switches, routers, gateway and transmission facilities, central office equipment, land and buildings related to our operations. Our gross property, plant and equipment consisted of the following components:

	As of December 31,	
	2023[5]	2022[5]
Land	2 %	2 %
Fiber, conduit and other outside plant [1]	37 %	37 %
Central office and other network electronics [2]	38 %	39 %
Support assets[3]	16 %	17 %
Construction in progress[4]	7 %	5 %
Gross property, plant and equipment	100 %	100 %

(1) Fiber, conduit and other outside plant consists of fiber and metallic cable, conduit, poles and other supporting structures.

(2) Central office and other network electronics consists of circuit and packet switches, routers, transmission electronics and electronics providing service to customers.

(3) Support assets consist of buildings, data centers, computers and other administrative and support equipment.

(4) Construction in progress includes inventory held for construction and property of the aforementioned categories that is under construction and has not yet been placed in service.

(5) These values exclude assets classified as held for sale.

We own a substantial portion of our telecommunications equipment required for our business. However, we also lease from third parties certain facilities, plant and equipment under various finance and operating lease arrangements. We also own or lease administrative offices in major metropolitan locations both in the United States and internationally. Substantially all of our network electronics equipment is located in buildings or on land that we own or lease, typically within our local service area. Outside of our local service area, our assets are generally located on real property pursuant to an agreement with the property owner or another person with rights to the property. It is possible that we may lose our rights under one or more of these agreements, due to their termination or expiration or in connection with legal challenges to our rights under such agreements.

Our net property, plant and equipment was approximately $19.8 billion and $19.2 billion at December 31, 2023 and 2022, respectively, excluding assets held for sale. Substantial portions of our property, plant and equipment are pledged to secure the long-term debt of our subsidiaries or the guarantee obligations of our subsidiary guarantors. For additional information, see Note 9—Property, Plant and Equipment to our consolidated financial statements in Item 8 of Part II of this report.

We have entered into various agreements regarding our unused office and technical space to reduce our ongoing operating expenses regarding such space.

ITEM 3. LEGAL PROCEEDINGS

The information contained under the subheadings "Principal Proceedings" and "Other Proceedings, Disputes and Contingencies" in Note 18—Commitments, Contingencies and Other Items to our consolidated financial statements included in Item 8 of Part II of this report is incorporated herein by reference.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Our common stock is listed on the New York Stock Exchange ("NYSE") and the Berlin Stock Exchange and is traded under the symbol LUMN and CYTH, respectively.

At February 20, 2024, there were approximately 78,000 stockholders of record, although there were significantly more beneficial holders of our common stock.

Issuer Purchases of Equity Securities

During the fourth quarter of 2022, our Board of Directors authorized a two-year program to repurchase up to an aggregate of $1.5 billion of our outstanding common stock. During the three months ended December 31, 2022, we repurchased and retired 33 million shares of our outstanding common stock in the open market for an aggregate market price of $200 million, or an average purchase price of $6.07 per share. Since then, we have not repurchased any shares of our outstanding common stock under this program. For additional information, see Note 20—Repurchases of Lumen Common Stock to our consolidated financial statements included in Item 8 of Part II of this report.

The following table contains information about shares of our previously-issued common stock that we withheld from employees upon vesting of their stock-based awards during the fourth quarter of 2023 to satisfy the related tax withholding obligations:

Period	Total Number of Shares Withheld for Taxes	Average Price Paid Per Share
October 2023	28,853	$ 1.29
November 2023	95,290	1.37
December 2023	117,524	1.55
Total	241,667	

Equity Compensation Plan Information

See Item 12 of this report.

ITEM 6. [Reserved]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

All references to "Notes" in this Item 7 of Part II refer to the Notes to Consolidated Financial Statements included in Item 8 of Part II of this report. Certain statements in this report constitute forward-looking statements. See "Special Note Regarding Forward-Looking Statements" immediately prior to Item 1 of Part I of this report for factors relating to these statements and "Risk Factors" in Item 1A of Part I of this report for a discussion of certain risk factors applicable to our business, financial condition, results of operations, liquidity or prospects.

Overview

We are a facilities-based technology and communications company that provides a broad array of integrated products and services to our domestic and global business customers and our domestic mass markets customers. We operate one of the world's most interconnected networks. Our platform empowers our customers to swiftly adjust digital programs to meet immediate demands, create efficiencies, accelerate market access and reduce costs, which allows our customers to rapidly evolve their IT programs to address dynamic changes. With approximately 170,000 on-net buildings and 350,000 route miles of fiber optic cable globally, we are among the largest providers of communications services to domestic and global enterprise customers. Our long-haul network throughout North America and Asia Pacific connects to metropolitan fiber networks that we operate.

Divestitures of the Latin American, ILEC and EMEA Businesses

On August 1, 2022, affiliates of Level 3 Parent, LLC, an indirect wholly-owned subsidiary of Lumen Technologies, Inc., sold Lumen’s Latin American business for pre-tax cash proceeds of approximately $2.7 billion.

On October 3, 2022, we and certain of our affiliates sold the portion of our ILEC business conducted primarily within 20 Midwestern and Southeastern states. In exchange, we received $7.5 billion of consideration, which was reduced by approximately $0.4 billion of closing adjustments and partially paid through purchaser's assumption of approximately $1.5 billion of our long-term consolidated indebtedness, resulting in pre-tax cash proceeds of approximately $5.6 billion.

On November 1, 2023, we and certain of our affiliates sold Lumen's operations in Europe, the Middle East and Africa (the "EMEA business") to Colt Technology Services Group Limited, a portfolio company of Fidelity Investments, for pre-tax cash proceeds of $1.7 billion after certain closing adjustments and transaction costs. This consideration is further subject to certain other post-closing adjustments and indemnities set forth in the Purchase Agreement, as amended and supplemented to date.

For more information, see (i) Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses to our consolidated financial statements in Item 8 of Part II of this report and (ii) the risk factors included in Item 1A of Part I of this report.

Changes in the Macroeconomic, Industry and Work Environments

Societal, governmental and macroeconomic changes have impacted us, our customers and our business in several ways since the onset of the COVID-19 pandemic in the U.S. in March 2020. Beginning in the second half of 2020 and continuing into 2023, we rationalized our lease footprint and ceased using 42 underutilized leased property locations. These lease cancellations resulted in accelerated lease costs, including $35 million and $8 million of such costs recognized during the years ended December 31, 2021 and 2023, respectively, but will lower our future operating costs. We did not incur material accelerated lease costs during the year ended December 31, 2022. During the second quarter of 2023, we also donated our Monroe, Louisiana campus and leased back a portion thereof. This donation resulted in a loss recognized during the second and fourth quarters, as discussed further below. In conjunction with our plans to continue to reduce costs, we expect to continue our real estate rationalization efforts and expect to incur additional accelerated real estate costs in future periods.

Additionally, as discussed further elsewhere herein, the pandemic and macroeconomic changes arising therefrom have resulted in (i) increases in certain revenue streams and decreases in others, (ii) operational challenges resulting from inflation and, to a lesser extent, shortages of certain components and other supplies that we use in our business, (iii) delays in our cost transformation initiatives and (iv) delayed decision-making by certain of our customers. None of these effects, individually or in the aggregate, have to date materially impacted our financial performance or financial position.

Industry developments over the past couple years have increased fiber construction demand. The resulting increase in construction labor rates increased the cost of enabling units to be capable of receiving our fiber broadband services. In 2022 and 2021, we believe these factors contributed to a delay in attaining our Quantum Fiber buildout targets.

Continued inflationary pressures, supply constraints or business uncertainty could materially impact our financial results in a variety of ways, including by increasing our expenses, decreasing our revenues, further delaying our network expansion plans or otherwise interfering with our ability to deliver products and services. For additional information on the impacts of the pandemic and the macroeconomic changes arising therefrom, see (i) the remainder of this item, including "—Liquidity and Capital Resources—Overview of Sources and Uses of Cash" and (ii) Item 1A of this report.

Reporting Segments

Our reporting segments are currently organized as follows, by customer focus:

- *Business Segment:* Under our Business segment, we provide our products and services under four sales channels:
    - Large Enterprise: Under our large enterprise sales channel, we provide our products and services to large enterprises, including multinational and global enterprise customers and carriers.
    - Mid-Market Enterprise: Under our mid-market enterprise sales channel, we provide our products and services to medium-sized enterprises directly and through our indirect channel partners.
    - Public Sector: Under our public sector sales channel, we provide our products and services to the public sector, including the U.S. Federal government, state and local governments and research and education institutions.
    - Wholesale: Under our wholesale sales channel, we provide our products and services to a wide range of other communication companies providing wireline, wireless, cable, voice and data center services.
- *Mass Markets Segment.* Under our Mass Markets segment, we provide products and services to residential and small business customers. At December 31, 2023, we served 2.8 million broadband subscribers under our Mass Markets segment.

See Note 17—Segment Information to our consolidated financial statements in Item 8 of Part II of this report for additional information.

We categorize our Business segment revenue among the following products and services categories:

- *Grow*, which includes products and services that we anticipate will grow, including our dark fiber, Edge Cloud services, IP, managed security, software-defined wide area networks ("SD WAN"), secure access service edge ("SASE"), Unified Communications and Collaboration ("UC&C") and wavelengths services;
- *Nurture*, which includes our more mature offerings, including ethernet and VPN data networks services;
- *Harvest*, which includes our legacy services managed for cash flow, including Time Division Multiplexing ("TDM") voice, private line and other legacy services; and
- *Other*, which includes equipment sales, IT solutions and other services.

We categorize our Mass Markets products and services revenue among the following categories:

- *Fiber Broadband*, under which we provide high speed broadband services to residential and small business customers utilizing our fiber-based network infrastructure;
- *Other Broadband*, under which we provide primarily lower speed broadband services to residential and small business customers utilizing our copper-based network infrastructure; and
- *Voice and Other,* under which we derive revenues from (i) providing local and long-distance voice services, professional services, and other ancillary services, and (ii) federal broadband and state support programs.

Trends Impacting Our Operations

In addition to the above-described impact of the pandemic and its aftermath, our consolidated operations have been, and will continue to be, impacted by the following company-wide trends:

- Customers' demand for automated products and services and competitive pressures will require that we continue to invest in new technologies and automated processes to improve the customer experience and reduce our operating expenses.

- The increased use of digital applications, online video, gaming and artificial intelligence has substantially increased demand for robust, scalable network services. We are continuing to enhance our product capabilities and simplify our product portfolio based on demand and profitability to enable customers to have access to greater bandwidth.

- Businesses continue to adopt distributed, global operating models. We are expanding and enhancing our fiber network, connecting more buildings to our network to generate revenue opportunities and reducing our reliance upon other carriers.

- Changes in customer preferences and in the regulatory, technological and competitive environment are (i) significantly reducing demand for our more mature service offerings, commoditizing certain of our other offerings, or resulting in volume or rate reductions for other of our offerings and (ii) also creating certain opportunities for us arising out of increased demand for lower latency provided by Edge computing and for faster and more secure data transmissions.

- The operating margins of several of our newer, more technologically advanced services, some of which may connect to customers through other carriers, are lower than the operating margins on our traditional, on-net wireline services.

- Uncertainties regarding our financial performance, leverage and debt covenant compliance have caused, and may continue to cause, certain of our customers and other third parties to reduce or cease transacting business with us.

- Our expenses will be impacted by higher vendor costs, reduced economies of scale and other dis-synergies due to our completed 2022 and 2023 divestitures and any future divestitures.

- Declines in our traditional wireline services and other more mature offerings have necessitated right-sizing our cost structures to remain competitive.

Inflation has placed downward pressure on our margins and macroeconomic uncertainties have likely contributed to delayed decision-making by certain of our customers, which are trends that will likely continue to impact us as long as inflation rates remain elevated. These and other developments and trends impacting our operations are discussed elsewhere in this Item 7.

Results of Operations

In this section, we discuss our overall results of operations and highlight special items that are not included in our segment results. In "Segment Results" we review the performance of our two reporting segments in more detail. Results in this section include the results of our Latin American, ILEC and EMEA businesses prior to their sale on August 1, 2022, October 3, 2022 and November 1, 2023 respectively.

Revenue

The following table summarizes our consolidated operating revenue recorded under each of our two segments and in our four revenue sales channels within the Business segment described above:

	Years Ended December 31,			2023 vs 2022 % Change	2022 vs 2021 % Change
	2023	2022	2021		
	(Dollars in millions)				
Business Segment:					
Large Enterprise	$ 4,616	5,377	5,918	(14) %	(9) %
Mid-Market Enterprise	2,011	2,212	2,398	(9) %	(8) %
Public Sector	1,783	1,861	2,111	(4) %	(12) %
Wholesale	3,125	3,591	3,692	(13) %	(3) %
Business Segment Revenue	11,535	13,041	14,119	(12) %	(8) %
Mass Markets Segment Revenue	3,022	4,437	5,568	(32) %	(20) %
Total operating revenue	$ 14,557	17,478	19,687	(17) %	(11) %

Our consolidated operating revenue decreased by $2.9 billion for the year ended December 31, 2023 as compared to the year ended December 31, 2022, $2.1 billion of which is attributable to the sale of our Latin American and ILEC businesses in the second half of 2022, and the sale of our EMEA business on November 1, 2023. Our consolidated revenue decreased by $2.2 billion for the year ended December 31, 2022 compared to the year ended December 31, 2021 due to revenue declines in all of our revenue categories listed above, in addition to an approximately $1.0 billion decrease attributable to the sale of our Latin American and ILEC businesses in the second half of 2022. See our segment results below for additional information.

Operating Expenses

The following table summarizes our operating expenses for the years ended December 31, 2023 and 2022. For information regarding expenses for the year ended December 31, 2021, see "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7 of Part II of our Annual Report Form 10-K for the year ended December 31, 2022.

	Years Ended December 31,		% Change
	2023	2022	
	(Dollars in millions)		
Cost of services and products (exclusive of depreciation and amortization)	$ 7,144	7,868	(9) %
Selling, general and administrative	3,198	3,078	4 %
Net loss (gain) on sale of businesses	121	(113)	nm
Loss on disposal group held for sale	—	40	nm
Depreciation and amortization	2,985	3,239	(8) %
Goodwill impairment	10,693	3,271	nm
Total operating expenses	$ 24,141	17,383	39 %

nm Percentages greater than 200% and comparisons between positive and negative values or to/from zero values are considered not meaningful.

Cost of Services and Products (exclusive of depreciation and amortization)

Cost of services and products (exclusive of depreciation and amortization) decreased by $724 million for the year ended December 31, 2023 as compared to the year ended December 31, 2022. This decrease was primarily due to a decrease of $718 million due to the sale of the Latin American and ILEC businesses in the second half of 2022 and the sale of the EMEA business on November 1, 2023, as well as reduction of $108 million in employee-related expense from lower headcount. These decreases were partially offset by higher network and maintenance expense of $130 million.

Selling, General and Administrative

Selling, general and administrative expenses increased by $120 million for the year ended December 31, 2023 as compared to the year ended December 31, 2022. This increase in selling, general and administrative expenses was primarily driven by a $101 million loss incurred as a result of our donation of our Monroe, Louisiana campus and a $73 million net loss as a result of the sale of select CDN contracts. Additionally, employee-related expense increased $97 million and marketing and advertising expense increased $50 million. A portion of these and other expense increases were driven by our ongoing growth and optimization initiatives. These increases were partially offset by a $185 million decrease due to the sale of the Latin American and ILEC businesses in the second half of 2022 as well as the sale of the EMEA business on November 1, 2023 and $21 million from gains on the sale of property and assets.

Loss (Gain) on Sale of Businesses

For a discussion of the loss on the sale of the EMEA business and gain on the sales of the Latin American and ILEC businesses that we recognized for the years ended December 31, 2023 and December 31, 2022, see Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses.

Depreciation and Amortization

The following table provides detail of our depreciation and amortization expense:

	Years Ended December 31,		
	2023	2022	% Change
	(Dollars in millions)		
Depreciation	$ 1,932	2,133	(9) %
Amortization	1,053	1,106	(5) %
Total depreciation and amortization	$ 2,985	3,239	(8) %

Depreciation expense decreased by $201 million for the year ended December 31, 2023 as compared to the year ended December 31, 2022 primarily due to the discontinuation during the fourth quarter of 2022 of the depreciation of the tangible EMEA assets we divested, resulting in a decrease of $152 million of depreciation expense during the year ended December 31, 2023 compared to the year ended December 31, 2022. In addition, for the year ended December 31, 2023 as compared to the year ended December 31, 2022, depreciation expense decreased $137 million due to the impact of annual rate depreciable life changes, The decreases were partially offset by higher depreciation expense of $92 million associated with net growth in depreciable assets.

Amortization expense decreased by $53 million for the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to the discontinuation during the fourth quarter of 2022 of the amortization of the intangible EMEA assets we divested, resulting in a decrease of $30 million of amortization expense during the year ended December 31, 2023 compared to the year ended December 31, 2022. In addition, for the year ended December 31, 2023, as compared to the year ended December 31, 2022, amortization expense decreased $18 million resulting from certain customer relationship intangible assets becoming fully amortized in the second quarter of 2023.

Further analysis of our segment operating expenses by segment is provided below in "Segment Results."

Goodwill Impairments

We are required to perform impairment tests related to our goodwill annually, which we perform as of October 31, or sooner if an indicator of impairment occurs. We considered the sustained decline in our share price during the second quarter of 2023 an event or change in circumstance requiring evaluation of goodwill impairment.

We report under two segments: Business and Mass Markets. As of December 31, 2023, we had three reporting units for goodwill impairment testing, which are (i) Mass Markets, (ii) North America Business ("NA Business") and (iii) Asia Pacific ("APAC") region. Prior to the divestiture of the EMEA business, the EMEA region was also a reporting unit and was tested for impairment in the pre-classification test as of October 31, 2022 discussed elsewhere herein. Prior to its August 1, 2022 divestiture, the Latin American ("LATAM") region was also a reporting unit.

During the second and fourth quarters of 2023, we determined circumstances existed indicating it was more likely than not that the carrying value of one or more of our reporting units exceeded its fair value. When we performed an impairment test, we concluded that the estimated fair value of certain of our reporting units was less than their carrying value of equity as of our testing date. As a result, we recorded non-cash, non-tax-deductible goodwill impairment charges aggregating to $10.7 billion for the year ended December 31, 2023. When we performed our impairment tests during the fourth quarter of 2022, we concluded that the estimated fair value of certain of our reporting units was less than our carrying value of equity as of our testing date. As a result, we recorded non-cash, non-tax-deductible goodwill impairment charges aggregating to $3.3 billion in the fourth quarter of 2022. When we performed our annual impairment test in the fourth quarter of 2021, we concluded it was more likely than not that the fair value of each of our reporting units exceeded the carrying value of equity of our reporting units. Therefore, we concluded no impairment existed as of our annual assessment date in the fourth quarter of 2021.

We are currently experiencing competitive, macroeconomic and financial pressures, including dis-synergies resulting from our 2022 and 2023 divestitures and concerns about our ability to refinance debt in the future. In 2023, we also experienced a sustained decline in our share price. These and other factors contributed to us recognizing the above-described goodwill impairments. If these pressures continue, we may experience additional deterioration in our projected cash flows or market capitalization, or make significant changes to our assumptions of discount rates and market multiples. Any of these could result in additional goodwill impairments in future quarters.

See Note 3—Goodwill, Customer Relationships and Other Intangible Assets to our consolidated financial statements in Item 8 of Part II of this report for further details on these tests and impairment charges.

Other Consolidated Results

The following tables summarize our total other expense, net and income tax expense:

	Years Ended December 31,		
	2023	2022	% Change
	(Dollars in millions)		
Interest expense	$ (1,158)	(1,332)	(13) %
Net gain on early retirement of debt	618	214	189 %
Other (expense) income, net	(113)	32	nm
Total other expense, net	$ (653)	(1,086)	(40) %
Income tax expense	$ 61	557	(89) %

nm Percentages greater than 200% and comparisons between positive and negative values or to/from zero values are considered not meaningful.

Interest Expense

Interest expense decreased by $174 million for the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decline was primarily due to a $4.5 billion decrease in average outstanding long-term debt (inclusive of debt classified as held for sale), which was partially offset by an increase in the average interest rate from 5.14% to 5.9%.

Net Gain on Early Retirement of Debt

For a discussion of the exchange offers that resulted in the net gain on debt that we recognized for the years ended December 31, 2023 and 2022, see Note 7—Long-Term Debt and Credit Facilities.

Other (Expense) Income, Net

Other (expense) income, net reflects certain items not directly related to our core operations, including (i) components of net periodic pension and post-retirement benefit costs, (ii) foreign currency gains and losses, (iii) our share of income from partnerships we do not control, (iv) interest income, (v) gains and losses from non-operating asset dispositions, (vi) income from transition and separation services provided by us to the purchasers of our divested businesses and (vii) other non-core items.

	Years Ended December 31,	
	2023	2022
	(Dollars in millions)	
Pension and post-retirement net periodic (expense) income	$ (158)	1
Foreign currency (loss) gain	(10)	12
Loss on investment in limited partnership	(75)	(83)
Loss on investment in equity securities	(22)	(109)
Transition and separation services	186	152
Interest income	41	24
Other	(75)	35
Other (expense) income, net	$ (113)	32

See Note 14—Fair Value of Financial Instruments for more information regarding the losses recognized on our investments in equity securities and a limited partnership.

Income Tax Expense

For the years ended December 31, 2023 and 2022, our effective income tax rate was (0.6)% and (56.2)%, respectively. The effective tax rate for the year ended December 31, 2023 includes a $2.2 billion unfavorable aggregate impact of non-deductible goodwill impairments and a $137 million favorable impact as a result of utilizing available capital losses generated by the sale of our Latin American business in 2022. The effective tax rate for the year ended December 31, 2022 includes a $682 million unfavorable impact of non-deductible goodwill impairments and $128 million unfavorable impact as a result of the sale of our Latin American business. For additional information, see Note 16—Income Taxes to our consolidated financial statements in Item 8 of Part II of this report and "Critical Accounting Policies and Estimates—Income Taxes".

Segment Results

General

Reconciliation of segment revenue to total operating revenue is below. The results presented in this section include results of our Latin American, ILEC and EMEA businesses prior to their sale on August 1, 2022, October 3, 2022 and November 1, 2023 respectively:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Operating revenue			
Business	$ 11,535	13,041	14,119
Mass Markets	3,022	4,437	5,568
Total operating revenue	$ 14,557	17,478	19,687

Reconciliation of segment EBITDA to total adjusted EBITDA is below:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Net (loss) income	$ (10,298)	(1,548)	2,033
Income tax expense	61	557	668
Total other expense, net	653	1,086	1,584
Depreciation and amortization expense	2,985	3,239	4,019
Goodwill impairment	10,693	3,271	—
Stock-based compensation expense	52	98	120
Total adjusted EBITDA	$ 4,146	6,703	8,424
Business segment adjusted EBITDA	$ 7,165	8,569	9,358
Mass Markets segment adjusted EBITDA	1,589	2,690	3,730
Other unallocated expense	(4,608)	(4,556)	(4,664)

For additional information on our reportable segments and product and services categories, see Note 4—Revenue Recognition and Note 17—Segment Information to our consolidated financial statements in Item 8 of Part II of this report.

Business Segment

	Years Ended December 31,			Percent Change	
	2023	2022	2021	2023 vs 2022	2022 vs 2021
	(Dollars in millions)				
Business Segment Product Categories:					
Grow	$ 4,469	4,595	4,687	(3)%	(2)%
Nurture	3,465	4,094	4,540	(15)%	(10)%
Harvest	2,785	3,557	4,069	(22)%	(13)%
Other	816	795	823	3 %	(3)%
Total Business Segment Revenue	11,535	13,041	14,119	(12)%	(8)%
Expenses:					
Total expense	4,370	4,472	4,761	(2)%	(6)%
Total adjusted EBITDA	$ 7,165	8,569	9,358	(16)%	(8)%

Year ended December 31, 2023 compared to the years ended December 31, 2022 and December 31, 2021

Business segment revenue decreased $1.5 billion for the year ended December 31, 2023 compared to December 31, 2022 and decreased $1.1 billion for the year ended December 31, 2022 compared to December 31, 2021. Approximately $1.0 billion of the decrease for the year ended December 31, 2023 compared to December 31, 2022 was due to the sale of the Latin American, ILEC and EMEA businesses in late 2022 and 2023. For the year ended December 31, 2022 compared to December 31, 2021, the sale of the Latin American and ILEC businesses in the second half of 2022 caused a decrease of $511 million. More specifically, within each product category for the year ended December 31, 2023 compared to December 31, 2022 and for the year ended December 31, 2022 compared to December 31, 2021:

- Grow decreased $126 million and $92 million, reflecting decreases of approximately $370 million and $176 million associated with the sale of the divested businesses. This was offset by increases of approximately $244 million primarily in our IP, wavelengths, dark fiber, enterprise broadband and colocation products for the year ended December 31, 2023 compared to December 31, 2022 and approximately $84 million primarily in our IP, enterprise broadband, and wavelengths products for the year ended December 31, 2022 compared to December 31, 2021;

- Nurture decreased $629 million and $446 million, approximately $262 million and $119 million of which was attributable to the sale of the divested businesses. The remainder of these declines are principally attributable to declines in traditional VPN networks services of $261 million and $245 million and declines in Ethernet services of $112 million and $87 million;

- Harvest decreased by $772 million and $512 million, approximately $370 million and $209 million of which was attributable to the sale of the divested businesses. The remainder of the decline is principally attributable to a $265 million and $211 million decline in legacy voice services;

- Other increased by $21 million for the year ended December 31, 2023 compared to December 31, 2022 and decreased $28 million for the year ended December 31, 2022 compared to December 31, 2021. Equipment revenue drove both the increase for the year ended December 31, 2023 and the decrease for the year ended December 31, 2022 with an increase of $35 million and a decrease of $20 million, respectively.

The decrease in Business segment revenue for the year ended December 31, 2023 was also driven by $31 million of unfavorable foreign currency adjustments as compared to December 31, 2022. The decrease in Business segment revenue for the year ended December 31, 2022 was also driven by $54 million of unfavorable foreign currency adjustments for the year ended December 31, 2022 as compared to December 31, 2021.

Business segment expense decreased by $102 million for the year ended December 31, 2023 compared to December 31, 2022 primarily due to the sale of the divested businesses of $230 million. This decrease was partially offset by an increase of $80 million in cost of sales and $48 million in selling, general and administrative expenses, primarily $31 million of increased professional fees and $17 million of increased employee-related costs. A portion of these and other expense increases were driven by our ongoing growth and optimization initiatives. Business segment expenses decreased by $289 million for the year ended December 31, 2022 compared to December 31, 2021, primarily due to a $138 million decrease in expenses from the divested businesses, as well as decreases in cost of sales of $105 million and $46 million in selling, general and administrative expenses, primarily in reduced employee-related costs of $38 million.

Business segment adjusted EBITDA as a percentage of revenue was 62%, 66% and 66% for the years ended December 31, 2023, 2022 and 2021.

Mass Markets Segment

	Years Ended December 31,			Percent Change	
	2023	2022	2021	2023 vs 2022	2022 vs 2021
	(Dollars in millions)				
Mass Markets Product Categories:					
Fiber Broadband	$ 636	604	524	5 %	15 %
Other Broadband	1,394	2,164	2,507	(36)%	(14)%
Voice and Other	992	1,669	2,537	(41)%	(34)%
Total Mass Markets Segment Revenue	3,022	4,437	5,568	(32)%	(20)%
Expenses:					
Total expense	1,433	1,747	1,838	(18)%	(5)%
Total adjusted EBITDA	$ 1,589	2,690	3,730	(41)%	(28)%

Year ended December 31, 2023 compared to the years ended December 31, 2022 and December 31, 2021

Mass Markets segment revenue decreased by $1.4 billion for the year ended December 31, 2023 compared to December 31, 2022 and decreased $1.1 billion for the year ended December 31, 2022 compared to December 31, 2021. Approximately $1.1 billion and $448 million of these decreases are due to the sale of our ILEC business in the fourth quarter of 2022. More specifically, within each product category for the year ended December 31, 2023 compared to December 31, 2022 and for the year ended December 31, 2022 compared to December 31, 2021:

- Fiber Broadband revenue increased $32 million and $80 million, primarily driven by $73 million and $83 million of increases driven by growth in the number of fiber customers associated with our continued increase in enabled locations from our Quantum Fiber buildout, which was partially offset by respective decreases of $41 million and $3 million due to the sale of the ILEC business relating to such periods;

- Other Broadband revenue decreased $770 million and $343 million. Approximately $563 million and $230 million of this decrease was due to the ILEC divestiture and $207 million and $113 million was due to fewer customers for our lower speed copper-based broadband services;

- Voice and Other decreased $677 million and $868 million. These declines were principally due to (i) a $472 million and $215 million decrease due to the sale of the ILEC business, (ii) a $146 million and $222 million decrease due to the continued loss of copper-based voice customers and (iii) for the year ended December 31, 2023 compared to December 31, 2022, the recognition in the first quarter of 2022 of $59 million of previously deferred revenue related to the CAF II program, which lapsed on December 31, 2021. The decrease for the year ended December 31, 2022 compared to December 31, 2021 was additionally driven by the net reduction in CAF II revenue of $431 million.

Mass Markets segment expense decreased by $314 million for the year ended December 31, 2023 compared to December 31, 2022 primarily due to a $295 million decrease due to the ILEC divestiture and a $37 million decrease in professional fees, partially offset by an increase of $42 million in employee-related costs and $27 million in marketing and advertising costs. Mass Markets segment expense decreased $91 million for the year ended December 31, 2022 compared to December 31, 2021 primarily due to a $176 million decrease due to the divestiture of the ILEC business, partially offset by $107 million of increased bad debt expense, network expense and professional fees.

Mass Markets segment adjusted EBITDA as a percentage of revenue was 53%, 61% and 67% for the years ended December 31, 2023, 2022 and 2021, respectively.

Critical Accounting Policies and Estimates

Our consolidated financial statements are prepared in accordance with accounting principles that are generally accepted in the United States. The preparation of these consolidated financial statements requires management to make estimates and assumptions that affect the reported amounts of our assets, liabilities, revenue and expenses. We have identified certain policies and estimates as critical to our business operations and the understanding of our past or present results of operations related to (i) goodwill, customer relationships and other intangible assets; (ii) pension and post-retirement benefits; (iii) loss contingencies and litigation reserves and (iv) income taxes. These policies and estimates are considered critical because they had a material impact, or they have the potential to have a material impact, on our consolidated financial statements and because they require us to make significant judgments, assumptions or estimates. We believe that our estimates, judgments and assumptions made when accounting for the items described below were reasonable, based on information available at the time they were made. However, actual results may differ from those estimates, and these differences may be material.

Goodwill, Customer Relationships and Other Intangible Assets

We have a significant amount of goodwill and indefinite-lived intangible assets that are assessed at least annually for impairment. At December 31, 2023, goodwill and intangible assets totaled $7.4 billion, or 22%, of our total assets. The impairment analyses of these assets are considered critical because of their significance to us and our segments and the subjective nature of certain assumptions used to estimate fair value.

We have assigned our goodwill balance to our segments at December 31, 2023 as follows:

	Business	Mass Markets	Total
		(Dollars in millions)	
As of December 31, 2023	$ —	1,964	1,964

Intangible assets arising from business combinations, such as goodwill, customer relationships, capitalized software, trademarks and tradenames, are initially recorded at estimated fair value. We amortize customer relationships primarily over an estimated life of 7 to 14 years, using the straight-line method, depending on the customer. Certain customer relationship intangible assets became fully amortized at the end of the first quarter 2021 using the sum-of-years-digits method, which we no longer use for any of our remaining intangible assets. We amortize capitalized software using the straight-line method primarily over estimated lives ranging up to 7 years. We amortize our other intangible assets using the straight-line method over an estimated life of 9 to 20 years. Other intangible assets not arising from business combinations are initially recorded at cost. Where there are no legal, regulatory, contractual or other factors that would reasonably limit the useful life of an intangible asset, we classify them as indefinite-lived intangible assets and such intangible assets are not amortized.

We assess our long-lived intangible assets, other than goodwill, with indefinite lives for impairment annually, or, under certain circumstances, more frequently, such as when events or changes in circumstances indicate there may be an impairment. These assets are carried at the estimated fair value at the time of acquisition and assets not acquired in acquisitions are recorded at historical cost. However, if their estimated fair value is less than their carrying amount, we recognize an impairment charge for the amount by which the carrying amount of these assets exceeds their estimated fair value. For the year ended December 31, 2023 and 2022, we concluded it was more likely than not that our indefinite-lived intangible assets were not impaired; thus we recorded no impairment charge for these assets.

We are required to assess our goodwill for impairment annually, or more frequently if an event occurs or circumstances change that indicates it is more likely than not the fair values of any of our reporting units were less than their carrying values. In assessing goodwill for impairment, we may first assess qualitative factors to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value.

Our annual impairment assessment date for goodwill is October 31, at which date we assess our reporting units. We report two segments: Business and Mass Markets. At October 31, 2023, we had three reporting units for goodwill impairment testing, which are (i) Mass Markets (ii) North America Business ("NA Business") and (iii) Asia Pacific ("APAC") region. Prior to their divestitures in 2023 and 2022, the EMEA and Latin American ("LATAM") regions were also each considered their own reporting unit.

Our reporting units are not discrete legal entities with discrete full financial statements. Our assets and liabilities are employed in and relate to the operations of multiple reporting units and are allocated to individual reporting units based on their relative revenue or earnings before interest, taxes depreciation and amortization ("EBITDA"). For each reporting unit, we compare its estimated fair value of equity to its carrying value of equity that we assign to the reporting unit. If the estimated fair value of the reporting unit is greater than its carrying value, we conclude that no impairment exists. If the estimated fair value of the reporting unit is less than the carrying value, we record a non-cash impairment charge equal to the excess amount. Depending on the facts and circumstances, we typically estimate the fair value of our reporting units by considering either or both of (i) a discounted cash flow method, which is based on the present value of projected cash flows over a discrete projection period and a terminal value, which is based on the expected normalized cash flows of the reporting units following the discrete projection period, and (ii) a market approach, which includes the use of multiples of publicly-traded companies whose services are comparable to ours. With respect to our analysis using the discounted cash flow method, the timing and amount of projected cash flows under these forecasts require estimates developed from our long-range plan, which is informed by wireline industry trends, the competitive landscape, product lifecycles, operational initiatives, capital allocation plans and other company-specific and external factors that influence our business. These projected cash flows consider recent historical results and are consistent with our short-term financial forecasts and long-term business strategies. The development of these projected cash flows, and the discount rate applied to such cash flows, is subject to inherent uncertainties, and actual results could vary significantly from such estimates. Our determination of the discount rate is based on a weighted average cost of capital approach, which uses a market participant’s cost of equity and after-tax cost of debt and reflects certain risks inherent in the projected cash flows. With respect to our analysis using the market approach, we estimate the fair value of a reporting unit based upon a market multiple applied to the reporting unit’s revenue and EBITDA, adjusted for an appropriate control premium based on recent market transactions. We weigh these revenue and EBITDA market multiples depending on the characteristics of the individual reporting unit. We also reconcile the estimated fair values of the reporting units to our market capitalization to conclude whether the indicated control premium is reasonable in comparison to recent transactions in the marketplace.

Declines in our stock price have in the past caused an impairment of our goodwill, and future declines in our stock price could potentially cause additional impairments of our goodwill. Changes in the underlying assumptions that we use in allocating the assets and liabilities to reporting units under either the discounted cash flow or market approach method can result in materially different determinations of fair value. We perform sensitivity analyses that consider a range of discount rates and a range of EBITDA market multiples and we believe the estimates, judgments, assumptions and allocation methods used by us are reasonable. Nonetheless, changes in any of them can significantly affect whether we must incur impairment charges, as well as the size of such charges.

For additional information on our goodwill balances by segment and results of our impairment analyses, see Note 3—Goodwill, Customer Relationships and Other Intangible Assets to our consolidated financial statements in Item 8 of Part II of this report.

Pension and Post-retirement Benefits

We sponsor a noncontributory qualified defined benefit pension plan (referred to herein as our qualified pension plan, the "Lumen Combined Pension Plan" or the "Combined Pension Plan") for a substantial portion of our current and former employees in the United States. As of January 1, 2022, we spun off a new pension plan (the "Lumen Pension Plan") from the Combined Pension Plan in anticipation of the sale of a portion of our ILEC business on October 3, 2022. We recognized pension costs related to both plans through the sale of the ILEC business, at which time balances related to the Lumen Pension Plan were reflected in the calculation of our gain on the sale of the ILEC business and the pension obligation and assets of the Lumen Pension Plan were transferred to the purchaser.

In addition to the Lumen Combined Pension Plan, we also maintain several non-qualified pension plans for certain eligible highly compensated employees. Due to the insignificant impact of these non-qualified plans on our consolidated financial statements, we have excluded them from the following pension and post-retirement benefits disclosures for 2023, 2022 and 2021. See Note 11—Employee Benefits for additional information.

We also maintain post-retirement benefit plans that provide health care and life insurance benefits primarily for certain eligible retirees.

In 2023, approximately 62% of the Combined Pension Plan's January 1, 2023 net actuarial loss balance of $1.4 billion was subject to amortization as a component of net periodic expense over the average remaining service period of 14 years for participating employees expected to receive benefits under the plan. We treated the other 38% of the Combined Pension Plan's beginning net actuarial loss balance as indefinitely deferred during 2023. In 2023, approximately 56% of the beginning net actuarial gain of $371 million at January 1, 2023 for the post-retirement benefit plans was subject to amortization as a component of net periodic expense, with the other 44% of the beginning net actuarial gain balance for the post-retirement benefit plans treated as indefinitely deferred.

In 2022, approximately 62% of the Combined Pension Plan's January 1, 2022 net actuarial loss balance of $2.2 billion was subject to amortization as a component of net periodic expense over the average remaining service period of 14 years for participating employees expected to receive benefits under the plan. We treated the other 38% of the Combined Pension Plan's beginning net actuarial loss balance as indefinitely deferred during 2022. Additionally, upon the sale of the ILEC business on October 3, 2022, we recognized $564 million of net actuarial loss, pre-tax, related to the Lumen Pension Plan, which partially offset our gain on the sale of the business. We treated the entire beginning net actuarial loss of $217 million at January 1, 2022 for the post-retirement benefit plans as indefinitely deferred during 2022.

As of January 1, 2021, our qualified pension plan had a net actuarial loss balance of approximately $3.0 billion. A portion of this balance was subject to amortization as a component of net periodic expense over the average remaining service period for participating employees expected to receive benefits under the plan. During 2021, our lump sum pension settlement payments exceeded the settlement threshold and as a result we recognized a non-cash settlement charge of $383 million, accelerating previously unrecognized actuarial losses from our net actuarial loss balance. For our post-retirement benefit plans, the majority of the beginning net actuarial loss balance of $346 million at January 1, 2021 continued to be deferred during 2021.

In computing our pension and post-retirement health care and life insurance benefit obligations, our most significant assumptions are the discount rate and mortality rates. In computing our periodic pension expense, our most significant assumptions are the discount rate and the expected rate of return on plan assets. In computing our post-retirement benefit expense, our most significant assumption is the discount rate.

The discount rate for each plan is the rate at which we believe we could effectively settle the plan's benefit obligations as of the end of the year. We selected each plan's discount rate based on a cash flow matching analysis using hypothetical yield curves from high-quality U.S. corporate bonds and projections of the future benefit payments that constitute the projected benefit obligation for the plans. This process establishes the uniform discount rate that produces the same present value of the estimated future benefit payments as is generated by discounting each year's benefit payments by a spot rate applicable to that year. The spot rates used in this process were derived from a yield curve created from yields on the 60th to 90th percentile of U.S. high quality bonds.

The table below illustrates hypothetical changes in our benefit obligation for the qualified pension plan and the post-retirement benefit plans obligation if we had selected a higher or lower discount rate.

	Percentage point change		Increase/(decrease) at December 31, 2023
			(Dollars in millions)
Combined Pension Plan discount rate	1	% $	(451)
	(1)	%	373
Post-retirement benefit plans discount rate	1	%	(158)
	(1)	%	158

Published mortality rates help predict the expected life of plan participants and are based on historical demographic studies by the Society of Actuaries ("SOA"). The SOA publishes new mortality tables and projection scales on a regular basis which reflect updates to projected life expectancies in North America. Historically, we have adopted the new projection tables immediately after publication. The SOA did not release any revised mortality tables or projection scales in 2022 or 2023.

The expected rate of return on plan assets is the long-term rate of return we expect to earn on the plans' assets in the future, net of administrative expenses paid from plan assets. The rate of return is determined by the strategic allocation of plan assets and the long-term risk and return forecast for each asset class. The forecasts for each asset class are generated primarily from an analysis of the long-term expectations of various third-party investment management organizations, to which we then add a factor of 50 basis points to reflect the benefit we expect to result from our active management of the assets. The expected rate of return on plan assets is reviewed annually by management and our Board of Directors and is revised, as necessary, to reflect changes in the financial markets and our investment strategy.

Changes in any of the above factors could significantly impact operating expenses in our consolidated statements of operations and other comprehensive loss in our consolidated statements of comprehensive income (loss), as well as the amount of the liability and accumulated other comprehensive loss of stockholders' equity on our consolidated balance sheets.

Loss Contingencies and Litigation Reserves

We are involved in several potentially material legal proceedings, as described in more detail in Note 18—Commitments, Contingencies and Other Items. On a quarterly basis, we assess potential losses in relation to these and other pending or threatened tax and legal matters. For matters not related to income taxes, if a loss is considered probable and the amount can be reasonably estimated, we recognize an expense for the estimated loss. To the extent these estimates are more or less than the actual liability incurred upon resolving these matters, our earnings will be increased or decreased accordingly. If the differences are material, our consolidated financial statements could be materially impacted.

For matters related to income taxes, if we determine in our judgment that the impact of an uncertain tax position is more likely than not to be sustained upon audit by the relevant taxing authority, then we recognize in our financial statements a benefit for the largest amount that is more likely than not to be sustained. We do not recognize any portion of an uncertain tax position if we determine in our judgment that the position has less than a 50% likelihood of being sustained. Though the validity of any tax position is a matter of tax law, the body of statutory, regulatory and interpretive guidance on the application of the law is complex and often ambiguous, particularly in certain of the non-U.S. jurisdictions in which we operate. As such, our tax positions may not be sustained, which could materially impact our consolidated financial statements.

Income Taxes

Our provision for income taxes includes amounts for tax consequences deferred to future periods. We record deferred income tax assets and liabilities reflecting future tax consequences attributable to (i) tax credit carryforwards, (ii) differences between the financial statement carrying value of assets and liabilities and the tax basis of those assets and liabilities and (iii) tax NOLs. Deferred taxes are computed using enacted tax rates expected to apply in the year in which the differences are expected to affect taxable income. The effect of a change in tax rate on deferred income tax assets and liabilities is recognized in earnings in the period that includes the enactment date.

The measurement of deferred taxes often involves the exercise of considerable judgment related to the realization of tax basis. Our deferred tax assets and liabilities reflect our assessment that tax positions taken in filed tax returns and the resulting tax basis are more likely than not to be sustained if they are audited by taxing authorities. Assessing tax rates that we expect to apply and determining the years when the temporary differences are expected to affect taxable income requires judgment about the future apportionment of our income among the states in which we operate. Any changes in our practices or judgments involved in the measurement of deferred tax assets and liabilities could materially impact our financial condition or results of operations.

In connection with recording deferred income tax assets and liabilities, we establish valuation allowances when necessary to reduce deferred income tax assets to amounts that we believe are more likely than not to be realized. We evaluate our deferred tax assets quarterly to determine whether adjustments to our valuation allowances are appropriate in light of changes in facts or circumstances, such as changes in tax law, interactions with taxing authorities and developments in case law. In making this evaluation, we rely on our recent history of pre-tax earnings. We also rely on our forecasts of future earnings and the nature and timing of future deductions and benefits represented by the deferred tax assets, all of which involve the exercise of significant judgment. At December 31, 2023, we established a valuation allowance of $399 million primarily related to state NOLs, based on our determination that it was more likely than not that this amount of these NOLs would expire unused. If forecasts of future earnings and the nature and estimated timing of future deductions and benefits change in the future, we may determine that existing valuation allowances must be revised or eliminated or new valuation allowances created, any of which could materially impact our financial condition or results of operations. See Note 16—Income Taxes to our consolidated financial statements in Item 8 of Part II of this report.

Liquidity and Capital Resources

Overview of Sources and Uses of Cash

We are a holding company that is dependent on the capital resources of our subsidiaries to satisfy our parent company liquidity requirements. Several of our significant operating subsidiaries have borrowed funds either on a standalone basis or as part of a separate restricted group with certain of its subsidiaries or affiliates. The terms of the instruments governing the indebtedness of these borrowers or borrowing groups may restrict our ability to access their accumulated cash. In addition, our ability to access the liquidity of these and other subsidiaries may be constrained by tax, legal and other limitations.

At December 31, 2023, we held cash and cash equivalents of $2.2 billion. As of December 31, 2023, we had approximately $1.8 billion of borrowing capacity available under our $2.2 billion revolving credit facility, net of undrawn letters of credit and borrowings issued to us thereunder. We typically use our revolving credit facility as a source of liquidity for operating activities and our other cash requirements. We had approximately $61 million of cash and cash equivalents outside the United States at December 31, 2023. We currently believe that there are no material restrictions on our ability to repatriate cash and cash equivalents into the United States, and that we may do so without paying or accruing U.S. taxes. We do not currently intend to repatriate to the United States any of our foreign cash and cash equivalents from operating entities.

Our executive officers and our Board of Directors review our sources and potential uses of cash in connection with our annual budgeting process and throughout the year as circumstances warrant. Generally speaking, our principal funding source is cash from operating activities, and our principal cash requirements include operating expenses, capital expenditures, income taxes, debt repayments, periodic securities repurchases, periodic pension contributions and other benefits payments. The impact of the sales of our Latin American, ILEC and EMEA businesses is further described below.

Based on our current capital allocation objectives, during 2024 we project expending approximately $2.7 billion to $2.9 billion of capital expenditures.

For the 12-month period ending December 31, 2024, we project our fixed commitments will include (i) $125 million of scheduled term loan amortization payments and (ii) $32 million of finance lease and other fixed payments.

In January 2024, we received a cash federal income tax refund of $729 million, including interest. For additional information, see Note 24—Subsequent Events, to our consolidated financial statements included under Item 8 of Part II of this annual report.

As discussed elsewhere herein, we recorded an aggregate of $14.0 billion of non-cash, non tax-deductible goodwill impairment charges in 2023 and 2022, which has substantially reduced our total stockholders' equity. See "Results of Operations—Goodwill Impairments" discussing the potential for additional goodwill impairments in future quarters.

We will continue to monitor our future sources and uses of cash, and anticipate that we will make adjustments to our capital allocation strategies when, as and if determined by our Board of Directors. We may also draw on our revolving credit facility as a source of liquidity for operating activities and to give us additional flexibility to finance our capital investments, repayments of debt, pension contributions and other cash requirements.

For additional information, see "Risk Factors—Financial Risks" in Item 1A of Part I of this report.

Impact of the Divestitures of the Latin American, ILEC and EMEA Businesses

As discussed in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses to our consolidated financial statements in Item 8 of Part II of this report, we sold our Latin American, ILEC and EMEA businesses on August 1, 2022, October 3, 2022 and November 1, 2023, respectively. As further described elsewhere herein, these transactions have provided us with a substantial amount of cash proceeds but have also reduced our base of income-generating assets that generate our recurring cash from operating activities. For a discussion of the impact of our divestitures upon our federal income taxes, see "Liquidity and Capital Resources–Federal Income Tax Obligations."

Capital Expenditures

We incur capital expenditures on an ongoing basis to expand and improve our service offerings, enhance and modernize our networks and compete effectively in our markets. We evaluate capital expenditure projects based on a variety of factors, including expected strategic impacts (such as forecasted impact on revenue growth, productivity, expenses, service levels and customer retention) and our expected return on investment. The amount of capital investment is influenced by, among other things, current and projected demand for our services and products, our network requirements, cash flow generated by operating activities, cash required for debt services and other purposes, regulatory considerations (such as governmentally-mandated infrastructure buildout requirements) and the availability of requisite supplies, labor and permits.

Our capital expenditures continue to be focused on enhancing network operating efficiencies, supporting new service developments, and expanding our fiber network, including our Quantum Fiber buildout plan. A portion of our 2023 capital expenditures will also be focused on restoring network assets destroyed or damaged by Hurricane Ian in Florida during 2022 or replacing aged network assets. For more information on our capital spending, see (i) "—Overview of Sources and Uses of Cash" above, (ii) "Cash Flow Activities—Investing Activities" below and (iii) Item 1 of Part 1 of this report.

Debt Instruments and Financing Arrangements

Debt Instruments

On January 22, 2024, Lumen entered into an amended and restated transaction support agreement with a group of creditors representing over $12.5 billion of the outstanding indebtedness of the Company and its subsidiaries to, among other things, extend maturities of the debt instruments of the Company and Level 3 Financing, Inc. and provide access to a new revolving credit facility in an amount expected to be approximately $1.0 billion. In addition, the creditors have committed to provide $1.325 billion of financing to the Company through new long-term debt. The consummation of the transactions contemplated by the amended and restated transaction support agreement is subject to the satisfaction of various closing conditions. For more information, see Note 24—Subsequent Events, to our consolidated financial statements included under Item 8 of Part II of this annual report.

Pursuant to exchange offers commenced on March 16, 2023 (the "Exchange Offers"), (i) on March 31, 2023, Level 3 Financing, Inc. issued $915 million of its 10.500% Senior Secured Notes due 2030 in exchange for $1.535 billion of Lumen's outstanding senior unsecured notes and (ii) on April 17, 2023, Level 3 Financing, Inc. issued an additional $9 million of its 10.500% Notes in exchange for $19 million of Lumen's outstanding senior unsecured notes. All exchanged notes were concurrently cancelled. These transactions resulted in a net reduction in the aggregate principal amount of Lumen's consolidated indebtedness of approximately $630 million and, along with a repurchase of notes in the first quarter of 2023, a gain of $618 million during the year ended December 31, 2023.

At December 31, 2023, we had $11.4 billion of outstanding consolidated secured indebtedness, $8.5 billion of outstanding consolidated unsecured indebtedness (excluding (i) finance lease obligations, (ii) unamortized premiums, net and (iii) unamortized debt issuance costs) and approximately $1.8 billion of unused borrowing capacity under our revolving credit facility, as discussed further below.

Under our amended and restated credit agreement dated as of January 31, 2020 (the "Amended Credit Agreement"), we maintained at December 31, 2023 (i) a $2.2 billion senior secured revolving credit facility, under which we owed $200 million and had $218 million of letters of credit issued and undrawn as of such date, and (ii) $5.1 billion of senior secured term loan facilities. For additional information, see (i) "—Overview of Sources and Uses of Cash," and (ii) Note 7—Long-Term Debt and Credit Facilities to our consolidated financial statements in Item 8 of Part II of this report.

At December 31, 2023, we had $260 million undrawn letters of credit outstanding, $218 million of which were issued under our revolving line of credit, $40 million of letters of credit outstanding under our $225 million uncommitted letter of credit facility and $2 million of which were issued under a separate facility maintained by one of our subsidiaries (the full amount of which is collateralized by cash that is reflected on our consolidated balance sheets as restricted cash within other assets).

In addition to its indebtedness under our Amended Credit Agreement, Lumen Technologies is indebted under its outstanding senior notes, and several of its subsidiaries are indebted under separate credit facilities or senior notes.

For additional information on the terms and conditions of other debt instruments of ours and our subsidiaries, including financial and operating covenants, see (i) Note 7—Long-Term Debt and Credit Facilities to our consolidated financial statements in Item 8 of Part II of this report and (ii) "—Other Matters" below.

Future Financings and Debt Reduction Transactions

Subject to market conditions, we plan to continue to issue debt securities from time to time in the future to refinance a substantial portion of our maturing debt, including issuing debt securities of certain of our subsidiaries to refinance their maturing debt to the extent permitted under our debt covenants and consistent with our capital allocation strategies. The availability, interest rate and other terms of any new borrowings will depend on the ratings assigned by credit rating agencies, among other factors.

As of the filing date of this report, the credit ratings for the senior secured and unsecured debt of Lumen Technologies, Inc., Level 3 Financing, Inc. and Qwest Corporation were as follows:

Borrower	Moody's Investors Service, Inc.	Standard & Poor's	Fitch Ratings
Lumen Technologies, Inc.:			
Unsecured	Ca	CCC-/C	CCC-
Secured	Caa3	B/CC	B-
Level 3 Financing, Inc.:			
Unsecured	Caa2	CC	CCC+
Secured	B3	B	B-
Qwest Corporation:			
Unsecured	Caa3	B	B+

Our credit ratings are reviewed and adjusted from time to time by the rating agencies. Any future changes in the senior unsecured or secured debt ratings of us or our subsidiaries could impact our access to capital or borrowing costs. We cannot provide any assurances that we will be able to borrow additional funds on favorable terms, or at all. See "Risk Factors—Financial Risks" in Item 1A of Part I of this report.

From time to time over the past couple of years, we have engaged in various refinancings, redemptions, tender offers, open market purchases and other transactions designed to reduce our consolidated indebtedness, improve our financial flexibility or otherwise enhance our debt profile. Subject to market conditions, restrictions under our debt covenants, and other limitations, we may pursue similar transactions in the future to the extent feasible. See Note 7—Long-Term Debt and Credit Facilities to our consolidated financial statements in Item 8 of Part II of this report for additional information.

Federal Income Tax Obligations

As of December 31, 2023, Lumen Technologies had approximately $800 million of federal NOLs which, for U.S. federal income tax purposes, may be used to offset future taxable income. These NOLs are primarily related to federal NOLs we acquired through the Level 3 acquisition on November 1, 2017 and are subject to limitations under Section 382. We maintain a Section 382 rights agreement designed to safeguard through late 2026 our ability to use those NOLs. We utilized a substantial portion of our previously available NOLs to offset taxable gains generated by the completion of our 2022 divestitures. As a result, we anticipate that our cash income tax liability will increase in future periods. The amounts of our near-term future tax payments will depend upon many factors, including our future earnings and tax circumstances and the impact of any corporate tax reform or taxable transactions.

Although we expect to use substantially all of our remaining NOLs in future periods in accordance with Section 382's annual limitations, we cannot assure this. See "Risk Factors—Financial Risks—We may not be able to fully utilize our NOLs" in Item 1A of Part I of this report.

Stock Repurchases

Effective November 2, 2022, our Board of Directors authorized a new two-year program to repurchase up to an aggregate of $1.5 billion of our outstanding common stock (the "November 2022 stock repurchase program"). During the year ended December 31, 2023, we did not repurchase any shares of our outstanding common stock under this program. As of December 31, 2023, we were authorized to purchase up to an aggregate of $1.3 billion of our outstanding common stock under this program. We currently do not plan to purchase any shares of our outstanding common stock under this program in the near term.

Pension and Post-retirement Benefit Obligations

We are subject to material obligations under our existing defined benefit pension plans and post-retirement benefit plans. At December 31, 2023, the accounting unfunded status of our qualified and non-qualified defined benefit pension plans and our qualified post-retirement benefit plans was $769 million and $1.9 billion, respectively. For additional information about our pension and post-retirement benefit arrangements, see "Critical Accounting Policies and Estimates—Pension and Post-retirement Benefits" in Item 7 of Part II of this report and Note 11—Employee Benefits to our consolidated financial statements in Item 8 of Part II of this report.

On October 19, 2021, we, as sponsor of the Lumen Combined Pension Plan ("Combined Pension Plan"), along with the Plan's independent fiduciary, entered into an agreement committing the Plan to use a portion of its plan assets to purchase an annuity from an insurance company (the "Insurer") to transfer $1.4 billion of the Plan's pension liabilities. This agreement irrevocably transferred to the Insurer future Plan benefit obligations for approximately 22,600 U.S. Lumen participants ("Transferred Participants") effective on December 31, 2021. This annuity transaction was funded entirely by existing Plan assets and was intended to provide equivalent benefits to the Transferred Participants. The Insurer is committed to assume responsibility for administrative and customer service support, including distribution of payments to the Transferred Participants.

As of January 1, 2022, we spun off the Lumen Pension Plan from the Combined Pension Plan in anticipation of the sale of the ILEC business, as described further in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses to our consolidated financial statements in Item 1 of Part I of this report. At the time of the spin-off we transferred $2.5 billion of pension benefit obligation and $2.2 billion of plan assets to the Lumen Pension Plan. Following a revaluation of the pension obligation and pension assets for the Lumen Pension Plan in preparation for the closing of the ILEC business divestiture, we contributed approximately $319 million of cash in September 2022 to satisfy our contractual obligations to the purchaser of the divested business. This plan was subsequently assumed by the purchaser as part of our ILEC business divestiture on October 3, 2022. Upon sale of the ILEC business, we recognized $403 million of net actuarial loss and prior service cost, net of tax impact, related to the Lumen Pension Plan, which partially offset our gain on sale of the business.

Benefits paid by our Combined Pension Plan are paid through the trust that holds the Combined Pension Plan's assets. Based on current laws and circumstances, we do not expect any contributions to be required for our Combined Pension Plan during 2024. The amount of required contributions to our Combined Pension Plan in 2025 and beyond will depend on a variety of factors, most of which are beyond our control, including earnings on plan investments, prevailing interest rates, demographic experience, changes in plan benefits and changes in funding laws and regulations. We occasionally make voluntary contributions to our plans in addition to required contributions and reserve the right to do so in the future. We last made a voluntary contribution to the trust for our Combined Pension Plan during 2018. We currently do not expect to make a voluntary contribution in 2024.

Substantially all of our post-retirement health care and life insurance benefits plans are unfunded and are paid by us with available cash. As described further in Note 11—Employee Benefits, aggregate benefits paid by us under these plans (net of participant contributions and direct subsidy receipts) were $194 million, $210 million and $203 million for the years ended December 31, 2023, 2022 and 2021, respectively. For additional information on our expected future benefits payments for our post-retirement benefit plans, see Note 11—Employee Benefits to our consolidated financial statements in Item 8 of Part II of this report.

For 2023, our expected annual long-term rate of return on the pension plan assets, net of administrative expenses, was 6.5%. For 2024, our expected annual long-term rate of return on these assets is 6.5%. However, actual returns could be substantially different.

Our pension plan contains provisions that allow us, from time to time, to offer lump sum payment options to certain former employees in settlement of their future retirement benefits. We record an accounting settlement charge, consisting of the recognition of certain deferred costs of the pension plan, associated with these lump sum payments only if, in the aggregate, they exceed the sum of the annual service and interest costs for the plan's net periodic pension benefit cost, which represents the settlement accounting threshold. As of December 31, 2021, lump sum pension settlement payments exceeded the settlement threshold. As a result, for the year ended December 31, 2021 we recognized a non-cash settlement charge of $383 million to accelerate the recognition of a portion of the previously unrecognized actuarial losses in the qualified pension plan, which was allocated and reflected in other (expense) income, net in our consolidated statement of operations for the year ended December 31, 2021. The settlement threshold was not exceeded for the years ended December 31, 2023 or December 31, 2022. The amount of any future non-cash settlement charges will be dependent on several factors, including the total amount of our future lump sum benefit payments.

Future Contractual Obligations

Our estimated future obligations as of December 31, 2023 include both current and long term obligations. At December 31, 2023, we had a current obligation of $157 million and a long-term obligation of $20 billion of long-term debt (excluding unamortized premiums, net and unamortized debt issuance costs). Under our operating leases, at December 31, 2023 we had a current obligation of $350 million and a long-term obligation of $1.5 billion. As of such date, we had current obligations related to right-of-way agreements and purchase commitments of $587 million and a long-term obligation of $1.5 billion. Additionally, as of such date we had a current obligation for asset retirement obligation of $26 million and a long-term obligation of $131 million. Finally, at December 31, 2023 our pension and post-retirement benefit plans had an unfunded benefit obligation, of which $197 million is classified as current and $2.5 billion is classified as long-term. For additional information, see Note 7—Long-Term Debt and Credit Facilities, Note 5—Leases, Note 18—Commitments, Contingencies and Other Items, Note 9—Property, Plant and Equipment and Note 11—Employee Benefits, respectively.

Federal Broadband Support Programs

In January 2020, the FCC created the Rural Digital Opportunity Fund ("RDOF") program, a federal support program designed to fund broadband deployment in rural America. For the first phase of this program, RDOF Phase I, the FCC ultimately awarded $6.4 billion in support payments to be paid in equal monthly installments over 10 years. We were awarded RDOF funding in several of the states in which we operate and began receiving monthly support payments during the second quarter of 2022. We received approximately $17 million in annual RDOF Phase I support payments during 2023 and 2022 and expect to receive this same amount each year thereafter during the program period.

For additional information on these programs, see (i) Note 4—Revenue Recognition to our consolidated financial statements in Item 8 of Part II of this report, (ii) "Business—Regulation of Our Business" in Item 1 of Part I of this report and (iii) "Risk Factors—Legal and Regulatory Risks" in Item 1A of Part I of this report.

Federal officials have proposed changes to current programs and laws that could impact us, including proposals designed to increase broadband access, increase competition among broadband providers, lower broadband costs and re-adopt "net neutrality" rules similar to those adopted under a prior administration. In late 2021, the U.S. Congress enacted legislation that appropriated $65 billion to improve broadband affordability and access, primarily through federally funded state grants. As of the date of this report, various state and federal agencies are continuing to take steps to make this funding available to eligible applicants, including us. Although it remains premature to speculate on the ultimate impact of this legislation on us, we anticipate that the release of this funding would increase competition for broadband customers in newly-served areas.

Cash Flow Activities

The following table summarizes our consolidated cash flow activities for the years ended December 31, 2023 and 2022. For information regarding cash flow activities for the year ended December 31, 2021, see "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7 of Part II of our Annual Report Form 10-K for the year ended December 31, 2022.

	Years Ended December 31,		(Decrease) / Increase
	2023	2022	
	(Dollars in millions)		
Net cash provided by operating activities	$ 2,160	4,735	(2,575)
Net cash (used in) provided by investing activities	(1,201)	5,476	(6,677)
Net cash used in financing activities	(18)	(9,313)	(9,295)

Operating Activities

Net cash provided by operating activities decreased by $2.6 billion for the year ended December 31, 2023 as compared to the year ended December 31, 2022 primarily due to lower net income adjusted for non-cash expenses and gains, largely due to the impacts of the 2022 and 2023 divestitures discussed elsewhere herein. Cash provided by operating activities is subject to variability period over period as a result of timing differences, including with respect to the collection of receivables and payments of interest expense, accounts payable, income taxes and bonuses.

For additional information about our operating results, see "Results of Operations" above.

Investing Activities

Net cash (used in) provided by investing activities changed by $6.7 billion for the year ended December 31, 2023 as compared to the year ended December 31, 2022 primarily due to substantial pre-tax cash proceeds from the sales of our Latin American and ILEC businesses in 2022, partially offset by pre-tax cash proceeds from the sale of the EMEA business in 2023.

Financing Activities

Net cash used in financing activities decreased by $9.3 billion for the year ended December 31, 2023 as compared to the year ended December 31, 2022 primarily due to substantially higher debt repayments and dividends paid in 2022.

See Note 7—Long-Term Debt and Credit Facilities to our consolidated financial statements in Item 8 of Part II of this report for additional information on our outstanding debt securities.

Other Matters

We have cash management and loan arrangements with a majority of our income-generating subsidiaries, in which a substantial portion of the aggregate cash of those subsidiaries' is periodically advanced or loaned to us or our service company affiliate. Although we periodically repay these advances to fund the subsidiaries' cash requirements throughout the year, at any given point in time we may owe a substantial sum to our subsidiaries under these arrangements. In accordance with generally accepted accounting principles, these arrangements are reflected in the balance sheets of our subsidiaries, but are eliminated in consolidation and therefore not recognized on our consolidated balance sheets.

Our network includes some residual lead-sheathed copper cables installed years ago. These lead-sheathed cables constitute a small portion of our network. Due to recent media coverage of potential health and environmental risks associated with these cables, we anticipate incurring certain investigative costs. We also may incur other costs from related proceedings, including litigation, regulatory initiatives, and remediation. As of December 31, 2023, we had not accrued for any such potential costs and will only accrue when such costs are probable and reasonably estimable. For additional information about related litigation and potential risks, see Note 18—Commitments, Contingencies and Other Items to our consolidated financial statements in Item 1 of Part I of this report, and the risk factor disclosures incorporated by reference herein under “Risk Factors” in Item 1A of Part II of this report.

We are also involved in various legal proceedings that could substantially impact our financial position. See Note 18—Commitments, Contingencies and Other Items to our consolidated financial statements in Item 8 of Part II of this report for additional information.

Market Risk

As of December 31, 2023, we were exposed to market risk from changes in interest rates on our variable rate long-term debt obligations and fluctuations in certain foreign currencies.

Management periodically reviews our exposure to interest rate fluctuations and periodically implements strategies to manage the exposure. From time to time, we have used derivative instruments to swap our exposure to variable interest rates for fixed interest rates. We have established policies and procedures for risk assessment and the approval, reporting and monitoring of derivative instrument activities. As of December 31, 2023, we did not hold or issue derivative financial instruments for trading or speculative purposes.

As of December 31, 2023, we had approximately $7.9 billion of unhedged floating rate debt based on the secured overnight financing rate ("SOFR"). A hypothetical increase of 100 basis points in SOFR relating to our $7.9 billion of unhedged floating rate debt would, among other things, decrease our annual pre-tax earnings by approximately $79 million.

We conduct a portion of our business in currencies other than the U.S. dollar, the currency in which our consolidated financial statements are reported. Prior to the November 1, 2023 divestiture of our EMEA business, certain of our former European subsidiaries used the local currency as their functional currency, as the majority of their sales and purchases were transacted in their local currencies. Although we continue to evaluate strategies to mitigate risks related to the effect of fluctuations in currency exchange rates, we will likely recognize gains or losses from international transactions. Accordingly, changes in foreign currency rates relative to the U.S. dollar could positively or negatively impact our operating results.

Certain shortcomings are inherent in the method of analysis presented in the computation of exposures to market risks. Actual values may differ materially from those disclosed by us from time to time if market conditions vary from the assumptions used in the analyses performed. These analyses only incorporate the risk exposures that existed at December 31, 2023.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

The information in "Management's Discussion and Analysis of Financial Condition and Results of Operations—Market Risk" in Item 7 of Part II of this report is incorporated herein by reference.

ITEM 8. CONSOLIDATED FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors
Lumen Technologies, Inc.:

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheets of Lumen Technologies, Inc. and subsidiaries (the Company) as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive (loss) income, cash flows, and stockholders' equity for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the years in the three-year period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated February 22, 2024 expressed an unqualified opinion on the effectiveness of the Company's internal control over financial reporting.

Basis for Opinion

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee and that: (1) relate to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Testing of revenue

As discussed in Note 4 to the consolidated financial statements, the Company recorded $14.6 billion of operating revenues for the year ended December 31, 2023. The processing and recording of revenue are reliant upon multiple information technology (IT) systems.

We identified the evaluation of the sufficiency of audit evidence over revenue as a critical audit matter. Complex auditor judgment was required in evaluating the sufficiency of audit evidence over revenue due to the large volume of data and the number and complexity of the revenue accounting systems. Specialized skills and knowledge were needed to test the IT systems used for the processing and recording of revenue.

The following are the primary procedures we performed to address this critical audit matter. We applied auditor judgment to determine the nature and extent of procedures to be performed over the processing and recording of revenue, including the IT systems tested. We evaluated the design and tested the operating effectiveness of certain internal controls related to the processing and recording of revenue. This included manual and automated controls over the IT systems used for the processing and recording of revenue. For a selection of transactions, we compared the amount of revenue recorded to a combination of Company internal data, executed contracts, and other relevant third-party data. In addition, we involved IT professionals with specialized skills and knowledge who assisted in the design and performance of audit procedures related to certain IT systems used by the Company for the processing and recording of revenue. We evaluated the sufficiency of audit evidence obtained by assessing the results of procedures performed, including the relevance and reliability of evidence obtained.

Goodwill impairments for the North America Business and Mass Markets reporting units

As discussed in Note 3 to the consolidated financial statements, the goodwill balance at December 31, 2023 was $2.0 billion. The Company assesses goodwill for impairment at least annually, or more frequently, if events or circumstances indicate the carrying value of a reporting unit likely exceeds its fair value. During the second quarter of 2023, the Company determined circumstances related to the sustained decline in the Company's share price indicated it was more likely than not that the carrying value of their reporting units exceeded their fair value. Also, as of October 31, 2023, the Company performed their annual goodwill impairment test. For both the second quarter and annual impairment tests, the Company estimated the fair value of its reporting units using a market approach. The second quarter and annual impairment tests each determined the carrying value of the North America Business and Mass Markets reporting units exceeded their estimated fair value. As a result, the Company recorded non-cash impairment charges of $7.9 billion and $2.8 billion, respectively, reducing the carrying value of goodwill for the North America Business and Mass Markets reporting units.

We identified the assessment of the Company’s impairment testing of the goodwill of the North America Business and Mass Markets reporting units as a critical audit matter. Subjective auditor judgment was required in evaluating the earnings before interest, taxes, depreciation, and amortization (“EBITDA”) market multiple assumptions used to estimate the fair value of the reporting units. The evaluation of these assumptions was challenging as differences in judgment used to determine these assumptions could have had a significant effect on each reporting unit’s estimated fair value. Specialized skills and knowledge were required in the assessment of the EBITDA market multiple assumptions.

The following are the primary procedures we performed to address this critical audit matter. We evaluated the design and tested the operating effectiveness of certain internal controls related to the goodwill impairment tests. This included controls related to the Company’s determination of the EBITDA market multiple assumptions. We involved valuation professionals with specialized skills and knowledge, who assisted in evaluating the EBITDA market multiple assumptions by:

- comparing to EBITDA market multiple ranges developed using publicly available market data for comparable entities
- performing sensitivity analyses that considered a range of EBITDA market multiples.

/s/ KPMG LLP
We have served as the Company’s auditor since 1977.
Denver, Colorado
February 22, 2024

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors Lumen Technologies, Inc.:

Opinion on Internal Control Over Financial Reporting
We have audited Lumen Technologies, Inc. and subsidiaries' (the Company) internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive (loss) income, cash flows, and stockholders' equity for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements), and our report dated February 22, 2024 expressed an unqualified opinion on those consolidated financial statements.

Basis for Opinion
The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audit also included performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting
A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ KPMG LLP

Denver, Colorado
February 22, 2024

LUMEN TECHNOLOGIES, INC.

CONSOLIDATED STATEMENTS OF OPERATIONS

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions, except per share amounts, and shares in thousands)		
OPERATING REVENUE	$ 14,557	17,478	19,687
OPERATING EXPENSES			
Cost of services and products (exclusive of depreciation and amortization)	7,144	7,868	8,488
Selling, general and administrative	3,198	3,078	2,895
Net loss (gain) on sale of businesses	121	(113)	—
Loss on disposal groups held for sale	—	40	—
Depreciation and amortization	2,985	3,239	4,019
Goodwill impairment	10,693	3,271	—
Total operating expenses	24,141	17,383	15,402
OPERATING (LOSS) INCOME	(9,584)	95	4,285
OTHER EXPENSE			
Interest expense	(1,158)	(1,332)	(1,522)
Net gain on early retirement of debt (Note 7)	618	214	8
Other (expense) income, net	(113)	32	(70)
Total other expense, net	(653)	(1,086)	(1,584)
(LOSS) INCOME BEFORE INCOME TAXES	(10,237)	(991)	2,701
Income tax expense	61	557	668
NET (LOSS) INCOME	$ (10,298)	(1,548)	2,033
BASIC AND DILUTED (LOSS) EARNINGS PER COMMON SHARE			
BASIC	$ (10.48)	(1.54)	1.92
DILUTED	$ (10.48)	(1.54)	1.91
WEIGHTED AVERAGE COMMON SHARES OUTSTANDING			
BASIC	983,081	1,007,517	1,059,541
DILUTED	983,081	1,007,517	1,066,778

See accompanying notes to consolidated financial statements.

LUMEN TECHNOLOGIES, INC.

CONSOLIDATED STATEMENTS OF COMPREHENSIVE (LOSS) INCOME

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
NET (LOSS) INCOME	$ (10,298)	(1,548)	2,033
OTHER COMPREHENSIVE INCOME:			
Items related to employee benefit plans:			
Change in net actuarial loss, net of $ 20, $(205) and $(134) tax	(59)	631	424
Reclassification of net actuarial loss to (loss) gain on the sale of businesses, net of $—, $(142) and $— tax	(22)	422	—
Settlement charges recognized in net (loss) income, net of $ —, $— and $(93) tax	—	—	290
Change in net prior service cost, net of $ 4, $(9) and $(5) tax	(11)	30	14
Reclassification of prior service credit to (loss) gain on the sale of businesses, net of $—, $6 and $— tax	—	(19)	—
Reclassification of realized loss on interest rate swaps to net (loss) income, net of $—, $(5) and $(20) tax	—	17	63
Unrealized holding loss on interest rate swaps, net of $ —, $— and $— tax	—	—	(1)
Reclassification of realized loss on foreign currency translation to (loss) gain on the sale of businesses, net of $—, $— and $— tax	382	112	—
Foreign currency translation adjustment, net of $(3), $58 and $30 tax	(1)	(134)	(135)
Other comprehensive income	289	1,059	655
COMPREHENSIVE (LOSS) INCOME	$ (10,009)	(489)	2,688

See accompanying notes to consolidated financial statements.

LUMEN TECHNOLOGIES, INC.
CONSOLIDATED BALANCE SHEETS

	As of December 31,	
	2023	2022
	(Dollars in millions and shares in thousands)	
ASSETS		
CURRENT ASSETS		
Cash and cash equivalents	$ 2,234	1,251
Accounts receivable, less allowance of $67 and $85	1,318	1,508
Assets held for sale	104	1,889
Other	1,119	803
Total current assets	4,775	5,451
Property, plant and equipment, net of accumulated depreciation of $21,318 and $19,886	19,758	19,166
GOODWILL AND OTHER ASSETS		
Goodwill	1,964	12,657
Other intangible assets, net	5,470	6,166
Other, net	2,051	2,172
Total goodwill and other assets	9,485	20,995
TOTAL ASSETS	$ 34,018	45,612
LIABILITIES AND STOCKHOLDERS' EQUITY		
CURRENT LIABILITIES		
Current maturities of long-term debt	$ 157	154
Accounts payable	1,134	1,044
Accrued expenses and other liabilities		
Salaries and benefits	696	692
Income and other taxes	251	1,158
Current operating lease liabilities	268	344
Interest	168	181
Other	209	277
Liabilities held for sale	4	451
Current portion of deferred revenue	647	596
Total current liabilities	3,534	4,897
LONG-TERM DEBT	19,831	20,418
DEFERRED CREDITS AND OTHER LIABILITIES		
Deferred income taxes, net	3,127	3,163
Benefit plan obligations, net	2,490	2,391
Deferred revenue	1,969	1,758
Other	2,650	2,611
Total deferred credits and other liabilities	10,236	9,923
COMMITMENTS AND CONTINGENCIES (Note 18)		
STOCKHOLDERS' EQUITY		
Preferred stock — non-redeemable, $25.00 par value, authorized 2,000 and 2,000 shares, issued and outstanding 7 and 7 shares	—	—
Common stock, $1.00 par value, authorized 2,200,000 and 2,200,000 shares, issued and outstanding 1,008,486 and 1,001,688 shares	1,008	1,002
Additional paid-in capital	18,126	18,080
Accumulated other comprehensive loss	(810)	(1,099)
Accumulated deficit	(17,907)	(7,609)
Total stockholders' equity	417	10,374
TOTAL LIABILITIES AND STOCKHOLDERS' EQUITY	$ 34,018	45,612

See accompanying notes to consolidated financial statements.

LUMEN TECHNOLOGIES, INC.

CONSOLIDATED STATEMENTS OF CASH FLOWS

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
OPERATING ACTIVITIES			
Net (loss) income	$ (10,298)	(1,548)	2,033
Adjustments to reconcile net (loss) income to net cash provided by operating activities:			
Depreciation and amortization	2,985	3,239	4,019
Net loss (gain) on sale of businesses	121	(113)	—
Loss on disposal groups held for sale	—	40	—
Goodwill impairment	10,693	3,271	—
Deferred income taxes	8	(1,230)	598
Provision for uncollectible accounts	100	133	105
Net gain on early retirement and modification of debt	(618)	(214)	(8)
Unrealized loss (gain) on investments	97	191	(138)
Stock-based compensation	52	98	120
Changes in current assets and liabilities:			
Accounts receivable	102	(158)	(8)
Accounts payable	(97)	98	(261)
Accrued income and other taxes	(1,185)	972	(69)
Other current assets and liabilities, net	(549)	(372)	(353)
Retirement benefits	(1)	46	163
Changes in other noncurrent assets and liabilities, net	730	258	283
Other, net	20	24	17
Net cash provided by operating activities	2,160	4,735	6,501
INVESTING ACTIVITIES			
Capital expenditures	(3,100)	(3,016)	(2,900)
Proceeds from sale of businesses	1,746	8,369	—
Proceeds from sale of property, plant and equipment, and other assets	165	120	135
Other, net	(12)	3	53
Net cash (used in) provided by investing activities	(1,201)	5,476	(2,712)
FINANCING ACTIVITIES			
Net proceeds from issuance of long-term debt	—	—	1,881
Payments of long-term debt	(185)	(8,093)	(3,598)
Net proceeds from (payments on) revolving line of credit	200	(200)	50
Dividends paid	(11)	(780)	(1,087)
Repurchases of common stock	—	(200)	(1,000)
Other, net	(22)	(40)	(53)
Net cash used in financing activities	(18)	(9,313)	(3,807)

Net increase (decrease) in cash, cash equivalents and restricted cash	941	898	(18)
Cash, cash equivalents and restricted cash at beginning of period	1,307	409	427
Cash, cash equivalents and restricted cash at end of period	$ 2,248	1,307	409
Supplemental cash flow information:			
Income taxes paid, net	$ (1,303)	(76)	(112)
Interest paid (net of capitalized interest of $111, $66 and $53)	$ (1,138)	(1,365)	(1,487)
Supplemental non-cash information regarding investing activities:			
Sale of property, plant and equipment in exchange for note receivable	$ —	—	56
Supplemental non-cash information regarding financing activities:			
Purchase of software subscription in exchange for installment debt	$ —	—	77
Cancellation of senior unsecured notes as part of exchange offers (Note 7)	$ (1,554)	—	—
Issuance of senior secured notes as part of exchange offers (Note 7)	$ 924	—	—
Cash, cash equivalents and restricted cash:			
Cash and cash equivalents	$ 2,234	1,251	354
Cash and cash equivalents and restricted cash included in Assets held for sale	—	44	40
Restricted cash included in Other current assets	4	—	2
Restricted cash included in Other, net noncurrent assets	10	12	13
Total	$ 2,248	1,307	409

See accompanying notes to consolidated financial statements.

LUMEN TECHNOLOGIES, INC.

CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions except per share amounts)		
COMMON STOCK			
Balance at beginning of period	$ 1,002	1,024	1,097
Issuance of common stock through dividend reinvestment, incentive and benefit plans	6	11	8
Repurchases of common stock	—	(33)	(81)
Balance at end of period	1,008	1,002	1,024
ADDITIONAL PAID-IN CAPITAL			
Balance at beginning of period	18,080	18,972	20,909
Repurchases of common stock	—	(167)	(919)
Shares withheld to satisfy tax withholdings	(5)	(30)	(45)
Stock-based compensation and other, net	50	96	122
Dividends declared	1	(791)	(1,095)
Balance at end of period	18,126	18,080	18,972
ACCUMULATED OTHER COMPREHENSIVE LOSS			
Balance at beginning of period	(1,099)	(2,158)	(2,813)
Other comprehensive income	289	1,059	655
Balance at end of period	(810)	(1,099)	(2,158)
ACCUMULATED DEFICIT			
Balance at beginning of period	(7,609)	(6,061)	(8,094)
Net (loss) income	(10,298)	(1,548)	2,033
Balance at end of period	(17,907)	(7,609)	(6,061)
TOTAL STOCKHOLDERS' EQUITY	$ 417	10,374	11,777
DIVIDENDS DECLARED PER COMMON SHARE	$ —	0.75	1.00

See accompanying notes to consolidated financial statements.

References in the Notes to "Lumen Technologies" or "Lumen," "we," "us," the "Company," and "our" refer to Lumen Technologies, Inc. and its consolidated subsidiaries, unless the context otherwise requires. References in the Notes to "Level 3" refer to Level 3 Parent, LLC and its predecessor, Level 3 Communications, Inc., which we acquired on November 1, 2017.

(1) Background and Summary of Significant Accounting Policies

General

We are a facilities-based technology and communications company that provides a broad array of integrated products and services to our domestic and global business customers and our domestic mass markets customers. We operate one of the world's most interconnected networks. Our platform empowers our customers to swiftly adjust digital programs to meet immediate demands, create efficiencies, accelerate market access and reduce costs, which allows our customers to rapidly evolve their IT programs to address dynamic changes. Our specific products and services are detailed in Note 4—Revenue Recognition.

Basis of Presentation

The accompanying consolidated financial statements include our accounts and the accounts of our subsidiaries in which we have a controlling interest. Intercompany amounts and transactions with our consolidated subsidiaries have been eliminated.

To simplify the overall presentation of our consolidated financial statements, we report immaterial amounts attributable to noncontrolling interests in certain of our subsidiaries as follows: (i) income attributable to noncontrolling interests in other (expense) income, net, (ii) equity attributable to noncontrolling interests in additional paid-in capital and (iii) cash flows attributable to noncontrolling interests in other, net financing activities.

We reclassified certain prior period amounts to conform to the current period presentation, including the recategorization of our Business revenue by product category and sales channel in our segment reporting for 2022 and 2021. See Note 17—Segment Information for additional information. These changes had no impact on total operating revenue, total operating expenses or net (loss) income for any period.

Operating Expenses

Our current definitions of operating expenses are as follows:

- *Cost of services and products (exclusive of depreciation and amortization)* are expenses incurred in providing products and services to our customers. These expenses include: employee-related expenses directly attributable to operating and maintaining our network (such as salaries, wages, benefits and professional fees); facilities expenses (which include third-party telecommunications expenses we incur for using other carriers' networks to provide services to our customers); rents and utilities expenses; equipment sales expenses (such as data integration and modem expenses); and other expenses directly related to our operations; and

- *Selling, general and administrative expenses* are corporate overhead and other operating expenses. These expenses include: employee-related expenses (such as salaries, wages, internal commissions, benefits and professional fees) directly attributable to selling products or services and employee-related expenses for administrative functions; marketing and advertising; property and other operating taxes and fees; external commissions; litigation expenses associated with general matters; bad debt expense; and other selling, general and administrative expenses.

These expense classifications may not be comparable to those of other companies.

Summary of Significant Accounting Policies

Use of Estimates

Our consolidated financial statements are prepared in accordance with U.S. generally accepted accounting principles. These accounting principles require us to make certain estimates, judgments and assumptions. We believe that the estimates, judgments and assumptions we make when accounting for specific items and matters are reasonable, based on information available at the time they are made. These estimates, judgments and assumptions can materially affect the reported amounts of assets, liabilities and components of stockholders' equity as of the dates of the consolidated balance sheets, as well as the reported amounts of revenue, expenses and components of cash flows during the periods presented in our other consolidated financial statements. We also make estimates in our assessments of potential losses in relation to threatened or pending tax and legal matters. See Note 16—Income Taxes and Note 18—Commitments, Contingencies and Other Items for additional information.

For matters not related to income taxes, if a loss is considered probable and the amount can be reasonably estimated, we recognize an expense for the estimated loss. If we have the potential to recover a portion of the estimated loss from a third party, we make a separate assessment of recoverability and reduce the estimated loss if recovery is also deemed probable.

For matters related to income taxes, if we determine the impact of an uncertain tax position is more likely than not to be sustained upon audit by the relevant taxing authority, then we recognize a benefit for the largest amount that is more likely than not to be sustained. We do not recognize any portion of an uncertain tax position if the position has less than a 50% likelihood of being sustained. We recognize interest on the amount of unrecognized benefit from uncertain tax positions.

For all of these and other matters, actual results could differ materially from our estimates.

Assets Held for Sale

We classify assets and related liabilities as held for sale when: (i) management has committed to a plan to sell the assets, (ii) the net assets are available for immediate sale, (iii) there is an active program to locate a buyer and (iv) the sale and transfer of the net assets is probable within one year. Assets and liabilities held for sale are presented separately on our consolidated balance sheets with a valuation allowance, if necessary, to recognize the net carrying amount at the lower of cost or fair value, less costs to sell. Depreciation of property, plant and equipment and amortization of finite-lived intangible assets and right-of-use assets are not recorded while these assets are classified as held for sale. For each period that assets are classified as being held for sale, they are tested for recoverability. Unless otherwise specified, the amounts and information presented in the notes do not include assets and liabilities that were classified as held for sale as of December 31, 2023 and December 31, 2022. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses for additional information.

Revenue Recognition

We earn most of our consolidated revenue from contracts with customers, primarily through the provision of communications and other services. Revenue from contracts with customers is accounted for under Accounting Standards Codification ("ASC") 606. We also earn revenue from leasing arrangements (primarily fiber capacity and colocation agreements) and governmental subsidy payments, which are not accounted for under ASC 606.

Revenue is recognized when control of the promised goods or services is transferred to our customers, in an amount that reflects the consideration we expect to be entitled to receive in exchange for those goods or services. Revenue is recognized based on the following five-step model:

- Identification of the contract with a customer;
- Identification of the performance obligations in the contract;
- Determination of the transaction price;
- Allocation of the transaction price to the performance obligations in the contract; and
- Recognition of revenue when, or as, we satisfy a performance obligation.

We provide an array of communications services to business and residential customers, including local voice, VPN, Ethernet, data, broadband, private line (including special access), network access, transport, voice, information technology, video and other ancillary services. We provide these services to a wide range of businesses, including global, enterprise, wholesale, government, and small and medium business customers. Certain contracts also include the sale of equipment, which is not significant to our business.

We recognize revenue for services when we provide the applicable service or when control of a product is transferred. Recognition of certain payments received in advance of services being provided is deferred. These advance payments may include certain activation and certain installation charges. If the activation and installation charges are not separate performance obligations, we recognize them as revenue over the actual or expected contract term using historical experience, which typically ranges from one to five years depending on the service. In most cases, termination fees or other fees on existing contracts that are negotiated in conjunction with new contracts are deferred and recognized over the new contract term.

For access services, we generally bill fixed monthly charges one month in advance to customers and recognize revenue as service is provided over the contract term in alignment with the customer's receipt of service. For usage and other ancillary services, we generally bill in arrears and recognize revenue as usage or delivery occurs. In most cases, the amount invoiced for our service offerings constitutes the price that would be billed on a standalone basis.

In certain cases, customers may be permitted to modify their contracts. We evaluate the change in scope or price to identify whether the modification should be treated as a separate contract, as a termination of the existing contract and creation of a new contract, or as a change to the existing contract.

Customer contracts are evaluated to determine whether the performance obligations are separable. If the performance obligations are deemed separable and separate earnings processes exist, the total transaction price that we expect to receive with the customer is allocated to each performance obligation based on its relative standalone selling price. The revenue associated with each performance obligation is then recognized as earned.

We periodically sell transmission capacity on our network. These transactions are generally structured as indefeasible rights of use, commonly referred to as IRUs, which are the exclusive right to use a specified amount of capacity or fiber for a specified term, typically 20 years. In most cases, we account for the cash consideration received on transfers of transmission capacity as ASC 606 revenue which is adjusted for the time value of money and is recognized ratably over the term of the agreement. Cash consideration received on transfers of dark fiber is accounted for as non-ASC 606 lease revenue, which we also recognize ratably over the term of the agreement. We do not recognize revenue on any contemporaneous exchanges of our transmission capacity assets for other non-owned transmission capacity assets.

In connection with offering products and services provided to the end user by third-party vendors, we review the relationship between us, the vendor and the end user to assess whether revenue should be reported on a gross or net basis. In assessing whether revenue should be reported on a gross or net basis, we consider whether we act as a principal in the transaction and control the goods and services used to fulfill the performance obligations associated with the transaction.

We have service level commitments pursuant to contracts with certain of our customers. To the extent that we determine that such service levels were not achieved or may not have been achieved, we estimate the amount of credits to be issued and record a corresponding reduction to revenue in the period that the service level commitment was not met or may not be met.

Customer payments are made based on billing schedules included in our customer contracts, which is typically on a monthly basis.

We defer (or capitalize) incremental contract acquisition and fulfillment costs and recognize (or amortize) such costs over the average contract life. Our deferred contract costs for our customers have average amortization periods of approximately 36 months for mass markets customers and 33 months for business customers. These deferred costs are periodically monitored to reflect any significant change in assumptions.

See Note 4—Revenue Recognition for additional information.

Advertising Costs

Costs related to advertising are expensed as incurred and recorded as selling, general and administrative expenses in our consolidated statements of operations. Our advertising expense was $87 million, $62 million and $56 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Legal Costs

In the normal course of our business, we incur costs to hire and retain external legal counsel to advise us on finance, regulatory, litigation and other matters. Subject to certain exceptions, we expense these costs as the related services are received.

Income Taxes

We file a consolidated federal income tax return with our eligible subsidiaries. The provision for income taxes reflects taxes currently payable, tax consequences deferred to future periods and adjustments to our liabilities for uncertain tax positions. We record deferred income tax assets and liabilities reflecting future tax consequences attributable to tax NOLs, tax credit carryforwards and differences between the financial statement carrying value of assets and liabilities and the tax basis of those assets and liabilities. Deferred taxes are computed using enacted tax rates expected to apply in the year in which the differences are expected to affect taxable income. The effect on deferred income tax assets and liabilities of a change in tax rate is recognized in earnings in the period that includes the enactment date.

We establish valuation allowances when necessary to reduce deferred income tax assets to the amounts that we believe are more likely than not to be recovered. Each quarter we evaluate the need to retain or adjust each valuation allowance on our deferred tax assets. See Note 16—Income Taxes for additional information.

Cash and Cash Equivalents

Cash and cash equivalents include highly liquid investments that are readily convertible into cash and are not subject to significant risk from fluctuations in interest rates. As a result, the value at which cash and cash equivalents are reported in our consolidated financial statements approximates their fair value. In evaluating investments for classification as cash equivalents, we require that individual securities have original maturities of ninety days or less and that individual investment funds have dollar-weighted average maturities of ninety days or less. To preserve capital and maintain liquidity, we invest with financial institutions we deem to be of sound financial condition and in high quality and relatively risk-free investment products. Our cash investment policy limits the concentration of investments with specific financial institutions or among certain products and includes criteria related to credit worthiness of any particular financial institution.

Book overdrafts occur when we have issued checks but they have not yet been presented to our controlled disbursement bank accounts for payment. Disbursement bank accounts allow us to delay funding of issued checks until the checks are presented for payment. Until the issued checks are presented for payment, the book overdrafts are included in accounts payable on our consolidated balance sheets. This activity is included in the operating activities section in our consolidated statements of cash flows. There were no book overdrafts included in accounts payable at December 31, 2023 or 2022.

Restricted Cash

Restricted cash consists primarily of cash and investments that collateralize our outstanding letters of credit and certain performance and operating obligations. Restricted cash and securities are recorded as current or non-current assets in the consolidated balance sheets depending on the duration of the restriction and the purpose for which the restriction exists. Restricted securities are stated at cost which approximated their fair value as of December 31, 2023 and 2022.

Accounts Receivable and Allowance for Credit Losses

Accounts receivable are recognized based upon the amount due from customers for the services provided or at cost for purchased and other receivables, less an allowance for credit losses. We use a loss rate method to estimate our allowance for credit losses. For more information on our methodology for estimating our allowance for credit losses, see Note 6—Credit Losses on Financial Instruments.

We generally consider our accounts past due if they are outstanding over 30 days. Our past due accounts are written off against our allowance for credit losses when collection is considered to be not probable. Any recoveries of accounts previously written off are generally recognized as a reduction in bad debt expense in the period received. The carrying value of accounts receivable net of the allowance for credit losses approximates fair value. Accounts receivable balances acquired in a business combination are recorded at fair value for all balances receivable at the acquisition date and at the invoiced amount for those amounts invoiced after the acquisition date.

Property, Plant and Equipment

We record property, plant and equipment acquired in connection with our acquisitions based on its estimated fair value as of its acquisition date plus the estimated value of any associated legally or contractually required retirement obligations. We record purchased and constructed property, plant and equipment at cost, plus the estimated value of any associated legally or contractually required retirement obligations. We depreciate the majority of our property, plant and equipment using the straight-line group method over the estimated useful lives of groups of assets, but depreciate certain of our assets using the straight-line method over the estimated useful lives of the specific asset. Under the straight-line group method, assets dedicated to providing telecommunications services (which comprise the majority of our property, plant and equipment) that have similar physical characteristics, use and expected useful lives are pooled for purposes of depreciation and tracking. We use the equal life group procedure to establish each pool's average remaining useful life. Generally, under the straight-line group method, when an asset is sold or retired in the course of normal business activities, the cost is deducted from property, plant and equipment and charged to accumulated depreciation without recognition of a gain or loss. A gain or loss is recognized in our consolidated statements of operations only if a disposal is unusual. Leasehold improvements are amortized over the shorter of the useful lives of the assets or the expected lease term. Expenditures for maintenance and repairs are expensed as incurred. During the construction phase of network and other internal-use capital projects, we capitalize related employee and interest costs. Property, plant and equipment supplies used internally are carried at average cost, except for significant individual items which are carried at actual cost.

We perform annual internal reviews to evaluate the reasonableness of the depreciable lives for our property, plant and equipment. Our reviews utilize models that take into account actual usage, physical wear and tear, replacement history, assumptions about technology evolution and, in certain instances, actuarially determined probabilities to estimate the remaining useful life of our asset base. Our remaining useful life assessments evaluate the possible loss in service value of assets that may precede the physical retirement. Assets shared among many customers may lose service value as those customers reduce their use of the asset. However, the asset is not retired until all customers no longer utilize the asset and we determine there is no alternative use for the asset.

We have asset retirement obligations associated with the legally or contractually required removal of a limited group of property, plant and equipment assets from leased properties and the disposal of certain hazardous materials present in our owned properties. When an asset retirement obligation is identified, usually in association with the acquisition of the asset, we record the fair value of the obligation as a liability. The fair value of the obligation is also capitalized as property, plant and equipment and then amortized over the estimated remaining useful life of the associated asset. Where the removal obligation is not legally binding, we expense the net cost to remove assets in the period in which the costs are actually incurred.

We review long-lived tangible assets for impairment whenever facts and circumstances indicate that the carrying amounts of the assets may not be recoverable. For assessment purposes, long-lived assets are grouped with other assets and liabilities at the lowest identifiable level for which we generate cash flows independently of other groups of assets and liabilities, absent a material change in operations. An impairment loss is recognized only if the carrying amount of the asset group is not recoverable and exceeds its estimated fair value. Recoverability of the asset group to be held and used is assessed by comparing the carrying amount of the asset group to the estimated undiscounted future net cash flows expected to be generated by the asset group. If the asset group's carrying value is not recoverable, we recognize an impairment charge for the amount by which the carrying amount of the asset group exceeds its estimated fair value.

Goodwill, Customer Relationships and Other Intangible Assets

We initially record intangible assets arising from business combinations, such as goodwill, customer relationships, capitalized software, trademarks and trade names, at estimated fair value. We amortize customer relationships primarily over an estimated life of 7 to 14 years, using the straight-line method, depending on the type of customer. Certain customer relationship intangible assets became fully amortized at the end of the first quarter 2021 using the sum-of-years-digits method, which we no longer use for any of our remaining intangible assets. We amortize capitalized software using the straight-line method primarily over estimated lives ranging up to 7 years. We amortize our other intangible assets using the straight-line method over an estimated life of 9 to 20 years. Other intangible assets not arising from business combinations are initially recorded at cost. Where there are no legal, regulatory, contractual or other factors that would reasonably limit the useful life of an intangible asset, we classify them as indefinite-lived intangible assets and such intangible assets are not amortized.

Internally used software, whether purchased or developed by us, is capitalized and amortized using the straight-line method over its estimated useful life. We have capitalized certain costs associated with software such as costs of employees devoted to software development and external direct costs for materials and services. Costs associated with software to be used for internal purposes are expensed until the point at which the project has reached the development stage. Subsequent additions, modifications or upgrades to internal-use software are capitalized only to the extent that they allow the software to perform a task it previously did not perform. Software maintenance, data conversion and training costs are expensed in the period in which they are incurred. We review the remaining economic lives of our capitalized software annually. Capitalized software is included in other intangible assets, net, in our consolidated balance sheets.

Our long-lived intangible assets, other than goodwill, with indefinite lives are assessed for impairment annually, or, under certain circumstances, more frequently, such as when events or changes in circumstances indicate there may be an impairment. These assets are carried at the estimated fair value at the time of acquisition and assets not acquired in acquisitions are recorded at historical cost. However, if their estimated fair value is less than the carrying amount, we recognize an impairment charge for the amount by which the carrying amount of these assets exceeds their estimated fair value.

We are required to assess our goodwill for impairment annually, or more frequently if an event occurs or circumstances change that indicates it is more likely than not the fair values of any of our reporting units were less than their carrying values. We are required to write-down the value of goodwill of our reporting units in periods in which the recorded carrying value of any such unit exceeds its fair value of equity. Our reporting units are not discrete legal entities with discrete full financial statements. Therefore, we assess the equity carrying value and future cash flows each time we perform a goodwill impairment assessment on a reporting unit. To do so, we assign our assets, liabilities and cash flows to reporting units using allocation methodologies which we believe are reasonable and consistent. This process entails various estimates, judgments and assumptions.

We are required to reassign goodwill to reporting units whenever reorganizations of our internal reporting structure changes the composition of our reporting units. Goodwill is reassigned to the reporting units using a relative fair value approach. When the fair value of a reporting unit is available, we allocate goodwill based on the relative fair value of the reporting units. When fair value is not available, we utilize an alternative allocation methodology that we believe represents a reasonable approximation of the fair value of the operations being reorganized.

For more information, see Note 3—Goodwill, Customer Relationships and Other Intangible Assets.

Derivatives and Hedging

From time to time we have used derivative instruments to hedge exposure to interest rate risks arising from fluctuation in interest rates. We account for derivative instruments in accordance with ASC 815, *Derivatives and Hedging*, which establishes accounting and reporting standards for derivative instruments. We do not use derivative financial instruments for speculative purposes.

Derivatives are recognized in the consolidated balance sheets at their fair values. When we become a party to a derivative instrument and intend to apply hedge accounting, we formally document the hedge relationship and the risk management objective for undertaking the hedge, which includes designating the instrument for financial reporting purposes as a fair value hedge, a cash flow hedge, or a net investment hedge.

As of December 31, 2023, we were not party to any swap agreements. All of our variable-to-fixed interest rate swap agreements in place at the beginning of 2022 expired during the first half of 2022. While we held these agreements, we evaluated the effectiveness as described in Note 15—Derivative Financial Instruments (designated as cash-flow hedges) qualitatively on a quarterly basis. We reflected the change in the fair value of the interest rate swaps in accumulated other comprehensive loss and subsequently reclassified into earnings in the period the hedged transaction affects earnings, by virtue of qualifying as effective cash flow hedges. For more information see Note 15—Derivative Financial Instruments.

Pension and Post-Retirement Benefits

We recognize the funded status of our defined benefit and post-retirement plans as an asset or a liability on our consolidated balance sheets. Each year's actuarial gains or losses are a component of our other comprehensive income (loss), which is then included in our accumulated other comprehensive loss on our consolidated balance sheets. Pension and post-retirement benefit expenses are recognized over the period in which the employee renders service and becomes eligible to receive benefits. We make significant assumptions (including the discount rate, expected rate of return on plan assets, mortality and health care trend rates) in computing the pension and post-retirement benefits expense and obligations. See Note 11—Employee Benefits for additional information.

Foreign Currency

Local currencies of our foreign subsidiaries are the functional currencies for financial reporting purposes except for certain foreign subsidiaries, primarily in Latin America prior to the August 1, 2022 sale of our Latin American business. For operations outside the United States that have functional currencies other than the U.S. dollar, assets and liabilities are translated to U.S. dollars at period-end exchange rates, and revenue, expenses and cash flows are translated using average monthly exchange rates. Prior to the November 1, 2023 sale of our EMEA business and the August 1, 2022 sale of our Latin American business, a significant portion of our non-United States subsidiaries used the British pound, the Euro, or the Brazilian Real, as their functional currency, each of which experienced significant fluctuations against the U.S. dollar during the years ended December 31, 2023, 2022 and 2021. We recognize foreign currency translation gains and losses as a component of accumulated other comprehensive loss in stockholders' equity in our consolidated balance sheet and in our consolidated statements of comprehensive (loss) income in accordance with accounting guidance for foreign currency translation. Prior to the announcement of our divestitures as discussed in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses, we considered the majority of our investments in our foreign subsidiaries to be long-term in nature. Our foreign currency transaction gains (losses), including where transactions with our non-United States subsidiaries are not considered to be long-term in nature, are included within other (expense) income, net on our consolidated statements of operations.

Common Stock

As of December 31, 2023, we had 11 million shares authorized for future issuance under our equity incentive plans.

Preferred Stock

Holders of outstanding Lumen Technologies preferred stock are entitled to receive cumulative dividends, receive preferential distributions equal to $25 per share plus unpaid dividends upon Lumen's liquidation and vote as a single class with the holders of common stock.

Section 382 Rights Plan

We maintain a Section 382 Rights Plan to protect our U.S. federal net operating loss carryforwards from certain Internal Revenue Code Section 382 limitations. Under the plan, one preferred stock purchase right was distributed for each share of our outstanding common stock as of the close of business on February 25, 2019, and those rights currently trade in tandem with the common stock until they expire or detach under the plan. This plan was designed to deter trading that would result in a change of control (as defined in Code Section 382), and therefore protect our ability to use our historical federal NOLs in the future. The plan is scheduled to lapse in late 2026.

Dividends

The declaration and payment of dividends is at the discretion of our Board of Directors. On November 2, 2022, we announced that our Board had terminated our quarterly cash dividend program.

Correction of Immaterial Errors

During 2023, we identified errors in our previously reported consolidated financial statements related to accounts receivable and accounts payable. The errors are the result of understated revenues from one of our legacy mainframe billing systems and understated network expenses for periods prior to 2021. We have completed a quantitative and qualitative evaluation of the errors individually and in aggregate, and concluded the errors are immaterial to our previously issued consolidated financial statements. Notwithstanding this evaluation, we have revised certain line items on our December 31, 2022 consolidated balance sheet for these errors. The net effect of these adjustments was an increase in accounts receivable and total assets of $31 million and an increase of accounts payable and total liabilities of $94 million on our December 31, 2022 consolidated balance sheet. In addition, we recorded an adjustment to increase our January 1, 2021 accumulated deficit by $63 million, which represents the cumulative correction of the immaterial errors prior to January 1, 2021. The errors did not have an impact on our previously issued consolidated statements of operations, comprehensive (loss) income, or cash flows for the years ended December 31, 2022 or 2021, and did not, and are not expected to, have an impact on the economics of the Company's existing or future commercial arrangements.

Recently Adopted Accounting Pronouncements

Supplier Finance Programs

On January 1, 2023, we adopted Accounting Standards Update ("ASU") 2022-04, “Liabilities-Supplier Finance Program (Subtopic 405-50): Disclosure of Supplier Finance Program Obligations” (“ASU 2022-04”).These amendments require that a company that uses a supplier finance program in connection with the purchase of goods or services disclose sufficient information about the program to allow a user of financial statements to understand the program’s nature, program activity during the period, changes from period to period and the potential magnitude of program transactions. The adoption of ASU 2022-04 did not have a material impact to our consolidated financial statements.

Credit Losses

On January 1, 2023, we adopted ASU 2022-02, “Financial Instruments-Credit Losses (Topic 326): Troubled Debt Restructurings (“TDR”) and Vintage Disclosures” (“ASU 2022-02”). The ASU eliminates the TDR recognition and measurement guidance, enhances existing disclosure requirements and introduces new requirements related to certain modifications of receivables made to borrowers experiencing financial difficulty. The adoption of ASU 2022-02 did not have a material impact to our consolidated financial statements.

Government Assistance

On January 1, 2022, we adopted ASU 2021-10 "Government Assistance (Topic 832): Disclosures by Business Entities about Government Assistance” (“ASU 2021-10”) ASU 2021-10. This ASU requires business entities to disclose information about certain types of government assistance they receive. Please refer to Note 4—Revenue Recognition for more information.

Leases

On January 1, 2022, we adopted ASU 2021-05, “Leases (Topic 842): Lessors—Certain Leases with Variable Lease Payments” (“ASU 2021-05”). This ASU (i) amends the lease classification requirements for lessors to align them with practice under ASC Topic 840, (ii) provides criteria for lessors to classify and account for a lease with variable lease payments that do not depend on a reference index or a rate as an operating lease; and (iii) provides guidance with respect to net investments by lessors under operating leases and other related topics. The adoption of ASU 2021-05 did not have a material impact to our consolidated financial statements.

Debt

On January 1, 2021, we adopted ASU 2020-09, "Debt (Topic 470) Amendments to SEC Paragraphs Pursuant to SEC Release No. 33-10762" ("ASU 2020-09"). This ASU amends and supersedes various SEC guidance to reflect SEC Release No. 33-10762, which includes amendments to the financial disclosure requirements applicable to registered debt offerings that include credit enhancements, such as subsidiary guarantees. The adoption of ASU 2020-09 did not have a material impact to our consolidated financial statements.

Investments

On January 1, 2021, we adopted ASU 2020-01, "Investments - Equity Securities (Topic 321), Investments - Equity Method and Joint Ventures (Topic 323), and Derivatives and Hedging (Topic 815) - Clarifying the Interactions between Topic 321, Topic 323, and Topic 815)" ("ASU 2020-01"). This ASU, among other things, clarifies that a company should consider observable transactions that require a company to either apply or discontinue the equity method of accounting under Topic 323, Investments - Equity Method and Joint Ventures, for the purposes of applying the measurement alternative in accordance with Topic 321 immediately before applying or upon discontinuing the equity method. As of December 31, 2023, we determined there was no application or discontinuation of the equity method during the reporting periods covered in this report. The adoption of ASU 2020-01 did not have an impact to our consolidated financial statements.

Income Taxes

On January 1, 2021, we adopted ASU 2019-12, "Simplifying the Accounting for Income Taxes (Topic 740)" ("ASU 2019-12") . This ASU removes certain exceptions for investments, intra-period allocations and interim calculations, and adds guidance to reduce complexity in accounting for income taxes. The adoption of ASU 2019-12 did not have a material impact to our consolidated financial statements.

Recently Issued Accounting Pronouncements

In December 2023, the Financial Accounting Standards Board (“FASB”) issued ASU 2023-09, “Income Taxes (Topic 740): Improvements to Income Tax Disclosures” (“ASU 2023-09”). This ASU requires that public business entities must annually “(1) disclose specific categories in the rate reconciliation and (2) provide additional information for reconciling items that meet a quantitative threshold (if the effect of those reconciling items is equal to or greater than 5 percent of the amount computed by multiplying pretax income or loss by the applicable statutory income tax rate).” ASU 2023-09 will become effective for us in the annual period of fiscal 2025 and early adoption is permitted. We have chosen not to early adopt this ASU.

In December 2023, the FASB issued ASU 2023-08, “Intangibles—Goodwill and Other—Crypto Assets (Subtopic 350-60): Accounting for and Disclosure of Crypto Assets” (“ASU 2023-08”). This ASU is intended to improve the accounting for certain crypto assets by requiring an entity to measure those crypto assets at fair value each reporting period with changes in fair value recognized in net income. The amendments also improve the information provided to investors about an entity’s crypto asset holdings by requiring disclosure about significant holdings, contractual sale restrictions, and changes during the reporting period. This ASU will become effective for us in the first quarter of fiscal 2025 and early adoption is permitted. As of December 31, 2023, we do not hold crypto assets and do not expect ASU 2023-08 will have any impact to our consolidated financial statements.

In November 2023, the FASB issued ASU 2023-07, “Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures” (“ASU 2023-07”). This ASU is intended to improve reportable segment disclosure requirements primarily through enhanced disclosures about significant segment expenses. This ASU will become effective for us in annual period fiscal 2024 and early adoption is permitted. As of December 31, 2023, we are evaluating its impact on our consolidated financial statements.

In October 2023, the FASB issued ASU 2023-06, “Disclosure Improvements: Codification Amendments in Response to the SEC’s Disclosure Update and Simplification Initiative” (“ASU 2023-06”). This ASU incorporates certain SEC disclosure requirements into the FASB Accounting Standards Codification (“Codification”). The amendments in the ASU are expected to clarify or improve disclosure and presentation requirements of a variety of Codification Topics, allow users to more easily compare entities subject to the SEC’s existing disclosures with those entities that were not previously subject to the requirements, and align the requirements in the Codification with the SEC’s regulations. ASU 2023-06 will become effective for each amendment on the effective date of the SEC's corresponding disclosure rule changes. As of December 31, 2023, we do not expect ASU 2023-06 will have any impact to our consolidated financial statements.

In August 2023, the FASB issued ASU 2023-05, “Business Combinations – Joint Venture Formations (Subtopic 805-60): Recognition and initial Measurement” (“ASU 2023-05”). This ASU applies to the formation of entities that meet the definition of a joint venture (or a corporate joint venture). The amendments in the ASU require that a joint venture apply a new basis of accounting upon formation. ASU 2023-05 will become effective for us in the first quarter of fiscal 2025 and early adoption is permitted. As of December 31, 2023, we do not expect ASU 2023-05 will have any impact to our consolidated financial statements.

In August 2023, the FASB issued ASU 2023-04, “Liabilities (Topic 405): Amendments to SEC Paragraphs Pursuant to SEC Staff Accounting Bulletin No. 121” (“ASU 2023-04”). This ASU amends and adds various SEC paragraphs to the FASB Codification to reflect guidance regarding the accounting for obligations to safeguard crypto assets an entity holds for platform users. This ASU does not provide any new guidance. ASU 2023-04 became effective for us once the addition to the FASB Codification was made available. As of December 31, 2023, we do not expect ASU 2023-04 will have any impact to our consolidated financial statements.

In July 2023, the FASB issued ASU 2023-03, “Presentation of Financial Statements (Topic 205), Income Statement—Reporting Comprehensive Income (Topic 220), Distinguishing Liabilities from Equity (Topic 480), Equity (Topic 505), and Compensation—Stock Compensation (Topic 718): Amendments to SEC Paragraphs Pursuant to SEC Staff Accounting Bulletin No. 120, SEC Staff Announcement at the March 24, 2022 EITF Meeting, and Staff Accounting Bulletin Topic 6.B, Accounting Series Release 280—General Revision of Regulation S-X: Income or Loss Applicable to Common Stock” (“ASU 2023-03”). This ASU amends or supersedes various SEC paragraphs within the applicable codification to conform to past SEC staff announcements. This ASU does not provide any new guidance. ASU 2023-03 became effective for us once the addition to the FASB Codification was made available. As of December 31, 2023, we do not expect ASU 2023-03 will have any impact to our consolidated financial statements.

In March 2023, the FASB issued ASU 2023-02, “Investments-Equity Method and Joint Ventures (Topic 323): Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method” (“ASU 2023-02”). These amendments allow reporting entities to elect to account for qualifying tax equity investments using the proportional amortization method, regardless of the program giving rise to the related income tax credits. ASU 2023-02 will become effective for us in the first quarter of fiscal 2024 and early adoption is permitted. As of December 31, 2023, we do not expect ASU 2023-02 will have any impact to our consolidated financial statements.

In March 2023, the FASB issued ASU 2023-01, “Leases (Topic 842): Common Control Arrangements” (“ASU 2023-01”). These amendments require all entities to amortize leasehold improvements associated with common control leases over the useful life to the common control group. ASU 2023-01 will become effective for us in the first quarter of fiscal 2024 and early adoption is permitted. As of December 31, 2023, we do not expect ASU 2023-01 will have any impact to our consolidated financial statements.

In December 2022, the Financial Accounting Standards Board (“FASB”) issued ASU 2022-06, “Reference Rate Reform (Topic 848) – Deferral of the Sunset Date of Topic 848" ("ASU 2022-06"). These amendments extend the period of time preparers can utilize the reference rate reform relief guidance in Topic 848, which defers the sunset date from December 31, 2022, to December 31, 2024, after which entities will no longer be permitted to apply the relief in Topic 848. ASU 2022-06 is effective upon issuance. Based on our review of our key material contracts through December 31, 2023, ASU 2022-06 does not have a material impact to our consolidated financial statements.

In June 2022, the FASB issued ASU 2022-03, “Fair Value Measurement (Topic 820): Fair Value Measurement of Equity Securities Subject to Contractual Sale Restrictions” (“ASU 2022-03”). These amendments clarify that a contractual restriction on the sales of an investment in an equity security is not considered part of the unit of account of the equity security and, therefore, is not considered in measuring its fair value. ASU 2022-03 will become effective for us in the first quarter of fiscal 2024 and early adoption is permitted. As of December 31, 2023, we do not expect ASU 2022-03 will have any impact to our consolidated financial statements.

In January 2021, the FASB issued ASU 2021-01, "Reference Rate Reform (Topic 848): Scope" ("ASU 2021-01"), which clarifies that certain optional expedients and exceptions in Topic 848 for contract modifications and hedge accounting apply to derivatives that are affected by the discounting transition. ASU 2021-01 also amends the expedients and exceptions in Topic 848 to capture the incremental consequences of the scope clarification and to tailor the existing guidance to derivative instruments affected by the discounting transition. These amendments may be applied prospectively to contract modifications made and hedging relationships entered into or evaluated on or before December 31, 2022. ASU 2021-01 provides optional expedients for a limited time to ease the potential burden in accounting for reference rate reform. Based on our review of our key material contracts through December 31, 2023, ASU 2021-01 will not have a material impact to our consolidated financial statements.

(2) Divestitures of the Latin American, ILEC and EMEA Businesses

Latin American Business

On August 1, 2022, affiliates of Level 3 Parent, LLC, an indirect wholly-owned subsidiary of Lumen Technologies, Inc., sold Lumen’s Latin American business pursuant to a definitive agreement dated July 25, 2021, for pre-tax cash proceeds of approximately $2.7 billion.

For the year ended December 31, 2022, we recorded a $ 597 million net pre-tax gain on disposal associated with the sale of our Latin American business. This gain is reflected as operating income within the consolidated statements of operations.

In connection with the sale, we entered into a transition services agreement under which we provide the purchaser various support services. In addition, Lumen and the purchaser entered into commercial agreements whereby they provide each other various network and other commercial services. In addition, we agreed to indemnify the purchaser for certain matters for which future cash payments by Lumen could be required. Lumen has estimated the fair value of these indemnifications to be $86 million, which is included in other long-term liabilities in our consolidated balance sheet and has reduced our gain on the sale accordingly.

The Latin American business was included in our continuing operations and classified as assets and liabilities held for sale on our consolidated balance sheets through the closing of the transaction on August 1, 2022. As a result of closing the transaction, we derecognized net assets of $1.9 billion, primarily made up of (i) property, plant and equipment, net of accumulated depreciation, of $1.7 billion, (ii) goodwill of $245 million, (iii) other intangible assets, net of accumulated amortization, of $140 million, and (iv) deferred income tax liabilities, net, of $ 154 million. In addition, we reclassified $112 million of realized loss on foreign currency translation, net of tax, to partially offset the gain on sale of our Latin American business.

ILEC Business

On October 3, 2022, we and certain of our affiliates sold the portion of our incumbent local exchange ("ILEC") business primarily conducted within 20 Midwestern and Southeastern states to affiliates of funds advised by Apollo Global Management, Inc. In exchange, we received $ 7.5 billion of consideration, which was reduced by approximately $0.4 billion of closing adjustments and partially paid through purchaser's assumption of approximately $1.5 billion of our long-term consolidated indebtedness, resulting in pre-tax cash proceeds of approximately $ 5.6 billion.

For the year ended December 31, 2022, we recorded a $ 176 million net pre-tax gain on disposal associated with the sale of our ILEC business. This gain is reflected as operating income within the consolidated statements of operations.

In connection with the sale, we entered into a transition services agreement under which we provide the purchaser various support services. In addition, Lumen and the purchaser entered into commercial agreements whereby they provide each other various network and other commercial services. Under these agreements, we committed to ordering services of approximately $373 million from the purchaser over a period of three years and the purchaser has committed to ordering services of approximately $67 million from us over a period of three years. We indemnified the purchaser for certain matters for which, at the time of closing, future cash payments by Lumen were expected. Lumen had estimated the fair value of these indemnifications to be $89 million, which was included in other current liabilities in our consolidated balance sheet as of December 31, 2022 and increased our income tax expense accordingly as of December 31, 2022. As of the first quarter of 2023, the full $89 million payments had been made.

The ILEC business was included in our continuing operations and classified as assets and liabilities held for sale on our consolidated balance sheets through the closing of the transaction on October 3, 2022. As a result of closing the transaction, we derecognized net assets of $4.8 billion, primarily made up of (i) property, plant and equipment, net of accumulated depreciation, of $3.6 billion, (ii) goodwill of $2.6 billion and (iii) long-term debt, net of discounts, of $1.4 billion. In addition, we reclassified $ 403 million of net actuarial loss and prior service credit related to the Lumen Pension Plan, net of tax, conveyed to the purchaser to partially offset the gain on the sale of our ILEC business.

EMEA Business

On November 1, 2023, affiliates of Level 3 Parent, LLC, sold Lumen's operations in Europe, the Middle East and Africa (the "EMEA business") to Colt Technology Services Group Limited, a portfolio company of Fidelity Investments, for pre-tax cash proceeds of $1.7 billion after certain closing adjustments and transaction costs. This consideration is further subject to other post-closing adjustments and indemnities set forth in the purchase agreement, as amended and supplemented to date. In connection with the sale, we entered into a transition services agreement under which we provide the purchaser various support services. In addition, Lumen and the purchaser entered into commercial agreements whereby they provide each other various network and other commercial services.

The classification of the EMEA business as held for sale was considered an event or change in circumstance which requires an assessment of the goodwill of the disposal group for impairment each reporting period until disposal. We performed a pre-classification and post-classification goodwill impairment test of the disposal group as described further in Note 3—Goodwill, Customer Relationships and Other Intangible Assets. As a result of our impairment tests, we determined the EMEA business disposal group was impaired, resulting in a non-cash, non-tax-deductible goodwill impairment charge of $43 million in the fourth quarter of 2022. We evaluated the recoverability of the carrying value of the assets and liabilities held for sale relative to the agreed upon sales price, adjusted for costs to sell, and recorded an estimated loss on disposal of $660 million during the year ended December 31, 2022 in the consolidated statement of operations and a valuation allowance included in assets held for sale on the consolidated balance sheet as of December 31, 2022. For the year ended December 31, 2023, we recorded a $102 million net loss on disposal associated with the sale of our EMEA business. This loss is reflected as operating expense within the consolidated statements of operations.

The EMEA business was included in our continuing operations and classified as assets and liabilities held for sale on our consolidated balance sheets through the closing of the transaction on November 1, 2023. As a result of closing the transaction, we derecognized net assets of $2.1 billion, primarily made up of (i) property, plant and equipment, net of accumulated depreciation, of $2.0 billion and (ii) customer relationships and other intangible assets, net of accumulated amortization of $107 million. In addition, we reclassified $ 382 million of realized loss on foreign currency translation, net of tax, with an offset to the valuation allowance and loss on sale of the EMEA business.

Other Information

We do not believe these divestiture transactions represented a strategic shift for Lumen. Therefore, the divested businesses discussed above did not meet the criteria to be classified as discontinued operations. As a result, we continued to report our operating results for the Latin American, ILEC and EMEA businesses in our consolidated operating results through their respective disposal dates of August 1, 2022, October 3, 2022, and November 1, 2023, respectively.

(3) Goodwill, Customer Relationships and Other Intangible Assets

Goodwill, customer relationships and other intangible assets consisted of the following:

	As of December 31,	
	2023	2022[1]
	(Dollars in millions)	
Goodwill[2]	$ 1,964	12,657
Indefinite-lived intangible assets	$ 9	9
Other intangible assets subject to amortization:		
Customer relationships[3], less accumulated amortization of $ 4,248 and $3,606	3,811	4,574
Capitalized software, less accumulated amortization of $ 4,045[4] and $3,895	1,564	1,482
Trade names, patents and other, less accumulated amortization of $ 72[4] and $188	86	101
Total other intangible assets, net	$ 5,470	6,166

(1) These values exclude assets classified as held for sale.

(2) We recorded cumulative non-cash, non-tax-deductible goodwill impairment charges of $10.7 billion during the year ended December 31, 2023.

(3) For the year ended December 31, 2023, customer relationships decreased $121 million in conjunction with the sale of select CDN customer contracts in the fourth quarter of 2023 that resulted in a net loss of $73 million included in selling, general and administrative expenses in our consolidated statements of operations.

(4) Certain capitalized software with a gross carrying value of $183 million and trade names with a gross carrying value of $130 million became fully amortized during 2022 and were retired during the first quarter of 2023.

As of December 31, 2023, the gross carrying amount of goodwill, customer relationships, indefinite-lived and other intangible assets was $ 15.8 billion.

Our goodwill was derived from numerous acquisitions where the purchase price exceeded the fair value of the net assets acquired.

We are required to assess our goodwill and other indefinite-lived intangible assets for impairment annually, or, under certain circumstances, more frequently, such as when events or changes in circumstances indicate there may be impairment. Our annual impairment assessment date for indefinite-lived intangible assets other than goodwill is December 31. We completed our qualitative assessment of our indefinite-lived intangible assets other than goodwill as of December 31, 2023, 2022 and 2021 and concluded it is more likely than not that our indefinite-lived intangible assets are not impaired; thus, no impairment charge for these assets was recorded in 2023, 2022 or 2021. We are required to write down the value of goodwill only when our assessment determines the carrying value of equity of any of our reporting units exceeds its fair value. Our annual impairment assessment date for goodwill is October 31, at which date we assess our reporting units.

We report our results within two segments: Business and Mass Markets. See Note 17—Segment Information for more information on these segments and the underlying sales channels. As of December 31, 2023, we had three reporting units for goodwill impairment testing, which are (i) Mass Markets, (ii) North America Business ("NA Business") and (iii) Asia Pacific ("APAC") region. Prior to the divestiture of the EMEA business, the EMEA region was also a reporting unit and was tested for impairment in the pre-classification test as of October 31, 2022 discussed below. Prior to its August 1, 2022 divestiture, the Latin American ("LATAM") region was also a reporting unit.

Our reporting units are not discrete legal entities with discrete full financial statements. Our assets and liabilities are employed in and relate to the operations of multiple reporting units. For each reporting unit, we compare its estimated fair value of equity to its carrying value of equity that we assign to it. If the estimated fair value of the reporting unit is greater than the carrying value, we conclude that no impairment exists. If the estimated fair value of the reporting unit is less than its carrying value, we record a non-cash impairment charge equal to the excess amount. Depending on the facts and circumstances, we typically estimate the fair value of our reporting units by considering either or both of (i) a discounted cash flow method, which is based on the present value of projected cash flows over a discrete projection period and a terminal value, which is based on the expected normalized cash flows of the reporting units following the discrete projection period, and (ii) a market approach, which includes the use of market multiples of publicly-traded companies whose services and markets are comparable to ours.

2023 Goodwill Impairment Analyses

At October 31, 2023, we performed our annual impairment analysis of the goodwill of our three above-mentioned reporting units. Given the continued erosion in our market capitalization, we determined our quantitative impairment analysis would estimate the fair value of our reporting units using only the market approach. Applying this approach, we utilized company comparisons and analyst reports within the telecommunications industry which supported a range of fair values derived from annualized revenue and Earnings Before Interest, Tax, Depreciation and Amortization ("EBITDA") multiples between 1.5x and 3.5x and 4.8x and 8.4x, respectively. In determining the fair value of each reporting unit, we used revenue and EBITDA multiples below these comparable market multiples. We reconciled the estimated fair values of the reporting units to our market capitalization as of October 31, 2023 and concluded that the indicated control premium of approximately 2% was reasonable based on recent market transactions. Based on our assessments performed with respect to the reporting units as described above, we concluded the estimated fair value of certain of our reporting units was less than their carrying value of equity. As a result, we recorded a non-cash, non-tax-deductible goodwill impairment charge of $1.9 billion on October 31, 2023.

During the second quarter of 2023, we determined circumstances existed indicating it was more likely than not that the carrying value of our reporting units exceed their fair value. Given the continued erosion in our market capitalization, we determined our quantitative impairment analysis would estimate the fair value of our reporting units using only the market approach. Applying this approach, we utilized company comparisons and analyst reports within the telecommunications industry which supported a range of fair values derived from annualized revenue and EBITDA multiples between 1.5x and 4.3x and 4.6x and 10.5x, respectively. In determining the fair value of each reporting unit, we used revenue and EBITDA multiples below these comparable market multiples. The estimated fair values of the reporting units determined in connection with our impairment analysis in the second quarter of 2023 resulted in no control premium, which we determined to be reasonable based on our market capitalization relative to recent transactions. For the three months ended June 30, 2023, based on our assessments performed with respect to the reporting units as described above, we concluded the estimated fair value of certain of our reporting units was less than their carrying value of equity. As a result, we recorded a non-cash, non-tax-deductible goodwill impairment charge of $8.8 billion for the three months ended June 30, 2023.

The market approach that we used in the quarter ended June 30, 2023 and October 31, 2023 tests incorporated estimates and assumptions related to the forecasted results for the remainder of the year, including revenues, expenses, and the achievement of certain strategic initiatives. In developing the market multiples applicable to each reporting unit, we considered observed trends of our industry participants. Our assessment included many factors that required significant judgment. Alternative interpretations of these factors could have resulted in different conclusions regarding the size of our impairments.

2022 Goodwill Impairment Analyses

As of October 31, 2022, we estimated the fair value of our four above-mentioned reporting units by considering both a market approach and a discounted cash flow method. We discounted the projected cash flows for our Mass Markets, NA Business, EMEA and APAC reporting units using a rate that represented their weighted average cost of capital as of the assessment date, which comprised an after-tax cost of debt and a cost of equity, as disclosed in the table below. We utilized company comparisons and analyst reports within the telecommunications industry which at the time of assessment supported a range of fair values derived from annualized revenue and EBITDA multiples between 1.8x and 4.6x and 4.7x and 10.8x, respectively. We selected a revenue and EBITDA multiple for each of our reporting units, resulting in an overall company revenue and EBITDA multiple of 2.5x and 5.5x, respectively. We also reconciled the estimated fair values of the reporting units to our market capitalization as of October 31, 2022 and concluded that the indicated control premium of approximately 59% was reasonable based on recent market transactions, including our divestitures, and our depressed stock price. Due to the depressed trading price of our stock at October 31, 2022, and our assessment performed with respect to the reporting units described above, we concluded that the estimated fair value of our NA Business reporting unit was less than our carrying value of equity for that reporting unit, resulting in a non-cash, non-tax-deductible goodwill impairment charge of approximately $3.2 billion. See the goodwill rollforward by segment table below for the impairment charges by segment. As of October 31, 2022, the estimated fair value of equity exceeded the carrying value of equity for our Mass Markets, EMEA and APAC reporting units by 97%, 171% and 101%, respectively. Based on our assessments performed, we concluded that the goodwill assigned to our Mass Markets, EMEA and APAC reporting units was not impaired at October 31, 2022.

	As of October 31, 2022			
	Reporting Units			
	Mass Markets	NA Business	EMEA	APAC
Weighted average cost of capital	9.4 %	9.4 %	9.8 %	11.3 %
After-tax cost of debt	4.7 %	4.7 %	5.1 %	6.3 %
Cost of equity	14.0 %	14.0 %	14.4 %	16.2 %

Our classification of the EMEA Business as being held for sale as described in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses was considered an event or change in circumstance which required an assessment of our goodwill for impairment as of October 31, 2022. We performed a pre-announcement goodwill impairment test described above to determine whether there was an impairment prior to the classification of these assets as held for sale and to determine the November 2, 2022, fair values to be utilized for goodwill allocation regarding the disposal group to be classified as assets held for sale. We also performed a post-announcement goodwill impairment test using our estimated post-divestiture cash flows and carrying value of equity to evaluate whether the fair value of our NA Business, Mass Markets and APAC reporting units that will remain following the divestiture exceeds the carrying value of the equity of such reporting units after classification of assets held for sale. We concluded no impairment existed regarding our post-divestiture reporting units.

Separate from the annual, pre-announcement and post-announcement goodwill assessments discussed above, we performed an assessment of our EMEA business disposal group for impairment using the purchase price compared to the carrying value of the EMEA business net assets. As a result, the EMEA business disposal group was impaired, resulting in a non-cash, non-tax-deductible goodwill impairment charge of $43 million. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses for additional information regarding the purchase price, carrying value, and impairment for goodwill of the EMEA business. See the goodwill rollforward by segment table below for the impairment charges by segment.

2021 Goodwill Impairment Analyses

At October 31, 2021, we estimated the fair value of our five above-mentioned reporting units by considering both a market approach and a discounted cash flow method. As of October 31, 2021, we determined that the estimated fair value of equity exceeded the carrying value of equity for our Mass Markets, NA Business, EMEA, LATAM and APAC reporting units by 277%, 8%, 57%, 100% and 125%, respectively. Based on our assessments performed, we concluded it was more likely than not that the fair value of each of our reporting units exceeded the carrying value of equity of those reporting units at October 31, 2021. Therefore, we concluded no impairment existed as of our assessment date.

Our third quarter 2021 classification of held for sale assets related to the divestitures of the Latin American and ILEC businesses as described in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses, was considered an event or change in circumstance which required an assessment of our goodwill for impairment as of July 31, 2021. We performed a pre-classification goodwill impairment test to determine whether there was an impairment prior to the classification of these assets and to determine the July 31, 2021 fair values to be utilized for goodwill allocation regarding the Latin American and ILEC businesses classified as assets held for sale. We concluded it was more likely than not that the fair value of each of our reporting units exceeded the carrying value of equity of those reporting units at July 31, 2021. We also performed a post-classification goodwill impairment test using our estimated post-divestiture cash flows and carrying value of equity to evaluate whether the fair value of our reporting units that would remain following the divestitures exceeded the carrying value of the equity of such reporting units after classification of assets held for sale. At July 31, 2021, we estimated the fair value of our five above-mentioned reporting units as of such date by considering both a market approach and a discounted cash flow method. As of July 31, 2021, we determined that the estimated fair value of equity exceeded the carrying value of equity for our Mass Markets, NA Business, EMEA, LATAM and APAC reporting units by 150%, 24%, 58%,100% and 134%, respectively. Based on our assessments performed, we concluded it was more likely than not that the fair value of each of our reporting units exceeded the carrying value of equity of our reporting units at July 31, 2021. Therefore, we concluded no impairment existed as of our assessment date.

The January 2021 internal reorganization of our reporting structure was considered an event or change in circumstance which required an assessment of our goodwill for impairment. We performed a qualitative impairment assessment in the first quarter of 2021 and concluded it was more likely than not that the fair value of each of our reporting units exceeded the carrying value of equity of those reporting units at January 31, 2021. Therefore, we concluded no impairment existed as of our assessment date.

The following table shows the rollforward of goodwill assigned to our reportable segments from December 31, 2021 through December 31, 2023.

	Business	Mass Markets	Total
		(Dollars in millions)	
As of December 31, 2021	$ 11,235	4,751	15,986
Effect of foreign currency exchange rate change and other	$ (58)	—	(58)
Impairment	$ (3,271)	—	(3,271)
As of December 31, 2022 [1]	$ 7,906	4,751	12,657
Impairment	(7,906)	(2,787)	(10,693)
As of December 31, 2023 [1]	$ —	1,964	1,964

[1] Goodwill at December 31, 2023, December 31, 2022 and December 31, 2021 is net of accumulated impairment losses of $21.7 billion, $11.0 billion and $7.7 billion, respectively.

For additional information on our segments, see Note 17—Segment Information.

As of December 31, 2023, the weighted average remaining useful lives of our finite-lived intangible assets were approximately 6 years in total, approximately 7 years for customer relationships and 4 years for capitalized software.

Total amortization expense for finite-lived intangible assets for the years ended December 31, 2023, 2022 and 2021 was $ 1.1 billion, $1.1 billion and $1.3 billion, respectively.

We estimate that total amortization expense for finite-lived intangible assets for the years ending December 31, 2024 through 2028 will be as provided in the table below.

	(Dollars in millions)
2024	$ 922
2025	847
2026	803
2027	722
2028	657

(4) Revenue Recognition

Product and Service Categories

We categorize our products and services revenue among the following categories for the Business segment:

- *Grow*, which includes products and services that we anticipate will grow, including our dark fiber, Edge Cloud services, IP, managed security, software-defined wide area networks ("SD WAN"), secure access service edge ("SASE"), Unified Communications and Collaboration ("UC&C") and wavelengths services;
- *Nurture*, which includes our more mature offerings, including ethernet and VPN data networks services;
- *Harvest*, which includes our legacy services managed for cash flow, including Time Division Multiplexing ("TDM") voice, private line and other legacy services; and
- *Other*, which includes equipment sales, IT solutions and other services.

We categorize our products and services revenue among the following categories for the Mass Markets segment:

- *Fiber Broadband*, under which we provide high speed broadband services to residential and small business customers utilizing our fiber-based network infrastructure;
- *Other Broadband*, under which we provide primarily lower speed broadband services to residential and small business customers utilizing our copper-based network infrastructure; and
- *Voice and Other,* under which we derive revenues from (i) providing local and long-distance voice services, professional services, and other ancillary services, and (ii) federal broadband and state support programs.

Reconciliation of Total Revenue to Revenue from Contracts with Customers

The following tables provide total revenue by segment, sales channel and product category. They also provide the amount of revenue that is not subject to ASC 606, "*Revenue from Contracts with Customers*" ("ASC 606"), but is instead governed by other accounting standards. The amounts in the tables below include revenue for the Latin American, ILEC and EMEA businesses prior to their sales on August 1, 2022, October 3, 2022 and November 1, 2023, respectively:

	Year Ended December 31, 2023		
	Total Revenue	Adjustments for Non-ASC 606 Revenue [1]	Total Revenue from Contracts with Customers
	(Dollars in millions)		
Business Segment by Sales Channel and Product Category			
Large Enterprise			
Grow	$ 2,167	(294)	1,873
Nurture	1,450	—	1,450
Harvest	760	—	760
Other	239	(5)	234
Total Large Enterprise Revenue	4,616	(299)	4,317
Mid-Market Enterprise			
Grow	803	(28)	775
Nurture	797	—	797
Harvest	378	(4)	374
Other	33	(4)	29
Total Mid-Market Enterprise Revenue	2,011	(36)	1,975
Public Sector			
Grow	469	(81)	388
Nurture	398	—	398
Harvest	383	(1)	382
Other	533	—	533
Total Public Sector Revenue	1,783	(82)	1,701
Wholesale			
Grow	1,030	(251)	779
Nurture	820	(25)	795
Harvest	1,264	(165)	1,099
Other	11	—	11
Total Wholesale Revenue	3,125	(441)	2,684
Business Segment by Product Category			
Grow	4,469	(654)	3,815
Nurture	3,465	(25)	3,440
Harvest	2,785	(170)	2,615
Other	816	(9)	807
Total Business Segment Revenue	11,535	(858)	10,677
Mass Markets Segment by Product Category			
Fiber Broadband	636	(16)	620
Other Broadband	1,394	(126)	1,268
Voice and Other	992	(36)	956
Total Mass Markets Revenue	3,022	(178)	2,844
Total Revenue	$ 14,557	(1,036)	13,521
Timing of revenue			
Goods and services transferred at a point in time			$ 178
Services performed over time			13,343
Total revenue from contracts with customers			$ 13,521

	Year Ended December 31, 2022		
	Total Revenue	Adjustments for Non-ASC 606 Revenue [1]	Total Revenue from Contracts with Customers
	(Dollars in millions)		
Business Segment by Sales Channel and Product Category			
Large Enterprise			
Grow	$ 2,415	(352)	2,063
Nurture	1,685	—	1,685
Harvest	1,022	—	1,022
Other	255	(8)	247
Total Large Enterprise Revenue	5,377	(360)	5,017
Mid-Market Enterprise			
Grow	757	(32)	725
Nurture	915	—	915
Harvest	510	(7)	503
Other	30	(1)	29
Total Mid-Market Enterprise Revenue	2,212	(40)	2,172
Public Sector			
Grow	444	(103)	341
Nurture	490	—	490
Harvest	468	(4)	464
Other	459	(2)	457
Total Public Sector Revenue	1,861	(109)	1,752
Wholesale			
Grow	979	(271)	708
Nurture	1,004	(23)	981
Harvest	1,557	(215)	1,342
Other	51	—	51
Total Wholesale Revenue	3,591	(509)	3,082
Business Segment by Product Category			
Grow	4,595	(758)	3,837
Nurture	4,094	(23)	4,071
Harvest	3,557	(226)	3,331
Other	795	(11)	784
Total Business Segment Revenue	13,041	(1,018)	12,023
Mass Markets Segment by Product Category			
Fiber Broadband	604	(18)	586
Other Broadband	2,164	(200)	1,964
Voice and Other	1,669	(134)	1,535
Total Mass Markets Revenue	4,437	(352)	4,085
Total Revenue	$ 17,478	(1,370)	16,108
Timing of revenue			
Goods and services transferred at a point in time			$ 154
Services performed over time			15,954
Total revenue from contracts with customers			$ 16,108

	Year Ended December 31, 2021		
	Total Revenue	Adjustments for Non-ASC 606 Revenue [1]	Total Revenue from Contracts with Customers
	(Dollars in millions)		
Business Segment by Sales Channel and Product Category			
Large Enterprise			
Grow	$ 2,552	(427)	2,125
Nurture	1,906	—	1,906
Harvest	1,205	(2)	1,203
Other	255	(5)	250
Total Large Enterprise Revenue	5,918	(434)	5,484
Mid-Market Enterprise			
Grow	724	(29)	695
Nurture	1,026	—	1,026
Harvest	613	(7)	606
Other	35	(4)	31
Total Mid-Market Enterprise Revenue	2,398	(40)	2,358
Public Sector			
Grow	481	(84)	397
Nurture	528	—	528
Harvest	569	(3)	566
Other	533	(2)	531
Total Public Sector Revenue	2,111	(89)	2,022
Wholesale			
Grow	930	(279)	651
Nurture	1,080	(25)	1,055
Harvest	1,682	(228)	1,454
Other	—	—	—
Total Wholesale Revenue	3,692	(532)	3,160
Business Segment by Product Category			
Grow	4,687	(819)	3,868
Nurture	4,540	(25)	4,515
Harvest	4,069	(240)	3,829
Other	823	(11)	812
Total Business Segment Revenue	14,119	(1,095)	13,024
Mass Markets Segment by Product Category			
Fiber Broadband	524	—	524
Other Broadband	2,507	(227)	2,280
Voice and Other	2,537	(570)	1,967
Total Mass Markets Revenue	5,568	(797)	4,771
Total Revenue	$ 19,687	(1,892)	17,795
Timing of revenue			
Goods and services transferred at a point in time			$ 138
Services performed over time			17,657
Total revenue from contracts with customers			$ 17,795

(1) Includes regulatory revenue and lease revenue not within the scope of ASC 606.

Customer Receivables and Contract Balances

The following table provides balances of customer receivables, contract assets and contract liabilities, net of amounts classified as held for sale, as of December 31, 2023 and 2022:

	December 31, 2023	December 31, 2022
	(Dollars in millions)	
Customer receivables[1]	$ 1,256	1,424
Contract assets[2]	29	34
Contract liabilities[3]	698	656

(1) Reflects gross customer receivables of $1.3 billion and $1.5 billion, net of allowance for credit losses of $60 million and $73 million, at December 31, 2023 and December 31, 2022, respectively. At December 31, 2022 amounts exclude customer receivables, net, classified as held for sale of $76 million, related to the EMEA business which was sold November 1, 2023.

(2) At December 31, 2022 these amounts exclude contract assets classified as held for sale of $16 million, related to the EMEA business which was sold November 1, 2023.

(3) At December 31, 2022 these amounts exclude contract liabilities classified as held for sale of $59 million, related to the EMEA business which was sold November 1, 2023.

Contract liabilities are consideration we have received from our customers or billed in advance of providing goods or services promised in the future. We defer recognizing this consideration as revenue until we have satisfied the related performance obligation to the customer. Contract liabilities include recurring services billed one month in advance and installation and maintenance charges that are deferred and recognized over the actual or expected contract term, which typically ranges from 1 to 5 years depending on the service. Contract liabilities are included within deferred revenue in our consolidated balance sheets. During the years ended December 31, 2023 and December 31, 2022, we recognized $434 million and $539 million, respectively, of revenue that was included in contract liabilities of $715 million and $841 million as of January 1, 2023 and 2022, respectively, including contract liabilities that were classified as held for sale.

Performance Obligations

As of December 31, 2023, we expect to recognize approximately $ 6.8 billion of revenue in the future related to performance obligations associated with existing customer contracts that are partially or wholly unsatisfied. As of December 31, 2023, the transaction price related to unsatisfied performance obligation that are expected to be recognized in 2024, 2025 and thereafter was $2.8 billion, $1.7 billion and $2.3 billion, respectively.

These amounts exclude (i) the value of unsatisfied performance obligations for contracts for which we recognize revenue at the amount to which we have the right to invoice for services performed (for example, uncommitted usage or non-recurring charges associated with professional or technical services to be completed) and (ii) contracts that are classified as leasing arrangements or government assistance that are not subject to ASC 606.

Contract Costs

The following tables provide changes in our contract acquisition costs and fulfillment costs:

	Year Ended December 31, 2023	
	Acquisition Costs	Fulfillment Costs
	(Dollars in millions)	
Beginning of period balance	$ 202	192
Costs incurred	136	157
Amortization	(152)	(140)
Change in contract costs held for sale	(4)	(25)
End of period balance	$ 182	184

	Year Ended December 31, 2022	
	Acquisition Costs	Fulfillment Costs
	(Dollars in millions)	
Beginning of period balance	$ 222	186
Costs incurred	172	158
Amortization	(192)	(149)
Classified as held for sale [1]	—	(3)
End of period balance	$ 202	192

[1] Represents changes in amounts classified as held for sale related to the divestitures of our Latin American and ILEC businesses on August 1, 2022 and October 3, 2022, respectively, as well as changes of $6 million acquisition costs and no fulfillment costs classified as held for sale as of December 31, 2022 related to the divestiture of the EMEA business, held for sale as of December 31, 2022 and completed November 1, 2023. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses.

Acquisition costs include commission fees paid to employees as a result of obtaining contracts. Fulfillment costs include third party and internal costs associated with the provision, installation and activation of services to customers, including labor and materials consumed for these activities.

We amortize deferred acquisition and fulfillment costs based on the transfer of services on a straight-line basis over the average contract life of approximately 36 months for mass markets customers and 33 months for business customers. We include amortized fulfillment costs in cost of services and products and amortized acquisition costs are included in selling, general and administrative expenses in our consolidated statements of operations. We include the amount of these deferred costs that are anticipated to be amortized in the next 12 months in other current assets on our consolidated balance sheets. We include the amount of deferred costs expected to be amortized beyond the next twelve months in other non-current assets on our consolidated balance sheets. We assess deferred acquisition and fulfillment costs for impairment on a quarterly basis.

Governmental Funding

Lumen participates in various U.S. federal and state programs under which government support payments are received to offset costs associated with providing services in targeted locations such as unserved or underserved high-cost or rural areas, or for certain types of customers, including non-profit organizations, educational institutions and local governmental bodies. In certain instances, support payments are conditioned on specified infrastructure buildouts by milestone deadlines or provision of services at specified locations and speed requirements. Commitments may be made annually, on a multi-year basis ranging from one to ten years or be on-going subject to periodic change or termination. Consistent with customary practice and as referenced in ASC 832 *Government Assistance*, Lumen applies a grant model of accounting by which it accounts for these transactions as non-ASC 606 revenue over the periods in which the costs for which the funding is intended to compensate are incurred. This non-ASC 606 revenue is included in operating revenue in our consolidated statements of operations. Corresponding receivables are recorded when services have been provided to the customers and costs incurred, but the cash has not been received. These amounts are included in our accounts receivable, less allowance in our consolidated balance sheets. Certain programs are subject to audits of compliance with program commitments and, subject to the outcomes of those assessments, Lumen may be required to reimburse the government entity for cash previously received, or, in some cases, pay a penalty. Lumen evaluates each program and establishes a liability under the principles of ASC 450 if it is probable support payments will be recaptured or a penalty will be imposed.

For the years ended December 31, 2023 and 2022, Lumen recorded non-customer revenue of $85 million and $190 million, respectively, under government assistance programs, of which 17% and 31%, respectively, was associated with state universal service fund support programs.

Between 2015 and 2021, we received approximately $ 500 million annually through the Federal Communications Commission (the "FCC")'s Connect America Fund II ("CAF II"), a federal multi-year recurring subsidy program for more extensive broadband deployment in price-cap ILEC territories. For this program, which ended on December 31, 2021, we were required to meet certain specified infrastructure buildout requirements in 33 states by the end of 2021, which required substantial capital expenditures. In the first quarter of 2022, we recognized $59 million of previously deferred revenue related to the conclusion of the CAF II program based upon our final buildout and filing submissions. The government has the right to audit our compliance with the CAF II program and the ultimate outcome of any remaining examinations is unknown, but could result in a liability to us in excess of our reserve accruals established for these matters.

In early 2020, the FCC created the Rural Digital Opportunity Fund (the “RDOF”) program, a federal support program designed to fund broadband deployment in rural America. For the first phase of this program, RDOF Phase I, the FCC ultimately awarded $6.4 billion support payments to be paid in equal monthly installments over 10 years. We were awarded RDOF funding in several of the states in which we operate and began receiving monthly support payments during the second quarter of 2022. We received approximately $17 million in annual RDOF Phase I support payments for the years ended December 31, 2023 and 2022 and expect to receive this same amount each year thereafter during the program period.

Lumen participates in multiple state sponsored programs for broadband deployment in unserved and underserved areas for which the states have state universal service funds sourced from fees levied on telecommunications providers and passed on to consumers. During the years ending December 31, 2023 and 2022, Lumen participated in these types of programs primarily in the states of Nebraska, North Carolina, New Mexico, Minnesota, Virginia and Wisconsin.

(5) Leases

We primarily lease to or from third parties various office facilities, colocation facilities, equipment and transmission capacity. Leases with an initial term of 12 months or less are not recorded on our consolidated balance sheets; we recognize lease expense for these leases on a straight-line basis over the lease term.

We determine if an arrangement is a lease at inception and whether that lease meets the classification criteria of a finance or operating lease. Lease-related assets, or right-of-use assets, are recognized at the lease commencement date at amounts equal to the respective lease liabilities. Lease-related liabilities are recognized at the present value of the remaining contractual fixed lease payments, discounted using our incremental borrowing rates. As part of the present value calculation for the lease liabilities, we use an incremental borrowing rate as the rates implicit in the leases are not readily determinable. The incremental borrowing rates used for lease accounting are based on our unsecured rates, adjusted to approximate the rates at which we could borrow on a collateralized basis over a term similar to the recognized lease term. We apply the incremental borrowing rates to lease components using a portfolio approach based upon the length of the lease term and the reporting entity in which the lease resides. Operating lease expense is recognized on a straight-line basis over the lease term, while variable lease payments are expensed as incurred. Operating lease assets are included in other, net under goodwill and other assets on our consolidated balance sheets. Noncurrent operating lease liabilities are included in other under deferred credits and other liabilities on our consolidated balance sheets.

Some of our lease arrangements contain lease components, non-lease components (including common-area maintenance costs) and executory costs (including real estate taxes and insurance costs). We generally account for each component separately based on the estimated standalone price of each component. For colocation leases, we account for the lease and non-lease components as a single lease component.

Many of our lease agreements contain renewal options; however, we do not recognize right-of-use assets or lease liabilities for renewal periods unless we determine that we are reasonably certain of renewing the lease. Certain leases also include options to purchase the leased property. The depreciable life of assets and leasehold improvements are limited by the expected lease term, unless there is a transfer of title or purchase option reasonably certain to be exercised. Our lease agreements do not generally contain any material residual value guarantees or material restrictive covenants.

Lease expense consisted of the following:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Operating and short-term lease cost	$ 459	451	535
Finance lease cost:			
Amortization of right-of-use assets	32	37	37
Interest on lease liability	12	15	16
Total finance lease cost	44	52	53
Total lease cost	$ 503	503	588

We primarily lease from third parties various equipment, office facilities, retail outlets, switching facilities and other network sites or components. These leases, with few exceptions, provide for renewal options and rent escalations that are either fixed or based on the consumer price index. Any rent abatements, along with rent escalations, are included in the computation of rent expense calculated on a straight-line basis over the lease term. The lease term for most leases includes the initial non-cancelable term plus any term under renewal options that we believe are reasonably assured.

Beginning in the second half of 2020 and continuing into 2023, we rationalized our lease footprint and ceased using 42 underutilized leased property locations. We determined that we no longer needed the leased space and, due to the limited remaining term on the contracts, concluded that we had neither the intent nor ability to sublease the properties. For the years ended December 31, 2023 and 2021, we incurred accelerated lease costs of approximately $8 million and $35 million, respectively. We did not incur material accelerated lease costs during 2022. Additionally, during the second quarter of 2023, we also donated our Monroe, Louisiana campus and leased back a portion thereof. This donation resulted in a $101 million loss recognized for the year ended December 31, 2023. In conjunction with our plans to continue to reduce costs, we expect to continue our real estate rationalization efforts and expect to incur additional accelerated real estate costs in future periods.

For the years ended December 31, 2023, 2022 and 2021, our gross rental expense, including the accelerated lease costs discussed above, was $503 million, $503 million and $588 million, respectively. We also received sublease rental income of $25 million for each of the years ended December 31, 2023, 2022 and 2021.

Supplemental consolidated balance sheet information and other information related to leases is included below:

		As of December 31,	
Leases (Dollars in millions)	**Classification on the Balance Sheet**	**2023**	**2022**
Assets			
Operating lease assets	Other, net	$ 1,230	1,340
Finance lease assets	Property, plant and equipment, net of accumulated depreciation	260	317
Total leased assets		$ 1,490	1,657
Liabilities			
Current			
Operating	Current operating lease liabilities	$ 268	344
Finance	Current maturities of long-term debt	16	16
Noncurrent			
Operating	Other	1,040	1,088
Finance	Long-term debt	215	234
Total lease liabilities		$ 1,539	1,682
Weighted-average remaining lease term (years)			
Operating leases		8.2	7.7
Finance leases		11.3	12.0
Weighted-average discount rate			
Operating leases		7.59 %	5.98 %
Finance leases		4.98 %	4.96 %

At December 31, 2022, we classified certain operating and finance lease assets and liabilities related to the EMEA business, which was sold as of November 1, 2023, as held for sale and discontinued recording amortization on the related right-of-use assets upon this classification. These operating and finance lease assets and liabilities held for sale are not reflected in the above or throughout the disclosures within this note. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses for more information.

Supplemental consolidated cash flow statement information related to leases is included below:

	Years Ended December 31,	
	2023	2022
	(Dollars in millions)	
Cash paid for amounts included in the measurement of lease liabilities:		
Operating cash flows for operating leases	$ 461	462
Operating cash flows for finance leases	12	15
Financing cash flows for finance leases	25	89
Supplemental lease cash flow disclosures:		
Operating lease right-of-use assets obtained in exchange for new operating lease liabilities	$ 143	381
Right-of-use assets obtained in exchange for new finance lease liabilities	10	94

As of December 31, 2023, maturities of lease liabilities were as follows:

	Operating Leases	Finance Leases
	(Dollars in millions)	
2024	$ 350	26
2025	257	27
2026	204	28
2027	163	28
2028	130	28
Thereafter	698	166
Total lease payments	1,802	303
Less: interest	(494)	(72)
Total	1,308	231
Less: current portion	(268)	(16)
Long-term portion	$ 1,040	215

As of December 31, 2023, we had no material operating or finance leases that had not yet commenced.

Operating Lease Income

Lumen Technologies leases various dark fiber, office facilities, colocation facilities, switching facilities, other network sites and service equipment to third parties under operating leases. Lease and sublease income are included in operating revenue in the consolidated statements of operations. See "Revenue Recognition" in Note 1—Background and Summary of Significant Accounting Policies.

For the years ended December 31, 2023, 2022 and 2021, our gross rental income was $1.0 billion, $1.2 billion and $1.2 billion, respectively, which represents 7%, 7% and 6% respectively, of our operating revenue for the years ended December 31, 2023, 2022 and 2021.

(6) Credit Losses on Financial Instruments

To assess our expected credit losses on financial instruments, we aggregate financial assets with similar risk characteristics to monitor their credit quality or deterioration over the life of such assets. We periodically monitor certain risk characteristics within our aggregated financial assets and revise their composition accordingly, to the extent internal and external risk factors change. We separately evaluate financial assets that do not share risk characteristics with other financial assets. Our financial assets measured at amortized cost primarily consist of accounts receivable.

We use a loss rate method to estimate our allowance for credit losses. Our determination of the current expected credit loss rate begins with our review of historical loss experience as a percentage of accounts receivable. We measure our historical loss period based on the average days to recognize accounts receivable as credit losses. When asset specific characteristics and current conditions change from those in the historical period, due to changes in our credit and collections strategy, certain classes of aged balances, or credit loss and recovery policies, we perform a qualitative and quantitative assessment to adjust our historical loss rate. We use regression analysis to develop an expected loss rate using historical experience and economic data over a forecast period. We measure our forecast period based on the average days to collect payment on billed accounts receivable. To determine our current allowance for credit losses, we combine the historical and expected credit loss rates and apply them to our period end accounts receivable.

If there is an unexpected deterioration of a customer's financial condition or an unexpected change in economic conditions, including macroeconomic events, we assess the need to adjust the allowance for credit losses. Any such resulting adjustments would affect earnings in the period that adjustments are made.

The assessment of the correlation between historical observed default rates, current conditions and forecasted economic conditions requires judgment. Alternative interpretations of these factors could have resulted in different conclusions regarding our allowance for credit losses. The amount of credit loss is sensitive to changes in circumstances and forecasted economic conditions. Our historical credit loss experience, current conditions and forecast of economic conditions may also not be representative of the customers' actual default experience in the future, and we may use methodologies that differ from those used by other companies.

The following table presents the activity of our allowance for credit losses by accounts receivable portfolio for the years ended December 31, 2023 and December 31, 2022:

	Business	Mass Markets	Total
	(Dollars in millions)		
Beginning balance at January 1, 2021	$ 109	82	191
Provision for expected losses	50	55	105
Write-offs charged against the allowance	(76)	(101)	(177)
Recoveries collected	13	6	19
Classified as assets held for sale [1]	(8)	(16)	(24)
Balance at December 31, 2021	$ 88	26	114
Provision for expected losses	25	108	133
Write-offs charged against the allowance	(61)	(114)	(175)
Recoveries collected	10	6	16
Change in allowance in assets held for sale [2]	(5)	2	(3)
Balance at December 31, 2022	$ 57	28	85
Provision for expected losses	35	65	100
Write-offs charged against the allowance	(62)	(65)	(127)
Recoveries collected	6	3	9
Balance at December 31, 2023	$ 36	31	67

(1) Represents the amounts classified as held for sale related to the divestitures of our Latin American and ILEC businesses on August 1, 2022 and October 3, 2022, respectively. See Note 2—Divestitures of the Latin American and ILEC Businesses and Planned Divestiture of the EMEA Business.

(2) Represents changes in amounts classified as held for sale related to the divestitures of our Latin American and ILEC businesses on August 1, 2022 and October 3, 2022, respectively, and the inclusion of a $5 million allowance for credit losses classified as held for sale as of December 31, 2022 related to the divestiture of the EMEA business. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses.

(7) Long-Term Debt and Credit Facilities

The following table reflects the consolidated long-term debt of Lumen Technologies, Inc. and its subsidiaries as of the dates indicated below, including unamortized discounts and premiums and unamortized debt issuance costs:

	Interest Rates[1]	Maturities[1]	As of December 31, 2023	As of December 31, 2022
			(Dollars in millions)	
Senior Secured Debt: [2]				
Lumen Technologies, Inc.				
Revolving Credit Facility[3]	SOFR + 2.00%	2025	$ 200	—
Term Loan A[4]	SOFR + 2.00%	2025	933	991
Term Loan A-1[4]	SOFR + 2.00%	2025	266	283
Term Loan B[5]	SOFR + 2.25%	2027	3,891	3,941
Senior notes	4.000%	2027	1,250	1,250
Subsidiaries:				
Level 3 Financing, Inc.				
Tranche B 2027 Term Loan[6]	SOFR + 1.75%	2027	2,411	2,411
Senior notes	3.400% - 10.500%	2027 - 2030	2,425	1,500
Senior Notes and Other Debt:				
Lumen Technologies, Inc.				
Senior notes	4.500% - 7.650%	2025 - 2042	2,143	3,722
Subsidiaries:				
Level 3 Financing, Inc.				
Senior notes	3.625% - 4.625%	2027 - 2029	3,940	3,940
Qwest Corporation				
Senior notes	6.500% - 7.750%	2025 - 2057	1,986	1,986
Term loan[7]	SOFR + 2.50%	2027	215	215
Qwest Capital Funding, Inc.				
Senior notes	6.875% - 7.750%	2028 - 2031	192	192
Finance lease and other obligations[8]	Various	Various	285	317
Unamortized discounts, net			(4)	(7)
Unamortized debt issuance costs			(145)	(169)
Total long-term debt			19,988	20,572
Less current maturities			(157)	(154)
Long-term debt, excluding current maturities			$ 19,831	20,418

(1) As of December 31, 2023.

(2) See the remainder of this Note for a description of certain parent or subsidiary guarantees and liens securing this debt.

(3) Revolving Credit Facility had an interest rate of 7.464% as of December 31, 2023.

(4) Term Loans A and A-1 had interest rates of 7.470% and 6.384% as of December 31, 2023 and December 31, 2022, respectively.

(5) Term Loan B had interest rates of 7.720% and 6.634% as of December 31, 2023 and December 31, 2022, respectively.

(6) The Level 3 Tranche B 2027 Term Loan had interest rates of 7.220% and 6.134% as of December 31, 2023 and December 31, 2022, respectively.

(7) The Qwest Corporation Term Loan had interest rates of 7.970% and 6.640% as of December 31, 2023 and December 31, 2022, respectively.

(8) December 31, 2022 excludes finance lease obligations of our EMEA business that were classified as held for sale as of December 31, 2022 and sold on November 1, 2023. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses.

Long-Term Debt Maturities

Set forth below is the aggregate principal amount of our long-term debt as of December 31, 2023 (excluding unamortized discounts, net, and unamortized debt issuance costs) maturing during the following years.

	(Dollars in millions)
2024	$ 157
2025	1,864
2026	498
2027	9,386
2028	1,539
2029 and thereafter	6,693
Total long-term debt	$ 20,137

Debt of Lumen Technologies, Inc. and its Subsidiaries

At December 31, 2023, most of our outstanding consolidated debt had been incurred by Lumen Technologies, Inc. or one of the following three other primary borrowers or "borrowing groups," each of which has borrowed funds either on a standalone basis or as part of a separate restricted group with certain of its subsidiaries:

- Level 3 Financing, Inc., including its parent guarantor Level 3 Parent, LLC, and one or more subsidiary guarantors;
- Qwest Corporation; and
- Qwest Capital Funding, Inc., including its parent guarantor, Qwest Communications International Inc.

Each of these borrowers or borrowing groups has entered into one or more credit agreements with certain financial institutions or other institutional lenders, or issued senior notes. Certain of these debt instruments are described further below.

Amended and Restated Credit Agreement

On January 31, 2020, we amended and restated our credit agreement dated June 19, 2017 (as so amended and restated, the "Amended Credit Agreement"). At December 31, 2023, the Amended Credit Agreement consisted of the following facilities:

- a $2.2 billion senior secured revolving credit facility ("the Revolving Credit Facility"), against which $ 200 million of borrowings and $ 218 million of undrawn letters of credit were issued under this facility as of December 31, 2023, discussed further below;
- a $933 million senior secured Term Loan A credit facility;
- a $266 million senior secured Term Loan A-1 credit facility with CoBank, ACB; and
- a $3.9 billion senior secured Term Loan B credit facility (the term loan facilities and the Revolving Credit Facility being referred to collectively as the "Amended Secured Credit Facilities").

Loans under the Term Loan A and A-1 facilities and the Revolving Credit Facility bear interest at a rate equal to, at our option, the Secured Overnight Financing Rate ("SOFR") or the alternative base rate (each as defined in the Amended Credit Agreement) plus an applicable margin between 1.50% to 2.25% per annum for SOFR loans and 0.50% to 1.25% per annum for alternative base rate loans, depending on our then current total leverage ratio. Loans under the Term Loan B facility bear interest at SOFR plus 2.25% per annum or the alternative base rate plus 1.25% per annum. Loans under each of the term loan facilities require certain specified quarterly amortization payments and certain specified mandatory prepayments in connection with certain asset sales and debt issuances and out of excess cash flow, among other things, subject in each case to certain significant exceptions.

Borrowings under the Revolving Credit Facility and the Term Loan A and A-1 facilities mature on January 31, 2025. Borrowings under the Term Loan B facility mature on March 15, 2027.

All of Lumen's obligations under the Amended Secured Credit Facilities are guaranteed by certain of its subsidiaries. The guarantees by certain of those guarantors are secured by a first priority security interest in substantially all assets (including certain subsidiaries stock) directly owned by them, subject to certain exceptions and limitations.

A portion of the revolving credit facility in an amount not to exceed $ 250 million is available for swingline loans, and a portion in an amount not to exceed $800 million is available for the issuance of letters of credit. During the year ended December 31, 2023, we issued approximately $ 218 million of letters of credit under our revolving credit facility, which reduced our borrowing capacity available thereunder by the same amount. As of December 31, 2023, these issued letters of credit were undrawn.

Lumen Technologies is permitted under the Amended Credit Agreement to request certain incremental borrowings subject to the satisfaction of various conditions and to certain other limitations. Any incremental borrowings would be subject to the same terms and conditions under the Amended Credit Agreement.

Term Loans and Certain Other Debt of Subsidiaries

Qwest Corporation

On October 23, 2020, Qwest Corporation borrowed $ 215 million under a variable-rate term loan with CoBank ACB. The outstanding unpaid principal amount of this term loan plus any accrued and unpaid interest is due on October 23, 2027. Interest is paid at least quarterly based upon either SOFR or the base rate (as defined in the credit agreement) plus an applicable margin between 1.50% to 2.50% per annum for SOFR loans and 0.50% to 1.50% per annum for base rate loans depending on Qwest Corporation's then current senior unsecured long-term debt rating.

Level 3 Financing, Inc.

At December 31, 2023, Level 3 Financing, Inc. owed $ 2.4 billion under a senior secured Tranche B 2027 Term Loan, which matures on March 1, 2027. The Tranche B 2027 Term Loan carries an interest rate, in the case of base rate borrowings, equal to (i) the greater of the Prime Rate, the Federal Funds Effective Rate plus 50 basis points, or SOFR plus 100 basis points (with all such terms and calculations as defined or further specified in the credit agreement) plus (ii) 0.75% per annum. Any Eurodollar borrowings under the Tranche B 2027 Term Loan bear interest at SOFR plus 1.75% per annum.

The Tranche B 2027 Term Loan requires certain specified mandatory prepayments in connection with certain asset sales and other transactions, subject to certain significant exceptions. The obligations of Level 3 Financing, Inc. under the Tranche B 2027 Term Loan are, subject to certain exceptions, secured by certain assets of Level 3 Parent, LLC and certain of its material domestic telecommunication subsidiaries. Also, Level 3 Parent, LLC and certain of its subsidiaries have guaranteed the obligations of Level 3 Financing, Inc. under the Tranche B 2027 Term Loan.

Revolving Letters of Credit

We use various financial instruments in the normal course of business. These instruments include letters of credit, which are conditional commitments issued on our behalf in accordance with specified terms and conditions. Lumen Technologies maintains an uncommitted $225 million revolving letter of credit facility separate from the letter of credit facility included in the revolving credit facility noted above. Letters of credit issued under this uncommitted facility are backed by credit enhancements in the form of secured guarantees issued by certain of our subsidiaries. As of December 31, 2023 and 2022, we had (i) $40 million and $94 million, respectively, of letters of credit outstanding under our committed facility and various other facilities and (ii) $218 million and no letters of credit outstanding, respectively, under our revolving credit facility. As of December 31, 2023, these issued letters of credit were undrawn.

Senior Notes

Lumen's consolidated indebtedness at December 31, 2023 included (i) senior secured notes issued by Lumen Technologies, Inc. and Level 3 Financing, Inc. and (ii) senior unsecured notes issued by Lumen Technologies, Inc., Level 3 Financing, Inc., Qwest Corporation, and Qwest Capital Funding, Inc. All of these notes carry fixed interest rates and all principal is due on the notes' respective maturity dates, which rates and maturity dates are summarized in the table above. The Lumen Technologies, Inc. secured senior notes are guaranteed by the same domestic subsidiaries that guarantee the Amended Credit Agreement on substantially the same terms and conditions that govern the guarantees of the Amended Credit Agreement. The Level 3 Financing, Inc. secured senior notes are secured by a pledge of substantially all of its assets and guaranteed on a secured basis by the same domestic subsidiaries that guarantee its Term B 2027 Term Loan. The remaining senior notes issued by Level 3 Financing, Inc. are guaranteed on an unsecured basis by its parent, Level 3 Parent, LLC, and one of its subsidiaries. The senior notes issued by Qwest Capital Funding, Inc. are guaranteed by its parent, Qwest Communications International Inc. Except for a limited number of senior notes issued by Qwest Corporation, the issuer generally can redeem the notes, at its option, in whole or in part, (i) pursuant to a fixed schedule of pre-established redemption prices, (ii) pursuant to a "make whole" redemption price or (iii) under certain other specified limited conditions. Under certain circumstances in connection with a "change of control" of Lumen Technologies, it will be required to make an offer to repurchase each series of these senior notes (other than two of its older series of notes) at a price of 101% of the principal amount redeemed, plus accrued and unpaid interest. Also, under certain circumstances in connection with a "change of control" of Level 3 Parent, LLC or Level 3 Financing, Inc., Level 3 Financing will be required to make an offer to repurchase each series of its outstanding senior notes at a price of 101% of the principal amount redeemed, plus accrued and unpaid interest.

2023 Borrowings and Repayments

During 2023, Lumen borrowed $925 million from, and made repayments of $725 million to, its revolving credit facility.

2023 Exchange Offers and Repurchases

Pursuant to exchange offers that commenced on March 16, 2023 (the "Exchange Offers"), on March 31, 2023, Level 3 Financing, Inc. issued $915 million of its 10.500% Senior Secured Notes due 2030 (the " 10.500% Notes") in exchange for $1.535 billion of Lumen's outstanding senior unsecured notes. On April 17, 2023, in connection with the Exchange Offers, Level 3 Financing, Inc. issued an additional $9 million of its 10.500% Notes in exchange for $19 million of Lumen's outstanding senior unsecured notes. All exchanged notes were concurrently cancelled. These transactions resulted in a $630 million net reduction in the aggregate principal amount of Lumen's consolidated indebtedness. In addition to the above described exchange offers, we repurchased $24 million aggregate principal amount of Lumen's outstanding senior unsecured notes during the first quarter of 2023. These above-described transactions resulted in an aggregate net gain of $618 million for the year ended December 31, 2023.

The following table sets forth the aggregate principal amount of each series of Lumen's senior unsecured notes retired during the year ended December 31, 2023, in connection with the above-described exchange transactions:

Debt	Period of Reduction	Aggregate principal (amounts in millions)
5.625% Senior Notes, Series X, due 2025	Q1 2023	$ 48
7.200% Senior Notes, Series D, due 2025	Q1 2023	21
5.125% Senior Notes due 2026	Q1 2023	291
6.875% Debentures, Series G, due 2028	Q1 2023	52
5.375% Senior Notes due 2029	Q1 2023	275
4.500% Senior Notes due 2029	Q1 2023	556
7.600% Senior Notes, Series P, due 2039	Q1 2023	161
7.650% Senior Notes, Series U, due 2042	Q1 2023	131
5.625% Senior Notes, Series X, due 2025	Q2 2023	1
4.500% Senior Notes due 2029	Q2 2023	2
7.600% Senior Notes, Series P, due 2039	Q2 2023	3
7.650% Senior Notes, Series U, due 2042	Q2 2023	13
Total		$ 1,554

2022 Borrowings and Repayments

During 2022, Lumen borrowed $2.4 billion from, and made repayments of $ 2.6 billion to, its revolving credit facility. We used our net revolving credit draws and available cash to repay the following aggregate principal amounts of indebtedness through a combination of tender offers, redemptions, prepayments, amortization payments and payments at maturity. These transactions resulted in a net gain on the extinguishment of debt of $214 million.

Debt	Period of Repayment	(Dollars in millions)
Lumen Technologies, Inc.		
5.800% Senior Notes due 2022 (at maturity)	Q1 2022	$ 1,400
6.750% Senior Notes, Series W, due 2023	Q4 2022	750
7.500% Senior Notes, Series Y, due 2024	Q4 2022	982
7.500% Senior Notes, Series Y, due 2024	Q3 2022	18
5.625% Senior Notes, Series X, due 2025	Q4 2022	286
7.200% Senior Notes, Series D, due 2025	Q4 2022	34
5.125% Senior Notes due 2026	Q4 2022	520
5.125% Senior Notes due 2026	Q3 2022	11
6.875% Debentures, Series G, due 2028	Q4 2022	130
5.375% Senior Notes due 2029	Q4 2022	494
Term Loan B prepayment	Q4 2022	909
Scheduled term loan payments	Multiple	125
Level 3 Financing, Inc.		
Tranche B 2027 Term Loan	Q3 2022	700
5.375% Senior Notes due 2025	Q3 2022	800
5.250% Senior Notes due 2026	Q3 2022	775
Embarq Corporation Subsidiaries		
First Mortgage Bonds	Q4 2022	137
Qwest Capital Funding, Inc.		
Senior Notes	Q4 2022	63
Other	Q4 2022	68
Total debt repayments		$ 8,202

Interest Expense

Interest expense includes interest on total long-term debt. The following table presents the amount of gross interest expense, net of capitalized interest:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Interest expense:			
Gross interest expense	$ 1,269	1,398	1,575
Capitalized interest	(111)	(66)	(53)
Total interest expense	$ 1,158	1,332	1,522

Covenants

Lumen Technologies, Inc.

With respect to the Term Loan A and A-1 facilities and the Revolving Credit Facility, the Amended Credit Agreement requires us to maintain (i) a maximum total leverage ratio of not more than 4.75 to 1.00 and (ii) a minimum consolidated interest coverage ratio of at least 2.00 to 1.00, with such ratios being determined and calculated in the manner described in the Amended Credit Agreement.

The Amended Secured Credit Facilities contain various representations and warranties and extensive affirmative and negative covenants. Such covenants include, among other things and subject to certain significant exceptions, restrictions on our ability to declare or pay dividends, repurchase stock, repay certain other indebtedness, create liens, incur additional indebtedness, make investments, engage in transactions with our affiliates, dispose of assets and merge or consolidate with any other person.

The senior unsecured notes of Lumen Technologies, Inc. were issued under four separate indentures. These indentures restrict our ability to (i) incur, issue or create liens upon the property of Lumen Technologies, Inc. and (ii) consolidate with or merge into, or transfer or lease all or substantially all of our assets to any other party. These indentures do not contain any provisions that restrict the incurrence of additional indebtedness. The senior secured notes of Lumen Technologies, Inc. were issued under a separate indenture that contains a more restrictive set of covenants. As indicated above under "Senior Notes", Lumen Technologies, Inc. will be required to offer to purchase certain of its long-term debt securities issued under its indentures under certain circumstances in connection with a "change of control" of Lumen Technologies, Inc.

Level 3 Companies

The term loan, senior secured notes and senior unsecured notes of Level 3 Financing, Inc. contain various representations and extensive affirmative and negative covenants. Such covenants include, among other things and subject to certain significant exceptions, restrictions on their ability to declare or pay dividends, repay certain other indebtedness, create liens, incur additional indebtedness, make investments, dispose of assets and merge or consolidate with any other person. Also, as indicated above under "Senior Notes", Level 3 Financing, Inc. will be required to offer to repurchase or repay certain of its long-term debt under certain circumstances in connection with a "change of control" of Level 3 Financing or Level 3 Parent, LLC.

Qwest Companies

Under its term loan, Qwest Corporation must maintain a debt to EBITDA ratio of not more than 2.85 to 1.00, as determined and calculated in the manner described in the applicable term loan documentation. The term loan also contains a negative pledge covenant, which generally requires Qwest Corporation to secure equally and ratably any advances under the term loan if it pledges assets or permits liens on its property for the benefit of other debtholders.

The senior notes of Qwest Corporation were issued under indentures dated April 15, 1990 and October 15, 1999. These indentures contain restrictions on the incurrence of liens and the consummation of certain transactions substantially similar to the above-described covenants in Lumen's indentures (but contain no mandatory repurchase provisions). The senior notes of Qwest Capital Funding, Inc. were issued under an indenture dated June 29, 1998 containing terms substantially similar to those set forth in Qwest Corporation's indentures.

Impact of Covenants

The debt covenants applicable to Lumen Technologies, Inc. and its subsidiaries could have a material adverse impact on their ability to operate or expand their respective businesses, to pursue strategic transactions, or to otherwise pursue their plans and strategies. The covenants of the Level 3 companies may significantly restrict the ability of Lumen Technologies, Inc. to receive cash from the Level 3 companies, to distribute cash from the Level 3 companies to other of Lumen's affiliated entities, or to enter into other transactions among Lumen's wholly-owned entities.

Certain of the debt instruments of Lumen Technologies, Inc. and its subsidiaries contain cross payment default or cross acceleration provisions. When present, these provisions could have a wider impact on liquidity than might otherwise arise from a default or acceleration of a single debt instrument.

The ability of Lumen Technologies, Inc. and its subsidiaries to comply with the financial covenants in their respective debt instruments could be adversely impacted by a wide variety of events, including unforeseen contingencies, many of which are beyond their control.

Compliance

As of December 31, 2023, Lumen Technologies, Inc. believes it and its subsidiaries were in compliance with the provisions and financial covenants in their respective material debt agreements in all material respects.

Guarantees

Lumen Technologies does not guarantee the debt of any unaffiliated parties, but, as noted above, as of December 31, 2023 certain of its largest subsidiaries guaranteed (i) its debt outstanding under its Amended Secured Credit Facilities, its senior secured notes and its $225 million letter of credit facility and (ii) the outstanding term loans or senior notes issued by certain other subsidiaries. As further noted above, several of the subsidiaries guaranteeing these obligations have pledged substantially all of their assets to secure certain of their respective guarantees.

Subsequent Event

See Note 24—Subsequent Events, for information regarding certain debt restructuring transactions contemplated under our amended and restated transaction support agreement dated as of January 22, 2024.

(8) Accounts Receivable

The following table presents details of our accounts receivable balances:

	As of December 31,	
	2023	2022[1]
	(Dollars in millions)	
Trade and purchased receivables	$ 1,181	1,319
Earned and unbilled receivables	165	209
Other	39	65
Total accounts receivable	1,385	1,593
Less: allowance for credit losses	(67)	(85)
Accounts receivable, less allowance	$ 1,318	1,508

(1) Amounts have been adjusted to reflect the immaterial correction of accounts receivable. See Note 1—Background and Summary of Significant Accounting Policies under the header *Correction of Immaterial Errors*.

We are exposed to concentrations of credit risk from our customers. We generally do not require collateral to secure our receivable balances. We have agreements with other communications service providers whereby we agree to bill and collect on their behalf for services rendered by those providers to our customers within our local service area. We purchase accounts receivable from other communications service providers primarily on a recourse basis and include these amounts in our accounts receivable balance. We have not experienced any significant loss associated with these purchased receivables.

(9) Property, Plant and Equipment

Net property, plant and equipment is composed of the following:

	Depreciable Lives	As of December 31,	
		2023	2022[5]
		(Dollars in millions)	
Land	N/A	$ 646	651
Fiber, conduit and other outside plant [1]	15-45 years	15,217	14,451
Central office and other network electronics [2]	3-10 years	15,741	15,077
Support assets[3]	3-30 years	6,714	6,863
Construction in progress[4]	N/A	2,758	2,010
Gross property, plant and equipment		41,076	39,052
Accumulated depreciation		(21,318)	(19,886)
Net property, plant and equipment		$ 19,758	19,166

(1) Fiber, conduit and other outside plant consists of fiber and metallic cable, conduit, poles and other supporting structures.

(2) Central office and other network electronics consists of circuit and packet switches, routers, transmission electronics and electronics providing service to customers.

(3) Support assets consist of buildings, data centers, computers and other administrative and support equipment.

(4) Construction in progress includes inventory held for construction and property of the aforementioned categories that has not been placed in service as it is still under construction.

(5) At December 31, 2022, we had $1.9 billion of certain property, plant and equipment, net related to our EMEA business which was classified as held for sale at this date and which was sold on November 1, 2023. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses for more information.

We recorded depreciation expense of $1.9 billion, $2.1 billion and $2.7 billion for the years ended December 31, 2023, 2022 and 2021, respectively.

Asset Retirement Obligations

As of December 31, 2023 and 2022, our asset retirement obligations balance was primarily related to estimated future costs of removing equipment from leased properties and estimated future costs of properly disposing of asbestos and other hazardous materials upon remodeling or demolishing buildings. Asset retirement obligations are included in other long-term liabilities on our consolidated balance sheets.

Our fair value estimates were determined using the discounted cash flow method.

The following table provides asset retirement obligation activity:

	Years Ended December 31,	
	2023	2022
	(Dollars in millions)	
Balance at beginning of year	$ 156	182
Accretion expense	6	10
Liabilities settled	(9)	(10)
Change in estimate	4	4
Classified as held for sale [1]	—	(30)
Balance at end of year	$ 157	156

[1] Represents the amounts classified as held for sale related to our EMEA business. See Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses.

The changes in estimate referred to in the table above were offset against gross property, plant and equipment.

(10) Severance

Periodically, we reduce our workforce and accrue liabilities for the related severance costs. These workforce reductions result primarily from the progression or completion of our post-acquisition integration plans, increased competitive pressures, cost reduction initiatives, process improvements through automation and reduced workloads due to reduced demand for certain services.

During the fourth quarter of 2023 we reduced our global workforce by approximately 4% as part of our ongoing efforts to reorganize Lumen for growth by right-sizing our operations to improve our profitability. As a result of this plan, we incurred severance and related costs of approximately $53 million. We do not expect to incur any material impairment or exit costs related to this plan.

We report severance liabilities within accrued expenses and other liabilities - salaries and benefits in our consolidated balance sheets and report severance expenses in selling, general and administrative expenses in our consolidated statements of operations. As described in Note 17—Segment Information, we do not allocate these severance expenses to our segments.

Changes in our accrued liabilities for severance expenses were as follows:

	Severance
	(Dollars in millions)
Balance at December 31, 2021	$ 36
Accrued to expense	12
Payments, net	(37)
Balance at December 31, 2022	11
Accrued to expense	74
Payments, net	(67)
Balance at December 31, 2023	$ 18

(11) Employee Benefits

Pension, Post-Retirement and Other Post-Employment Benefits

We sponsor various defined benefit pension plans (qualified and non-qualified) which, in the aggregate, cover a substantial portion of our employees. Pension benefits for participants of the Lumen Combined Pension Plan ("Combined Pension Plan") and, through the October 3, 2022 sale of the ILEC business, the Lumen Pension Plan, who are represented by a collective bargaining agreement are based on negotiated schedules. All other participants' pension benefits are based on each individual participant's years of service and compensation. We also maintain non-qualified pension plans for certain current and former highly compensated employees. We maintain post-retirement benefit plans that provide health care and life insurance benefits for certain eligible retirees. We also provide other post-employment benefits for certain eligible former employees. We use a December 31 measurement date for all our plans.

On October 19, 2021, we, as sponsor of the Combined Pension Plan, along with the Plan's independent fiduciary, entered into an agreement committing the Plan to use a portion of its plan assets to purchase an annuity from an insurance company (the "Insurer") to transfer approximately $1.4 billion of the Plan's pension liabilities. This agreement irrevocably transferred to the Insurer future Plan benefit obligations for approximately 22,600 U.S. Lumen participants ("Transferred Participants") effective on December 31, 2021. This annuity transaction was funded entirely by existing Plan assets. The Insurer assumed responsibility for administrative and customer service support, including distribution of payments to the Transferred Participants. Transferred Participants' benefits were not reduced as a result of this transaction.

As of January 1, 2022, we spun off the Lumen Pension Plan from the Lumen Combined Pension Plan in anticipation of the sale of the ILEC business, as described further in Note 2—Divestitures of the Latin American, ILEC and EMEA Businesses. At the time of the spin-off, the Lumen Pension Plan covered approximately 2,500 active plan participants along with 19,000 other participants. At the time of the spin-off, the Lumen Pension Plan had a pension benefit obligation of $2.5 billion and assets of $ 2.2 billion. In addition, the December 31, 2021 actuarial (loss) gain and prior service cost included in accumulated other comprehensive loss was allocated between the Lumen Pension Plan and the Lumen Combined Pension Plan. Following a revaluation of the pension obligation and pension assets for the Lumen Pension Plan, in preparation for the closing of the sale of the ILEC business, we contributed approximately $319 million of Lumen's cash to the Lumen Pension Plan trust to fully fund the pension plan in September 2022. The amounts allocated to the Lumen Pension Plan were subject to adjustment up to the closing of the sale of the ILEC business on October 3, 2022, at which time the plan was transferred along with the rest of the assets and liabilities of the ILEC business. We recognized pension costs related to both plans through the sale of the ILEC business, at which time balances related to the Lumen Pension Plan were reflected in the calculation of our gain on the sale of the business.

Pension Benefits

United States funding laws require a company with a pension shortfall to fund the annual cost of benefits earned in addition to a seven-year amortization of the shortfall. Our funding policy for our Combined Pension Plan is to make contributions with the objective of accumulating ample assets to pay all qualified pension benefits when due under the terms of the plan. The accounting unfunded status of the Combined Pension Plan was $736 million and $580 million as of December 31, 2023 and 2022, respectively.

We made no voluntary cash contributions to the Combined Pension Plan in 2023 or 2022. As discussed above, we contributed approximately $319 million of cash to the Lumen Pension Plan trust to fully fund the pension plan in September 2022 in preparation for the closing of the sale of the ILEC business. We paid $5 million of benefits directly to participants of our non-qualified pension plans in both 2023 and 2022.

Benefits paid by the Combined Pension Plan are paid through a trust that holds all of the Plan's assets. The amount of required contributions to the Combined Pension Plan in 2024 and beyond will depend on a variety of factors, most of which are beyond our control, including earnings on plan investments, prevailing interest rates, demographic experience, changes in plan benefits and changes in funding laws and regulations. Based on current laws and circumstances, we do not believe we are required to make any contributions to the Combined Pension Plan in 2024 and we do not expect to make voluntary contributions to the trust for the Combined Pension Plan in 2024. We estimate that in 2024 we will pay $4 million of benefits directly to participants of our non-qualified pension plans.

We recognize in our consolidated balance sheets the funded status of the legacy Level 3 defined benefit post-retirement plans. These plans were fully funded as of December 31, 2023 and 2022. Additionally, as previously mentioned, we sponsor unfunded non-qualified pension plans for certain current and former highly-compensated employees. The net unfunded status of our non-qualified pension plans was $33 million and $35 million for the years ended December 31, 2023 and 2022, respectively. Due to the insignificant impact of these pension plans on our consolidated financial statements, we have predominantly excluded them from the remaining employee benefit disclosures in this Note, unless otherwise specifically stated.

Post-Retirement Benefits

Our post-retirement benefit plans provide post-retirement benefits to qualified retirees and allow (i) eligible employees retiring before certain dates to receive benefits at no or reduced cost and (ii) eligible employees retiring after certain dates to receive benefits on a shared cost basis. The post-retirement benefits not paid by the trusts are funded by us and we expect to continue funding these post-retirement obligations as benefits are paid. The accounting unfunded status of our qualified post-retirement benefit plan was $1.9 billion and $2.0 billion as of December 31, 2023 and 2022, respectively.

Assets in the post-retirement trusts were substantially depleted as of December 31, 2016; as of December 31, 2019 the Company ceased to pay certain post-retirement benefits through the trusts. No contributions were made to the post-retirement trusts in 2023, nor 2022. Benefits are paid directly by us with available cash. In 2023, we paid $194 million of post-retirement benefits, net of participant contributions and direct subsidies. In 2024, we currently expect to pay directly $193 million of post-retirement benefits, net of participant contributions and direct subsidies.

We expect our expected health care cost trend to range from 5.4% to 7.50% in 2024 and grading to 4.50% by 2031. Our post-retirement benefit cost, for certain eligible legacy Qwest retirees and certain eligible legacy CenturyLink retirees, is capped at a set dollar amount. Therefore, those health care benefit obligations are not subject to increasing health care trends after the effective date of the caps.

Expected Cash Flows

The Combined Pension Plan payments, post-retirement health care benefit payments and premiums, and life insurance premium payments are either distributed from plan assets or paid by us. The estimated benefit payments provided below are based on actuarial assumptions using the demographics of the employee and retiree populations and have been reduced by estimated participant contributions.

	Combined Pension Plan	Post-Retirement Benefit Plans	Medicare Part D Subsidy Receipts
		(Dollars in millions)	
Estimated future benefit payments:			
2024	$ 574	195	(2)
2025	493	191	(2)
2026	475	186	(2)
2027	458	181	(2)
2028	440	174	(2)
2029 - 2033	1,974	762	(6)

Net Periodic Benefit Expense

We utilize a full yield curve approach in connection with estimating the service and interest components of net periodic benefit expense by applying the specific spot rates along the yield curve used in the determination of the benefit obligation to the relevant projected cash flow.

The actuarial assumptions used to compute the net periodic benefit expense for our Combined Pension Plan and post-retirement benefit plans are based upon information available as of the beginning of the year, as presented in the following table.

	Combined Pension Plan			Post-Retirement Benefit Plans		
	2023	2022	2021	2023	2022	2021
Actuarial assumptions at beginning of year:						
Discount rate	5.45% - 5.69%	2.29% - 3.12%	1.70% - 2.88%	5.43% - 5.75%	2.19% - 5.78%	1.58% - 2.60%
Rate of compensation increase	3.25 %	3.25 %	3.25 %	N/A	N/A	N/A
Expected long-term rate of return on plan assets[1]	6.50 %	5.50 %	5.50 %	3.00 %	4.00 %	4.00 %
Initial health care cost trend rate	N/A	N/A	N/A	7.20% / 5.00%	5.00% / 5.75%	6.25% / 5.00%
Ultimate health care cost trend rate	N/A	N/A	N/A	4.50 %	4.50 %	4.50 %
Year ultimate trend rate is reached	N/A	N/A	N/A	2030	2025	2025

N/A - Not applicable

(1) Rates are presented net of projected fees and administrative costs.

Prior to the sale of the ILEC business on October 3, 2022, we realized pension costs related to the Lumen Pension Plan. Net periodic benefit expense (income) for our Combined Pension Plan and the Lumen Pension Plan (through October 3, 2022, together the "Pension Plans") includes the following components:

	Pension Plans Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Service cost	$ 25	44	56
Interest cost	270	194	201
Expected return on plan assets	(287)	(385)	(535)
Settlement charges	—	—	383
Realized to gain on sale of businesses	—	546	—
Special termination benefits charge	2	—	6
Recognition of prior service credit	(7)	(10)	(9)
Recognition of actuarial loss	104	122	184
Net periodic pension expense	$ 107	511	286

Net periodic benefit expense for our post-retirement benefit plans includes the following components:

	Post-Retirement Plans Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Service cost	$ 5	10	14
Interest cost	103	72	47
Realized to gain on sale of businesses	—	(32)	—
Recognition of prior service cost	(8)	8	15
Recognition of actuarial loss	(20)	(4)	4
Net periodic post-retirement benefit expense	$ 80	54	80

Service costs for our Combined Pension Plan and post-retirement benefit plans are included in the cost of services and products and selling, general and administrative line items on our consolidated statements of operations and all other costs listed above, except for amounts realized as part of the net gain on sale of businesses, are included in other (expense) income, net on our consolidated statements of operations for the years ended December 31, 2023, 2022 and 2021. Additionally, a portion of the service cost is also allocated to certain assets under construction, which are capitalized and reflected as part of property, plant and equipment in our consolidated balance sheets. As a result of ongoing efforts to reduce our workforce, we recognized a one-time charge in 2023 and in 2021 of $2 million and $6 million, respectively, for special termination benefit enhancements paid to certain eligible employees upon voluntary retirement.

Our pension plan contains provisions that allow us, from time to time, to offer lump sum payment options to certain former employees in settlement of their future retirement benefits. We record an accounting settlement charge, consisting of the recognition of certain deferred costs of the pension plan associated with these lump sum payments only if, in the aggregate, they exceed or are probable to exceed the sum of the annual service and interest costs for the plan's net periodic pension benefit cost, which represents the settlement accounting threshold. The lump sum pension settlement payments for 2021 exceeded the settlement threshold. In addition, during the fourth quarter of 2021, we executed an annuity purchase contract with a third party insurer that triggered additional settlement activity (see discussion above for further information). As a result, we recognized a non-cash settlement charge of $383 million as of December 31, 2021 to accelerate the recognition of a portion of the previously unrecognized actuarial losses in the qualified pension plan, which is reflected in other (expense) income, net in our consolidated statement of operations for the year ended December 31, 2021. This non-cash charge increased our recorded net loss and increased our recorded accumulated deficit, with an offset to accumulated other comprehensive loss in shareholders' equity for the year ended December 31, 2021. The amount of any future non-cash settlement charges will be dependent on several factors, including the total amount of our future lump sum benefit payments.

Benefit Obligations

The actuarial assumptions used to compute the funded status for the plans are based upon information available as of December 31, 2023 and 2022 and are as follows:

	Combined Pension Plan		Post-Retirement Benefit Plans	
	December 31,		December 31,	
	2023	2022	2023	2022
Actuarial assumptions at end of year:				
Discount rate	5.21 %	5.56 %	5.20 %	5.55 %
Rate of compensation increase	3.25 %	3.25 %	N/A	N/A
Initial health care cost trend rate	N/A	N/A	7.50% / 5.40%	7.20% / 5.00%
Ultimate health care cost trend rate	N/A	N/A	4.50 %	4.50 %
Year ultimate trend rate is reached	N/A	N/A	2031	2030

N/A - Not applicable

In 2021, we adopted the revised mortality tables and projection scales released by the Society of Actuaries, which increased the projected benefit obligation of our benefit plans by $37 million for 2021. The Society of Actuaries did not release any revised mortality tables or projection scales in 2022 or 2023.

The short-term and long-term interest crediting rates during 2023 for cash balance components of the Combined Pension Plan were 4.0% and 3.5%, respectively.

The following tables summarize the change in the benefit obligations for the Combined Pension Plan and post-retirement benefit plans:

	Combined Pension Plan Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Change in benefit obligation			
Benefit obligation at beginning of year	$ 5,295	9,678	12,202
Plan spin-off	—	(2,552)	—
Service cost	25	37	56
Interest cost	270	154	201
Plan amendments	—	—	(13)
Special termination benefits charge	2	—	6
Actuarial loss (gain)	114	(1,432)	(337)
Benefits paid from plan assets	(494)	(590)	(766)
Settlement payments and annuity purchase	—	—	(1,671)
Benefit obligation at end of year	$ 5,212	5,295	9,678

	Post-Retirement Benefit Plans Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Change in benefit obligation			
Benefit obligation at beginning of year	$ 1,995	2,781	3,048
Benefit obligation transferred to purchaser upon sale of business	—	(26)	—
Service cost	5	10	14
Interest cost	103	72	47
Participant contributions	32	37	41
Direct subsidy receipts	2	2	3
Plan amendments	—	(41)	—
Actuarial loss (gain)	14	(591)	(125)
Benefits paid by company	(228)	(249)	(247)
Benefits paid from plan assets	(4)	—	—
Benefit obligation at end of year	$ 1,919	1,995	2,781

Plan Assets

We maintain plan assets for our Combined Pension Plan and certain post-retirement benefit plans. As previously noted, assets in the post-retirement benefit plan trusts were substantially depleted as of December 31, 2016. The fair value of post-retirement benefit plan assets was $1 million, $5 million and $5 million at December 31, 2023, 2022 and 2021, respectively. Due to the insignificance of these assets on our consolidated financial statements, we have predominantly excluded them from the disclosures of plan assets in this Note, unless otherwise indicated.

The following table summarizes the change in the fair value of plan assets for the Combined Pension Plan:

	Combined Pension Plan Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Change in plan assets			
Fair value of plan assets at beginning of year	$ 4,715	8,531	10,546
Plan spin-off	—	(2,239)	—
Return on plan assets	255	(987)	422
Benefits paid from plan assets	(494)	(590)	(766)
Settlement payments and annuity purchase	—	—	(1,671)
Fair value of plan assets at end of year	$ 4,476	4,715	8,531

The expected rate of return on plan assets is the long-term rate of return we expect to earn on the plan's assets, net of administrative expenses paid from plan assets. It is determined annually based on the strategic asset allocation and the long-term risk and return forecast for each asset class.

Our investment objective for the Combined Pension Plan assets is to achieve an attractive risk-adjusted return over time that will provide for the payment of benefits and minimize the risk of large losses. We employ a liability-aware investment strategy designed to reduce the volatility of pension assets relative to pension liabilities. This strategy is evaluated frequently and is expected to evolve over time with changes in the funded status and other factors. Approximately 50% of plan assets is targeted to long-duration investment grade bonds and interest rate sensitive derivatives and 50% is targeted to diversified equity, fixed income and private market investments that are expected to outperform the liability with moderate funded status risk. At the beginning of 2024, our expected annual long-term rate of return on pension assets before consideration of administrative expenses is assumed to be 7.0%. Administrative expenses, including projected PBGC (Pension Benefit Guaranty Corporation) premiums, reduce the annual long-term expected return, net of administrative expenses, to 6.5%.

Permitted investments: Plan assets are managed consistent with the restrictions set forth by the Employee Retirement Income Security Act of 1974, as amended.

Fair Value Measurements: Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between independent and knowledgeable parties who are willing and able to transact for an asset or liability at the measurement date. We use valuation techniques that maximize the use of observable inputs and minimize the use of unobservable inputs when determining fair value and then we rank the estimated values based on the reliability of the inputs used following the fair value hierarchy set forth by the FASB. For additional information on the fair value hierarchy, see Note 14—Fair Value of Financial Instruments.

At December 31, 2023, we used the following valuation techniques to measure fair value for assets. There were no changes to these methodologies during 2023:

- Level 1—Assets were valued using the closing price reported in the active market in which the individual security was traded. U.S. Treasury securities are valued at the bid price reported in an active market in which the security is traded. Variation margin due from/(to) brokers is valued at the expected next day cash settlement amount.

- Level 2—Assets were valued using quoted prices in markets that are not active, broker dealer quotations, and other methods by which all significant inputs were observable at the measurement date. Fixed income securities primarily utilize observable market information and are based on a spread to U.S. Treasury securities and consider yields available on comparable securities of issuers with similar credit ratings, the new issue market for similar securities, secondary trading markets and dealer quotes. Option adjusted spread models are utilized to evaluate fixed income securities that have early redemption features. Derivative securities traded over the counter are valued based on gains or losses due to fluctuations in indices, interest rates, foreign currency exchange rates, security prices or other underlying factors. Repurchase agreements are valued based on expected settlement per the contract terms.

- Level 3—Assets were valued using unobservable inputs in which little or no market data exists as reported by the respective institutions at the measurement date. Valuation methods may consider a range of factors, including estimates based on the assumptions of the investment entity.

The Combined Pension Plan's assets are invested in various asset categories utilizing multiple strategies and investment managers. Interests in commingled funds are fair valued using a practical expedient to the net asset value ("NAV") per unit (or its equivalent) of each fund. The NAV reported by the fund manager is based on the market value of the underlying investments owned by each fund, minus its liabilities, divided by the number of shares outstanding. Commingled funds can be redeemed at NAV, with a frequency that includes daily, monthly, quarterly, semi-annually and annually. These commingled funds include redemption notice periods between same day and 180 days. Investments in private funds, primarily limited partnerships, represent long-term commitments with a fixed maturity date and are also valued at NAV. The plan has unfunded commitments related to certain private fund investments, which in aggregate are not material to the plan. Valuation inputs for these private fund interests are generally based on assumptions and other information not observable in the market. Underlying investments held in funds are aggregated and are classified based on the fund mandate. Investments held in separate accounts are individually classified.

The table below presents the fair value of plan assets by category and the input levels used to determine those fair values at December 31, 2023. It is important to note that the asset allocations do not include market exposures that are gained with derivatives. Investments include dividend and interest receivables, pending trades and accrued expenses.

	Fair Value of Combined Pension Plan Assets at December 31, 2023			
	Level 1	Level 2	Level 3	Total
	(Dollars in millions)			
Assets				
Investment grade bonds (a)	$ 390	1,838	—	2,228
High yield bonds (b)	—	32	4	36
Emerging market bonds (c)	57	57	—	114
U.S. stocks (d)	247	—	1	248
Non-U.S. stocks (e)	6	—	—	6
Multi-asset strategies (l)	28	—	—	28
Total investments, excluding investments valued at NAV	$ 728	1,927	5	2,660
Liabilities				
Repurchase agreements & other obligations (n)	$ —	(375)	—	(375)
Derivatives (m)	(1)	—	—	(1)
Investments valued at NAV				2,192
Total pension plan assets				$ 4,476

The table below presents the fair value of plan assets by category and the input levels used to determine those fair values at December 31, 2022. It is important to note that the asset allocations do not include market exposures that are gained with derivatives. Investments include dividend and interest receivable, pending trades and accrued expenses.

	Fair Value of Combined Pension Plan Assets at December 31, 2022			
	Level 1	Level 2	Level 3	Total
	(Dollars in millions)			
Assets				
Investment grade bonds (a)	$ 446	1,720	—	2,166
High yield bonds (b)	—	48	4	52
Emerging market bonds (c)	49	78	—	127
U.S. stocks (d)	214	—	1	215
Non-U.S. stocks (e)	149	1	—	150
Multi-asset strategies (l)	25	—	—	25
Cash equivalents and short-term investments (o)	—	1	—	1
Total investments, excluding investments valued at NAV	$ 883	1,848	5	2,736
Liabilities				
Repurchase agreements (n)	$ —	(269)	—	(269)
Derivatives (m)	(1)	(10)	—	(11)
Investments valued at NAV				2,259
Total pension plan assets				$ 4,715

The table below presents the fair value of plan assets valued at NAV by category for our Combined Pension Plan at December 31, 2023 and 2022.

	Fair Value of Plan Assets Valued at NAV	
	Combined Pension Plan at December 31,	
	2023	2022
	(Dollars in millions)	
Investment grade bonds (a)	$ 105	99
High yield bonds (b)	110	81
U.S. stocks (d)	51	79
Non-U.S. stocks (e)	412	270
Emerging market stocks (f)	10	15
Private equity (g)	272	326
Private debt (h)	421	438
Market neutral hedge funds (i)	77	135
Directional hedge funds (j)	124	166
Real estate (k)	265	333
Multi-asset strategies (l)	27	24
Cash equivalents and short-term investments (o)	318	293
Total investments valued at NAV	$ 2,192	2,259

Below is an overview of the asset categories and the underlying strategies used in the preceding tables:

(a) *Investment grade bonds* represent investments in U.S. Treasury securities, agencies, corporate bonds, mortgage-backed securities, asset-backed securities and commercial mortgage-backed securities.

(b) *High yield bonds* represent investments in below investment grade fixed income securities.

(c) *Emerging market bonds* represent investments issued by governments and other entities located in emerging countries.

(d) *U.S. stocks* represent investments in stocks of U.S. based companies.

(e) *Non-U.S. stocks* represent investments in companies based in developed countries outside the U.S.

(f) *Emerging market stocks* represent investments in stocks of companies located in emerging markets.

(g) *Private equity* represents non-public investments in domestic and foreign buy out and venture capital funds. Private equity funds are primarily structured as limited partnerships and are valued according to the valuation policy of each partnership, subject to prevailing accounting and other regulatory guidelines.

(h) *Private debt* represents non-public investments in distressed or mezzanine debt.

(i) *Market neutral hedge funds* hold investments in a diversified mix of instruments that are intended in combination to exhibit low correlations to market fluctuations. These investments are typically combined with futures to achieve uncorrelated excess returns over various markets.

(j) *Directional hedge funds*—This asset category represents investments that may exhibit somewhat higher correlations to market fluctuations than the market neutral hedge funds.

(k) *Real estate* represents investments in a diversified portfolio of real estate properties.

(l) *Multi-asset strategies* represent broadly diversified strategies that have the flexibility to tactically adjust exposures to different asset classes through time.

(m) *Derivatives* include exchange traded futures contracts as well as privately negotiated over the counter contracts. The market values represent gains or losses that occur due to differences between stated contract terms and fluctuations in underlying market instruments.

(n) *Repurchase agreements and other obligations* includes contracts where the security owner sells a security with the agreement to buy it back at a future date and price. Other obligations include obligations to repay cash collateral held by a plan, net liability for investment purchases pending settlement, and accrued plan expenses.

(o) *Cash equivalents and short-term investments* represent investments that are used in conjunction with derivatives positions or are used to provide liquidity for the payment of benefits or other purposes.

Derivative instruments: Derivative instruments are used to reduce risk as well as provide return. The gross notional exposure of the derivative instruments directly held by the Combined Pension Plan is shown below. The notional amount of the derivatives corresponds to market exposure but does not represent an actual cash investment.

	Gross Notional Exposure	
	Combined Pension Plan Years Ended December 31,	
	2023	2022
	(Dollars in millions)	
Derivative instruments:		
Exchange-traded U.S. equity futures	$ 60	70
Exchange-traded Treasury and other interest rate futures	1,136	1,256
Exchange-traded Foreign currency futures	1	2
Interest rate swaps	214	82
Credit default swaps	72	139
Index swaps	94	90
Foreign exchange forwards	57	50
Options	32	251

Concentrations of Risk: Investments, in general, are exposed to various risks, such as significant world events, interest rate, credit, foreign currency and overall market volatility risk. These risks are managed by broadly diversifying assets across numerous asset classes and strategies with differing expected returns, volatilities and correlations. Risk is also broadly diversified across numerous market sectors and individual companies. Financial instruments that potentially subject the plans to concentrations of counterparty risk consist principally of investment contracts with high quality financial institutions. These investment contracts are typically collateralized obligations and/or are actively managed, limiting the amount of counterparty exposure to any one financial institution. Although the investments are well diversified, the value of plan assets could change materially depending upon the overall market volatility, which could affect the funded status of the plan.

The table below presents a rollforward of the Combined Pension Plan assets valued using Level 3 inputs:

	Combined Pension Plan Assets Valued Using Level 3 Inputs		
	High Yield Bonds	U.S. Stocks	Total
	(Dollars in millions)		
Balance at December 31, 2021	$ 6	5	11
Dispositions	(1)	(4)	(5)
Actual return on plan assets	(1)	—	(1)
Balance at December 31, 2022	4	1	5
(Dispositions) acquisitions	(2)	—	(2)
Actual return on plan assets	2	—	2
Balance at December 31, 2023	$ 4	1	5

Certain gains and losses are allocated between assets sold during the year and assets still held at year-end based on transactions and changes in valuations that occurred during the year. These allocations also impact our calculation of net acquisitions and dispositions.

For the year ended December 31, 2023, the investment program produced actual gains on Combined Pension Plan assets of $ 255 million as compared to expected returns of $287 million, for a difference of $ 32 million. For the year ended December 31, 2022, the investment program produced actual losses on Combined Pension Plan assets of $987 million as compared to the expected returns of $329 million, for a difference of $ 1.3 billion. The short-term annual returns on plan assets will almost always be different from the expected long-term returns and the plans could experience net gains or losses, due primarily to the volatility occurring in the financial markets during any given year.

Unfunded Status

The following table presents the unfunded status of the Combined Pension Plan and post-retirement benefit plans:

	Combined Pension Plan		Post-Retirement Benefit Plans	
	Years Ended December 31,		Years Ended December 31,	
	2023	2022	2023	2022
	(Dollars in millions)			
Benefit obligation	$ (5,212)	(5,295)	(1,919)	(1,995)
Fair value of plan assets	4,476	4,715	1	5
Unfunded status	(736)	(580)	(1,918)	(1,990)
Current portion of unfunded status	—	—	(193)	(210)
Non-current portion of unfunded status	$ (736)	(580)	(1,725)	(1,780)

The current portion of our post-retirement benefit obligations is recorded on our consolidated balance sheets in accrued expenses and other current liabilities-salaries and benefits.

Accumulated Other Comprehensive Loss-Recognition and Deferrals

The following table presents cumulative items not recognized as a component of net periodic benefits expense as of December 31, 2022, items recognized as a component of net periodic benefits expense in 2023, additional items deferred during 2023 and cumulative items not recognized as a component of net periodic benefits expense as of December 31, 2023. The items not recognized as a component of net periodic benefits expense have been recorded on our consolidated balance sheets in accumulated other comprehensive loss:

	As of and for the Years Ended December 31,				
	2022	Recognition of Net Periodic Benefits Expense	Deferrals	Net Change in AOCL	2023
	(Dollars in millions)				
Accumulated other comprehensive (loss) income					
Pension plans:					
Net actuarial (loss) gain	$ (1,752)	80	(147)	(67)	(1,819)
Settlement charge	383	—	—	—	383
Prior service benefit (cost)	17	(7)	—	(7)	10
Deferred income tax benefit (expense)	367	(23)	37	14	381
Total pension plans	(985)	50	(110)	(60)	(1,045)
Post-retirement benefit plans:					
Net actuarial gain (loss)	371	(20)	(14)	(34)	337
Prior service benefit (cost)	37	(8)	—	(8)	29
Curtailment loss	4	—	—	—	4
Deferred income tax (expense) benefit	(104)	7	3	10	(94)
Total post-retirement benefit plans	308	(21)	(11)	(32)	276
Total accumulated other comprehensive (loss) income	$ (677)	29	(121)	(92)	(769)

The following table presents cumulative items not recognized as a component of net periodic benefits expense as of December 31, 2021, items recognized as a component of net periodic benefits expense in 2022, additional items deferred during 2022 and cumulative items not recognized as a component of net periodic benefits expense as of December 31, 2022. The items not recognized as a component of net periodic benefits expense have been recorded on our consolidated balance sheets in accumulated other comprehensive loss:

	As of and for the Years Ended December 31,				
	2021	Recognition of Net Periodic Benefits Expense	Deferrals	Net Change in AOCL	2022
	(Dollars in millions)				
Accumulated other comprehensive (loss) income					
Pension plans:					
Net actuarial (loss) gain	$ (2,564)	688	124	812	(1,752)
Settlement charge	383	—	—	—	383
Prior service benefit (cost)	45	(28)	—	(28)	17
Deferred income tax benefit (expense)	559	(166)	(26)	(192)	367
Total pension plans	(1,577)	494	98	592	(985)
Post-retirement benefit plans:					
Net actuarial (loss) gain	(217)	(3)	591	588	371
Prior service (cost) benefit	(5)	1	41	42	37
Curtailment loss	4	—	—	—	4
Deferred income tax benefit (expense)	54	1	(159)	(158)	(104)
Total post-retirement benefit plans	(164)	(1)	473	472	308
Total accumulated other comprehensive (loss) income	$ (1,741)	493	571	1,064	(677)

Medicare Prescription Drug, Improvement and Modernization Act of 2003

We sponsor post-retirement health care plans with several benefit options that provide prescription drug benefits that we deem actuarially equivalent to or exceeding Medicare Part D. We recognize the impact of the federal subsidy received under the Medicare Prescription Drug, Improvement and Modernization Act of 2003 in the calculation of our post-retirement benefit obligation and net periodic post-retirement benefit expense.

Other Benefit Plans

Health Care and Life Insurance

We provide health care and life insurance benefits to essentially all of our active employees. We are largely self-funded for the cost of the health care plan. Our health care benefit expense for current employees was $288 million, $296 million and $309 million for the years ended December 31, 2023, 2022 and 2021, respectively. Union-represented employee benefits are based on negotiated collective bargaining agreements. Employees contributed $89 million, $101 million, $120 million for the years ended December 31, 2023, 2022 and 2021, respectively. Our group basic life insurance plans are fully insured and the premiums are paid by us.

401(k) Plans

We sponsor a qualified defined contribution plan covering substantially all of our U.S. employees. Under this plan, employees may contribute a percentage of their annual compensation up to certain maximums, as defined by the plan and by the Internal Revenue Service. Currently, we match a percentage of employee contributions in cash. At December 31, 2023 and 2022, the assets of the plan included approximately 9 million and 10 million shares of our common stock, all of which were the result of the combination of previous employer match and participant directed contributions. We recognized expenses related to this plan of $87 million, $91 million and $96 million for the years ended December 31, 2023, 2022 and 2021, respectively.

Deferred Compensation Plans

We sponsor non-qualified deferred compensation plans for various groups that included certain of our current and former highly compensated employees. The value of liabilities related to these plans was not significant.

(12) Stock-based Compensation

We maintain an equity incentive program that allows our Board of Directors (through its Compensation Committee or a senior officer acting under delegated authority) to grant incentives to certain employees and outside directors in one or more forms, including: incentive and non-qualified stock options, stock appreciation rights, restricted stock awards, restricted stock units and market and performance shares.

Restricted Stock Awards and Restricted Stock Unit Awards

We grant equity based restricted stock and restricted stock units that contain service only conditions for vesting ("Service Awards"), awards that contain both service and market conditions for vesting ("Market Awards") and awards that contain both service and performance conditions for vesting ("Performance Awards"). The fair value of Service Awards is based upon the closing stock price on the accounting grant date and the awards generally vest over periods ranging from one to three years. The fair value of Market Awards is determined using Monte-Carlo simulations and the awards vest over periods up to three years. The number of shares ultimately earned for Market Awards is typically based upon our total shareholder return as compared to the return of selected peer companies and can range between 0% and 200% of the target number of shares for the award. The fair value of Performance Awards is based upon the closing stock price on the accounting grant date; however, the award value may increase, or decrease based upon the outcome of the performance conditions. Performance Awards vest over periods of up to three-years and specify a target number of shares for the award. The recipient ultimately can receive between 0% and 200% of the target number of shares depending upon the outcome of the performance conditions.

The following table summarizes activity involving restricted stock and restricted stock unit awards for the year ended December 31, 2023:

	Number of Shares	Weighted-Average Grant Date Fair Value
	(in thousands)	
Non-vested at December 31, 2022	27,279	$ 12.13
Granted	14,787	1.85
Vested	(7,170)	10.10
Forfeited	(6,844)	13.79
Non-vested at December 31, 2023	28,052	6.82

During 2023, we granted 14.8 million shares of restricted stock and restricted stock unit awards at a weighted-average price of $ 1.85. During 2022, we granted 18.8 million shares of restricted stock and restricted stock unit awards at a weighted-average price of $ 11.47. During 2021, we granted 13.9 million shares of restricted stock and restricted stock unit awards at a weighted-average price of $13.95. The total fair value of restricted stock and restricted stock unit awards that vested during 2023, 2022 and 2021, was $21 million, $98 million and $139 million, respectively. We do not estimate forfeitures but recognize them as they occur.

Compensation Expense and Tax Benefit

For Service Awards that vest ratably over the service period, we recognize compensation expense on a straight-line basis over the requisite service period for the entire award. For Service Awards that vest at the end of the service period and for Market Awards, we recognize compensation expense over the service period. For our Performance Awards, we recognize compensation expense over the service period and based upon the expected performance outcome, until the final performance outcome is determined. Total compensation expense for all stock-based payment arrangements for the years ended December 31, 2023, 2022 and 2021, was $52 million, $98 million and $120 million, respectively. Our tax benefit recognized in the consolidated statements of operations for our stock-based payment arrangements for the years ended December 31, 2023, 2022 and 2021, was $12 million, $25 million and $29 million, respectively. At December 31, 2023, there was $65 million of total unrecognized compensation expense related to our stock-based payment arrangements, which we expect to recognize over a weighted-average period of 1.5 years.

(13) (Loss) Earnings Per Common Share

Basic and diluted (loss) earnings per common share for the years ended December 31, 2023, 2022 and 2021 were calculated as follows:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions, except per share amounts, shares in thousands)		
(Loss) income (numerator)			
Net (loss) income	$ (10,298)	(1,548)	2,033
Net (loss) income applicable to common stock for computing basic (loss) earnings per common share	(10,298)	(1,548)	2,033
Net (loss) income as adjusted for purposes of computing diluted (loss) earnings per common share	$ (10,298)	(1,548)	2,033
Shares (denominator):			
Weighted average number of shares:			
Outstanding during period	1,006,787	1,028,069	1,077,393
Non-vested restricted stock	(23,706)	(20,552)	(17,852)
Weighted average shares outstanding for computing basic (loss) earnings per common share	983,081	1,007,517	1,059,541
Incremental common shares attributable to dilutive securities:			
Shares issuable under convertible securities	—	—	10
Shares issuable under incentive compensation plans	—	—	7,227
Number of shares as adjusted for purposes of computing diluted (loss) earnings per common share	983,081	1,007,517	1,066,778
Basic (loss) earnings per common share	$ (10.48)	(1.54)	1.92
Diluted (loss) earnings per common share [1]	$ (10.48)	(1.54)	1.91

[1] For the years ended December 31, 2023 and December 31, 2022, we excluded from the calculation of diluted loss per share0.3 million and 3.8 million shares, respectively, potentially issuable under incentive compensation plans or convertible securities, as their effect, if included, would have been anti-dilutive.

Our calculation of diluted (loss) earnings per common share excludes shares of common stock that are issuable upon exercise of stock options when the exercise price is greater than the average market price of our common stock. We also exclude unvested restricted stock awards that are antidilutive as a result of unrecognized compensation cost. Such shares were 22.5 million, 13.8 million and 3.2 million for 2023, 2022 and 2021, respectively.

(14) Fair Value of Financial Instruments

Our financial instruments consist of cash, cash equivalents, restricted cash, accounts receivable, accounts payable, long-term debt (excluding finance lease and other obligations), interest rate swap contracts, certain equity investments and certain indemnification obligations. Due primarily to their short-term nature, the carrying amounts of our cash, cash equivalents, restricted cash, accounts receivable and accounts payable approximate their fair values.

Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between independent and knowledgeable parties who are willing and able to transact for an asset or liability at the measurement date. We use valuation techniques that maximize the use of observable inputs and minimize the use of unobservable inputs when determining fair value and then we rank the estimated values based on the reliability of the inputs using the below-described fair value hierarchy.

We determined the fair values of our long-term debt, including the current portion, based on quoted market prices where available or, if not available, based on inputs other than quoted market prices in active markets that are either directly or indirectly observable such as discounted future cash flows using current market interest rates.

The three input levels in the hierarchy of fair value measurements are defined by the FASB generally as follows:

Input Level	Description of Input
Level 1	Observable inputs such as quoted market prices in active markets.
Level 2	Inputs other than quoted prices in active markets that are either directly or indirectly observable.
Level 3	Unobservable inputs in which little or no market data exists.

The following table presents the carrying amounts and estimated fair values of our following financial assets and liabilities as of December 31, 2023 and 2022:

		As of December 31, 2023		As of December 31, 2022	
	Input Level	Carrying Amount	Fair Value	Carrying Amount	Fair Value
		(Dollars in millions)			
Equity securities[1]	1	$ —	—	22	22
Long-term debt, excluding finance lease and other obligations	2	19,703	13,304	20,255	17,309
Indemnifications related to the sale of the Latin American business[2]	3	86	86	86	86

(1) For the years ended December 31, 2023 and 2022, we recognized a $22 million and a $109 million of loss on equity securities in other (expense) income, net in our consolidated statements of operations.

(2) Nonrecurring fair value is measured as of August 1, 2022.

Investment Held at Net Asset Value

We hold an investment in a limited partnership created as a holding company for various investments. The limited partnership has sole discretion as to the amount and timing of distributions of the underlying assets. As of December 31, 2023, the underlying investments held by the limited partnership were traded in active markets and as such, we account for our investment in the limited partnership using net asset value ("NAV"). Subject to restrictions imposed by law and other provisions of the limited partnership agreement, the general partner has the sole discretion as to the amounts and timing of distributions of partnership assets to partners. The following table summarizes the net asset value of our investment in this limited partnership.

	As of December 31, 2023	As of December 31, 2022
	Net Asset Value	
	(Dollars in millions)	
Investment in limited partnership [1]	$ 10	85

[1] For the years ended December 31, 2023 and December 31, 2022, we recognized $75 million and $83 million of loss on investment, respectively, reflected in other (expense) income, net in our consolidated statement of operations.

(15) Derivative Financial Instruments

From time to time, we use derivative financial instruments, primarily interest rate swaps, to manage our exposure to fluctuations in interest rates. Our primary objective in managing interest rate risk is to decrease the volatility of our earnings and cash flows affected by changes in the underlying rates. We have floating rate long-term debt (see Note 7—Long-Term Debt and Credit Facilities). These obligations expose us to variability in interest payments due to changes in interest rates. If interest rates increase, our interest expense increases. Conversely, if interest rates decrease, our interest expense also decreases. Through their expiration on June 30, 2022, we designated the interest rate swap agreements described below as cash flow hedges. Under these hedges, we received variable-rate amounts from a counterparty in exchange for us making fixed-rate payments over the lives of the agreements without exchange of the underlying notional amount. The change in the fair value of the interest rate swap agreements was reflected in accumulated other comprehensive loss and was subsequently reclassified into earnings in the period that the hedged transaction affected earnings by virtue of qualifying as effective cash flow hedges. We do not use derivative financial instruments for speculative purposes.

In 2019, we entered into variable-to-fixed interest rate swap agreements to hedge the interest on $ 4.0 billion notional amount of floating rate debt. As of December 31, 2021, we evaluated the effectiveness of our remaining hedges quantitatively and determined that hedges in effect on such dates qualified as effective hedge relationships. All remaining hedges were expired as of December 31, 2022.

Amounts accumulated in accumulated other comprehensive loss related to derivatives were indirectly recognized in earnings as periodic settlement payments were made throughout the term of the swaps.

The amount of unrealized losses recognized in accumulated other comprehensive loss consists of the following (in millions):

Derivatives designated as hedging instruments	
Cash flow hedging contracts	
Year Ended December 31, 2021	$ 1

The amount of realized losses reclassified from accumulated other comprehensive loss to the statement of operations consists of the following (in millions):

Derivatives designated as hedging instruments	2022	2021
Cash flow hedging contracts		
Years Ended December 31,	$ 22	83

For the year ended December 31, 2022, amounts included in accumulated other comprehensive loss at the beginning of the period were reclassified into earnings upon the settlement of the cash flow hedging contracts on March 31, 2022 and June 30, 2022. During the year ended December 31, 2022, $19 million of net losses on the interest rate swaps have been reflected in our consolidated statements of operations upon settlement of the agreements in the first half of 2022.

(16) Income Taxes

The components of the income tax expense are as follows:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Income tax expense:			
Federal			
Current	$ 7	838	5
Deferred	(2)	(332)	514
State			
Current	(6)	283	42
Deferred	55	(191)	72
Foreign			
Current	—	32	23
Deferred	7	(73)	12
Total income tax expense	$ 61	557	668

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Income tax expense was allocated as follows:			
Income tax expense in the consolidated statements of operations:			
Attributable to income	$ 61	557	668
Stockholders' equity:			
Tax effect of the change in accumulated other comprehensive loss	$ (21)	297	222

The following is a reconciliation from the statutory federal income tax rate to our effective income tax rate:

	Years Ended December 31,		
	2023	2022	2021
	(Percentage of pre-tax (loss) income)		
Statutory federal income tax rate	21.0 %	21.0 %	21.0 %
State income taxes, net of federal income tax benefit	(0.2)%	(8.8)%	3.3 %
Goodwill impairment	(21.9)%	(68.9)%	— %
Change in liability for unrecognized tax position	(0.1)%	(0.2)%	0.1 %
Nondeductible executive stock compensation	— %	(0.1)%	0.2 %
Change in valuation allowance	1.3 %	0.9 %	— %
Net foreign income taxes	— %	3.0 %	0.6 %
Research and development credits	0.1 %	1.1 %	(0.5)%
Divestitures of businesses[1]	(0.4)%	(4.0)%	— %
Other, net	(0.4)%	(0.2)%	— %
Effective income tax rate	(0.6)%	(56.2)%	24.7 %

(1) Includes GILTI (as defined below) incurred as a result of the sale of our Latin American business.

The effective tax rate for the year ended December 31, 2023 includes a $ 2.2 billion unfavorable impact of a non-deductible goodwill impairment and a $137 million favorable impact as a result of utilizing available capital losses generated by the sale of our Latin American business in 2022. The effective tax rate for the year ended December 31, 2022 includes a $682 million unfavorable impact of non-deductible goodwill impairments and $128 million unfavorable impact related to incurring tax on Global Intangible Low-Tax Income ("GILTI") as a result of the sale of our Latin American business.

The tax effects of temporary differences that gave rise to significant portions of the deferred tax assets and deferred tax liabilities were as follows:

	As of December 31,	
	2023	2022[1]
	(Dollars in millions)	
Deferred tax assets		
Post-retirement and pension benefit costs	$ 659	725
Net operating loss carryforwards	794	871
Other employee benefits	23	85
Other	511	519
Gross deferred tax assets	1,987	2,200
Less valuation allowance	(399)	(550)
Net deferred tax assets	1,588	1,650
Deferred tax liabilities		
Property, plant and equipment, primarily due to depreciation differences	(3,332)	(3,046)
Goodwill and other intangible assets	(1,271)	(1,634)
Gross deferred tax liabilities	(4,603)	(4,680)
Net deferred tax liability	$ (3,015)	(3,030)

(1) Excludes $138 million of deferred tax assets and $38 million of deferred tax liabilities related to the EMEA business sold November 1, 2023, that were classified as held for sale as of December 31, 2022.

Of the $3.0 billion net deferred tax liability at December 31, 2023 and 2022, respectively, $ 3.1 billion and $3.2 billion is reflected as a long-term liability and $112 million and $133 million is reflected as a net noncurrent deferred tax asset, in other, net on our consolidated balance sheets at December 31, 2023 and 2022, respectively.

Income taxes receivable as of December 31, 2023 was $ 273 million and income taxes payable as of December 31, 2022 was $ 943 million.

At December 31, 2023, we had federal NOLs of approximately $ 800 million, net of expirations from Section 382 limitations and uncertain tax positions, for U.S. federal income tax purposes. We expect to use substantially all of these tax attributes to reduce our future federal tax liabilities, although the timing of that use will depend upon our future earnings and future tax circumstances. Our ability to use these NOLs is subject to annual limits imposed by Section 382. As a result, we anticipate that our cash income tax liabilities will increase in future periods. If unused, the NOLs will expire between 2026 and 2029.

At December 31, 2023 we had state net operating loss carryforwards of $ 13 billion (net of uncertain tax positions). Our ability to use these NOLs is subject to annual limits imposed by Section 382.

We establish valuation allowances when necessary to reduce the deferred tax assets to amounts we expect to realize. As of December 31, 2023, we established a valuation allowance of $399 million as it is more likely than not that this amount of net operating loss will not be utilized prior to expiration. Our valuation allowance at December 31, 2023 and 2022 is primarily related to NOL carryforwards. This valuation allowance decreased by $151 million during 2023, primarily due to the impact of utilization of available capital losses.

A reconciliation of the change in our gross unrecognized tax benefits (excluding both interest and any related federal benefit) from January 1 to December 31 for 2023 and 2022 is as follows:

	2023	2022
	(Dollars in millions)	
Unrecognized tax benefits at beginning of year	$ 1,318	1,375
Decrease in tax positions of prior periods netted against deferred tax assets	(411)	(661)
(Decrease) increase in tax positions taken in the current year	(73)	634
Increase (decrease) in tax positions taken in the prior year	752	(3)
Decrease due to payments/settlements	(1)	—
Decrease from the lapse of statute of limitations	(52)	—
Decrease related to divestitures of businesses	$ (109)	(27)
Unrecognized tax benefits at end of year	$ 1,424	1,318

As of December 31, 2023 the total amount of unrecognized tax benefits that, if recognized, would impact the effective income tax rate was $280 million. The unrecognized tax benefits also includes tax positions that, if recognized, would result in adjustments to other tax accounts, primarily deferred taxes, that would not impact the effective tax rate but could impact cash tax amounts payable to taxing authorities.

Our policy is to reflect interest expense associated with unrecognized tax benefits in income tax expense. We had accrued interest (presented before related tax benefits) of approximately $100 million and $26 million at December 31, 2023 and 2022, respectively.

We, or at least one of our subsidiaries, file income tax returns in the U.S. federal jurisdiction and various states and foreign jurisdictions. With few exceptions, we are no longer subject to U.S. federal, state and local, or non-U.S. income tax examinations by tax authorities for years before 2004. The Internal Revenue Service and state and local taxing authorities reserve the right to audit any period where net operating loss carryforwards are available.

Based on our current assessment of various factors, including (i) the potential outcomes of these ongoing examinations, (ii) the expiration of statute of limitations for specific jurisdictions, (iii) the negotiated settlement of certain disputed issues, and (iv) the administrative practices of applicable taxing jurisdictions, it is reasonably possible that the related unrecognized tax benefits for uncertain tax positions previously taken may decrease by up to $676 million within the next 12 months. The actual amount of such decrease, if any, will depend on several future developments and events, many of which are outside our control.

In August 2022, the Inflation Reduction Act was signed into law and which, among other things, implemented a corporate alternative minimum tax (“CAMT”) on adjusted financial statement income effective for tax periods occurring after December 31, 2022. The CAMT had no material impact on our financial results as of December 31, 2023. In addition, the Organization for Economic Co-operation and Development has issued Pillar Two model rules introducing a new global minimum tax of 15% intended to be effective on January 1, 2024. While the US has not yet adopted the Pillar Two rules, various other governments around the world are enacting legislation, some of which are effective for tax periods after December 31, 2023. While the global minimum tax will increase our administrative and compliance burdens, it is expected to have an immaterial impact to our financial statements.

(17) Segment Information

We report our results within two segments: Business and Mass Markets.

Under our Business segment we provide products and services to meet the needs of our enterprise and wholesale customers under four distinct sales channels: Large Enterprise, Mid-Market Enterprise, Public Sector and Wholesale. For Business segment revenue, we report the following product categories: Grow, Nurture, Harvest and Other, in each case through the sales channels outlined above. The Business segment included the results of our Latin American, ILEC and EMEA businesses prior to their sales on August 1, 2022, October 3, 2022 and November 1, 2023, respectively.

Under our Mass Markets Segment, we provide products and services to residential and small business customers. We report the following product categories: Fiber Broadband, Other Broadband and Voice and Other. The Mass Markets segment included the results of our ILEC business prior to its sale on October 3, 2022.

See detailed descriptions of these product and service categories in Note 4—Revenue Recognition.

As described in more detail below, our segments are managed based on the direct costs of providing services to their customers and directly associated selling, general and administrative costs (primarily salaries and commissions). Shared costs are managed separately and included in "other unallocated expense" in the table included below "—Revenue and Expenses". As referenced above, we reclassified certain prior period amounts to conform to the current period presentation. See Note 1—Background and Summary of Significant Accounting Policies for additional detail on these changes.

The following tables summarize our segment results for 2023, 2022 and 2021 based on the segment categorization we were operating under at December 31, 2023.

	Year Ended December 31, 2023	
	Business	Mass Markets
	(Dollars in millions)	
Segment revenue	$ 11,535	3,022
Segment expense		
Cost of services and products	3,138	92
Selling, general and administrative	1,232	1,341
Total expense	4,370	1,433
Total segment adjusted EBITDA	$ 7,165	1,589

	Year Ended December 31, 2022	
	Business	Mass Markets
	(Dollars in millions)	
Segment revenue	$ 13,041	4,437
Segment expense		
Cost of services and products	3,257	124
Selling, general and administrative	1,215	1,623
Total expense	4,472	1,747
Total segment adjusted EBITDA	$ 8,569	2,690

	Year Ended December 31, 2021	
	Business	Mass Markets
	(Dollars in millions)	
Segment revenue	$ 14,119	5,568
Segment expense		
Cost of services and products	3,488	153
Selling, general and administrative	1,273	1,685
Total expense	4,761	1,838
Total segment adjusted EBITDA	$ 9,358	3,730

Revenue and Expenses

Our segment revenue includes all revenue from our two segments as described in more detail above. Our segment revenue is based upon each customer's classification. We report our segment revenue based upon all services provided to that segment's customers. Our segment expenses include specific cost of service expenses incurred as a direct result of providing services and products to segment customers, along with selling, general and administrative expenses that are directly associated with specific segment customers or activities. We have not allocated assets or debt to specific segments.

The following items are excluded from our segment results, because they are centrally managed and not monitored by or reported to our chief operating decision maker by segment:

- network expenses not incurred as a direct result of providing services and products to segment customers and centrally managed expenses such as Finance, Human Resources, Legal, Marketing, Product Management and IT, all of which are reported as "other unallocated expense" in the table below;
- depreciation and amortization expense;
- goodwill or other impairments;
- interest expense;
- stock-based compensation; and
- other income and expense items.

The following table reconciles total segment adjusted EBITDA to net (loss) income for the years ended December 31, 2023, 2022 and 2021:

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in millions)		
Total segment adjusted EBITDA	$ 8,754	11,259	13,088
Depreciation and amortization	(2,985)	(3,239)	(4,019)
Goodwill impairment	(10,693)	(3,271)	—
Other unallocated expense	(4,608)	(4,556)	(4,664)
Stock-based compensation	(52)	(98)	(120)
Operating (loss) income	(9,584)	95	4,285
Total other expense, net	(653)	(1,086)	(1,584)
(Loss) income before income taxes	(10,237)	(991)	2,701
Income tax expense	61	557	668
Net (loss) income	$ (10,298)	(1,548)	2,033

We do not have any single customer that comprises more than 10% of our consolidated total operating revenue.

The assets we hold outside of the U.S. represent less than 10% of our total assets. Revenue from sources outside of the U.S. comprises less than 10% of our total operating revenue.

(18) Commitments, Contingencies and Other Items

We are subject to various claims, legal proceedings and other contingent liabilities, including the matters described below, which individually or in the aggregate could materially affect our financial condition, future results of operations or cash flows. As a matter of course, we are prepared to both litigate these matters to judgment as needed, as well as to evaluate and consider reasonable settlement opportunities.

We review our litigation accrual liabilities on a quarterly basis, but in accordance with applicable accounting guidelines only establish accrual liabilities when losses are deemed probable and reasonably estimable and only revise previously-established accrual liabilities when warranted by changes in circumstances, in each case based on then-available information. As such, as of any given date we could have exposure to losses under proceedings as to which no liability has been accrued or as to which the accrued liability is inadequate. Subject to these limitations, at December 31, 2023 and December 31, 2022, we had accrued $84 million and $88 million, respectively, in the aggregate for our litigation and non-income tax contingencies, which is included in other current liabilities or other liabilities in our consolidated balance sheet as of such date. We cannot at this time estimate the reasonably possible loss or range of loss in excess of this $84 million accrual due to the inherent uncertainties and speculative nature of contested proceedings. The establishment of an accrual does not mean that actual funds have been set aside to satisfy a given contingency. Thus, the resolution of a particular contingency for the amount accrued could have no effect on our results of operations but nonetheless could have an adverse effect on our cash flows.

In this Note, when we refer to a class action as "putative" it is because a class has been alleged, but not certified, in that matter.

Principal Proceedings

Shareholder Class Action Suits

Lumen and certain Lumen Board of Directors members and officers were named as defendants in a putative shareholder class action lawsuit filed on June 12, 2018 in the Boulder County District Court of the state of Colorado, captioned Houser et al. v. CenturyLink, et al. The complaint asserted claims on behalf of a putative class of former Level 3 shareholders who became CenturyLink, Inc. shareholders as a result of our acquisition of Level 3. It alleged that the proxy statement provided to the Level 3 shareholders failed to disclose various material information of several kinds, including information about strategic revenue, customer loss rates, and customer account issues, among other items. The complaint seeks damages, costs and fees, rescission, rescissory damages, and other equitable relief. In May 2020, the court dismissed the complaint. Plaintiffs appealed that decision, and in March 2022, the appellate court affirmed the district court's order in part and reversed it in part. It then remanded the case to the district court for further proceedings. Plaintiff filed an amended complaint, and we filed a motion to dismiss. The court granted our motion to dismiss and the plaintiffs have appealed that dismissal.

On March 3, 2023, a purported shareholder of Lumen filed a putative class action complaint captioned Voigt v. Lumen Technologies, Inc., et al., Case 3:23-cv-00286-TAD-KDM, in the U.S. District Court for the Western District of Louisiana. The complaint alleges that Lumen and certain of its current or former officers violated the federal securities laws by omitting or misstating material information related to Lumen’s expansion of its Quantum Fiber business. The complaint seeks money damages, attorneys’ fees and costs, and other relief.

On September 15, 2023, a purported shareholder of Lumen filed a putative class action complaint captioned McLemore v. Lumen Technologies, Inc., et al., Case 3:23-cv-01290, in the U.S. District Court for the Western District of Louisiana. The complaint alleges that Lumen and certain of its current or former officers violated the federal securities laws by omitting or misstating material information related to Lumen’s responsibility for environmental degradation allegedly caused by the lead sheathing of certain telecommunications cables. The complaint seeks money damages, attorneys’ fees and costs, and other relief.

State Tax Suits

Since 2012, a number of Missouri municipalities have asserted claims in the Circuit Court of St. Louis County, Missouri, alleging that we and several of our subsidiaries have underpaid taxes. These municipalities are seeking, among other things, declaratory relief regarding the application of business license and gross receipts taxes and back taxes from 2007 to the present, plus penalties and interest. In a February 2017 ruling in connection with one of these pending cases, the court entered an order awarding the plaintiffs $4 million and broadening the tax base on a going-forward basis. We appealed that decision to the Missouri Supreme Court. In December 2019, it affirmed the circuit court's order in some respects and reversed it in others, remanding the case to the circuit court for further proceedings. The Missouri Supreme Court's decision reduced our exposure in the case. In a June 2021 ruling in one of the pending cases, another trial court awarded the cities of Columbia and Joplin approximately $ 55 million, plus statutory interest. On appeal, the Missouri Court of Appeals affirmed in part and reversed in part, vacated the judgment and remanded the case to the trial court with instructions for further proceedings consistent with the Missouri Supreme Court's decision. We continue to vigorously defend against these claims.

Billing Practices Suits

In June 2017, a former employee filed an employment lawsuit against us claiming that she was wrongfully terminated for alleging that we charged some of our retail customers for products and services they did not authorize. Thereafter, based in part on the allegations made by the former employee, several legal proceedings were filed, including consumer class actions in federal and state courts, a series of securities investor class actions in federal courts and several shareholder derivative actions in federal and Louisiana state courts. The derivative cases were brought on behalf of CenturyLink, Inc. against certain current and former officers and directors of the Company and seek damages for alleged breaches of fiduciary duties.

The consumer class actions, the securities investor class actions, and the federal derivative actions were transferred to the U.S. District Court for the District of Minnesota for coordinated and consolidated pretrial proceedings as In Re: CenturyLink Sales Practices and Securities Litigation. We have settled the consumer and securities investor class actions and the derivative actions.

We have engaged in discussions regarding related claims with a number of state attorneys general, and have entered into agreements settling certain of the consumer practices claims asserted by state attorneys general. While we do not agree with allegations raised in these matters, we have been willing to consider reasonable settlements where appropriate.

December 2018 Outage Proceedings

We experienced an outage on one of our transport networks that impacted voice, IP, 911, and transport services for some of our customers between the 27th and 29th of December 2018. We believe that the outage was caused by a faulty network management card from a third-party equipment vendor.

The FCC and four states (both Washington Utilities and Transportation Commission ("WUTC") and the Washington Attorney General; the Montana Public Service Commission; the Nebraska Public Service Commission; and the Wyoming Public Service Commission) initiated formal investigations. In November 2020, following the FCC's release of a public report on the outage, we negotiated a settlement which was released by the FCC in December 2020. The amount of the settlement was not material to our financial statements.

In December 2020, the Staff of the WUTC filed a complaint against us based on the December 2018 outage, seeking penalties of approximately $7 million for alleged violations of Washington regulations and laws. The Washington Attorney General's office sought penalties of $ 27 million. Following trial before the WUTC, it issued an order in June 2023 penalizing us for approximately $1 million. We and the Washington Attorney General's office have both filed for reconsideration. Those motions are pending.

Latin American Tax Litigation and Claims

In connection with the 2022 divestiture of our Latin American business, the purchaser assumed responsibility for the Peruvian tax litigation and Brazilian tax claims described in our prior periodic reports filed with the SEC. We agreed to indemnify the purchaser for amounts paid in respect of the Brazilian tax claims. The value of this indemnification is included in the indemnification amount as disclosed in Note 14—Fair Value of Financial Instruments.

Huawei Network Deployment Investigations

Lumen has received requests from the following federal agencies for information relating to the use of equipment manufactured by Huawei Technologies Company ("Huawei") in Lumen’s networks.

- DOJ. Lumen has received a civil investigative demand from the U.S. Department of Justice in the course of a False Claims Act investigation alleging that Lumen Technologies, Inc. and Lumen Technologies Government Solutions, Inc. failed to comply with the requirements in federal contracts concerning their use of Huawei equipment.

- FCC. The FCC’s Enforcement Bureau issued a Letter of Inquiry to Lumen Technologies, Inc. regarding its written certifications to the FCC that Lumen has complied with FCC rules governing the use of resources derived from the High Cost Program, Lifeline Program, Rural Health Care Program, E-Rate Program, Emergency Broadband Benefit Program, and the Affordable Connectivity Program. Under these programs, federal funds may not be used to facilitate the deployment or maintenance of equipment or services provided by Huawei, a company that the FCC has determined poses a national security threat to the integrity of communications networks or the communications supply chain.

- Team Telecom. The Committee for the Assessment of Foreign Participation in the United States Telecommunications Service Sector (comprised of the U.S. Attorney General, and the Secretaries of the Department of Homeland Security, and the Department of Defense), commonly referred to as Team Telecom, issued questions and requests for information relating to Lumen’s FCC licenses and its use of Huawei equipment.

We are cooperating with the investigations.

Marshall Fire Litigation

On December 30, 2021, a wildfire referred to as the Marshall Fire ignited near Boulder, Colorado. The Marshall Fire killed two people, and it burned thousands of acres, including entire neighborhoods. Approximately 300 lawsuits naming various defendants and asserting various claims for relief have been filed. To date, three of those name our affiliate Qwest Corporation as being at fault: Allstate Fire and Casualty Insurance Company, et al., v. Qwest Corp., et al., Case No. 2023-cv-3048, and Wallace, et al. v, Qwest Corp., et al, Case No. 2023-cv-30488, both of which have been consolidated with Kupfner et al v Public Service Company of Colorado, et al. Case No. 2022-cv-30195. The consolidated proceeding is pending in Colorado District Court, Boulder, Colorado, Preliminary estimates of potential damage claims exceed $2 billion. Qwest is vigorously defending the claims.

911 Surcharge

In June 2021, the Company was served with a complaint filed in the Santa Fe County District Court by Phone Recovery Services, LLC ("PRS"), acting on behalf of the State of New Mexico. The complaint claims Qwest Corporation and CenturyTel of the Southwest have violated the New Mexico Fraud Against Taxpayers Act since 2004 by failing to bill, collect and remit certain 911 surcharges from customers. Through pre-trial proceedings, the Court has narrowed the issues to be resolved by jury, ruling that Lumen bears the burden of proving that its actions were reasonable or known and approved by the State. Qwest is defending the New Mexico claims vigorously, as it has done successfully with other 911 claims involving PRS in other states.

Other Proceedings, Disputes and Contingencies

From time to time, we are involved in other proceedings incidental to our business, including patent infringement allegations, regulatory hearings relating primarily to our rates or services, actions relating to employee claims, various tax issues, environmental law issues, grievance hearings before labor regulatory agencies and miscellaneous third-party tort actions or commercial disputes.

We are currently defending several patent infringement lawsuits asserted against us by non-practicing entities, many of which are seeking substantial recoveries. These cases have progressed to various stages and one or more may go to trial within the next twelve months if they are not otherwise resolved. Where applicable, we are seeking full or partial indemnification from our vendors and suppliers. As with all litigation, we are vigorously defending these actions and, as a matter of course, are prepared to litigate these matters to judgment, as well as to evaluate and consider all reasonable settlement opportunities.

We are subject to various foreign, federal, state and local environmental protection and health and safety laws. From time to time, we are subject to judicial and administrative proceedings brought by various governmental authorities under these laws. Several such proceedings are currently pending, but none is reasonably expected to exceed $300,000 in fines and penalties. In addition, in the past we acquired companies that had installed lead-sheathed cables several decades earlier, or had operated certain manufacturing companies in the first part of the 1900s. Under applicable environmental laws, we could be named as a potentially responsible party for a share of the remediation of environmental conditions arising from the historical operations of our predecessors.

The outcome of these other proceedings described under this heading is not predictable. However, based on current circumstances, we do not believe that the ultimate resolution of these other proceedings, after considering available defenses and any insurance coverage or indemnification rights, will have a material adverse effect on us.

The matters listed in this Note do not reflect all of our contingencies. The ultimate outcome of the above-described matters may differ materially from the outcomes anticipated, estimated, projected or implied by us in certain of our statements appearing above in this Note, and proceedings currently viewed as immaterial by us may ultimately materially impact us.

Right-of-Way

At December 31, 2023, our future rental commitments and Right-of-Way ("ROW") agreements were as follows:

	Future Rental Commitments and ROW Agreements
	(Dollars in millions)
2024	$ 184
2025	64
2026	60
2027	59
2028	51
2029 and thereafter	676
Total future minimum payments	$ 1,094

Purchase Commitments

We have several commitments primarily for marketing activities and support services from a variety of vendors to be used in the ordinary course of business totaling $1.0 billion at December 31, 2023. Of this amount, we expect to purchase $ 403 million in 2024, $378 million in 2025 through 2026, $78 million in 2027 through 2028 and $127 million in 2029 and thereafter. These amounts do not represent our entire anticipated purchases in the future, but represent only those items for which we were contractually committed as of December 31, 2023.

(19) Other Financial Information

Other Current Assets

The following table presents details of other current assets reflected in our consolidated balance sheets:

	As of December 31,	
	2023	2022[1]
	(Dollars in millions)	
Prepaid expenses	$ 395	319
Income tax receivable	273	—
Materials, supplies and inventory	209	236
Contract assets	19	20
Contract acquisition costs	107	123
Contract fulfillment costs	102	100
Other	14	5
Total other current assets	$ 1,119	803

(1) Excludes $59 million of other current assets related to the EMEA business sold on November 1, 2023 that were classified as held for sale as of December 31, 2022.

Included in accounts payable at December 31, 2023 and 2022 were $ 274 million and $265 million, respectively, associated with capital expenditures.

(20) Repurchases of Lumen Common Stock

During the fourth quarter of 2022, our Board of Directors authorized a two-year program to repurchase up to an aggregate of $1.5 billion of our outstanding common stock. During the year ended December 31, 2023, we did not repurchase any shares of our outstanding common stock under this program. During the year ended December 31, 2022, we repurchased under this program 33 million shares of our outstanding common stock in the open market for an aggregate market price of $200 million, or an average purchase price of $6.07 per share. All repurchased common stock has been retired. As a result, common stock and additional paid-in capital were reduced as of December 31, 2022 by $33 million and $167 million, respectively.

On August 3, 2021, our Board of Directors authorized a 24-month program to repurchase up to an aggregate of $1.0 billion of our outstanding common stock. During the year ended December 31, 2021, we repurchased under this program 80.9 million shares of our outstanding common stock in the open market for an aggregate market price of $1.0 billion, or an average purchase price of $12.36 per share, thereby fully exhausting the program. All repurchased common stock has been retired. As a result, common stock and additional paid-in capital were reduced as of December 31, 2021 by $81 million and $919 million, respectively.

Any repurchases made in 2024 or thereafter will be subject to a non-deductible 1% excise tax on the fair market value of the stock under the Inflation Reduction Act of 2022.

(21) Accumulated Other Comprehensive Loss

Information Relating to 2023

The table below summarizes changes in accumulated other comprehensive loss recorded on our consolidated balance sheet by component for the year ended December 31, 2023:

	Pension Plans	Post-Retirement Benefit Plans	Foreign Currency Translation Adjustment and Other	Total
	(Dollars in millions)			
Balance at December 31, 2022	$ (985)	308	(422)	(1,099)
Other comprehensive loss before reclassifications	(110)	(11)	(1)	(122)
Amounts reclassified from accumulated other comprehensive loss	50	(21)	382	411
Net current-period other comprehensive (loss) income	(60)	(32)	381	289
Balance at December 31, 2023	$ (1,045)	276	(41)	(810)

The table below presents further information about our reclassifications out of accumulated other comprehensive loss by component for the year ended December 31, 2023:

Year Ended December 31, 2023	(Decrease) Increase in Net Loss	Affected Line Item in Consolidated Statement of Operations
	(Dollars in millions)	
Amortization of pension & post-retirement plans [1]		
Net actuarial loss	$ 82	Other (expense) income, net
Prior service cost	(15)	Other (expense) income, net
Total before tax	67	
Income tax benefit	(16)	Income tax expense
Net of tax	$ 51	

Year Ended December 31, 2023	Reclassification out of Accumulated Other Comprehensive Loss	Affected line item in Consolidated Balance Sheets and Consolidated Statement of Operations
Reclassification of realized loss on foreign currency translation to valuation allowance within assets held for sale[2]	$ 389	Assets held for sale
Reclassification of realized loss on foreign currency translation to loss on sale of business[3]	(7)	Net loss (gain) on sale of businesses
Subtotal reclassification of realized loss on foreign currency	382	
Reclassification of net actuarial loss to valuation allowance within assets held for sale[2]	(24)	Assets held for sale
Reclassification of net actuarial gain to loss on sale of business [3]	2	Net loss (gain) on sale of businesses
Subtotal reclassification of net actuarial loss	(22)	
Income tax benefit	—	Income tax expense
Net of tax	$ 360	

(1) See Note 11—Employee Benefits for additional information on our net periodic benefit (expense) income related to our pension and post-retirement plans.

(2) Recognized in net income through net loss (gain) on sale of business for the year ended December 31, 2022 and included in our valuation allowance in assets held for sale as of December 31, 2022.

(3) (Decrease) increase to net loss for the year ended December 31, 2023.

Information Relating to 2022

The table below summarizes changes in accumulated other comprehensive loss recorded on our consolidated balance sheet by component for the year ended December 31, 2022:

	Pension Plans	Post-Retirement Benefit Plans	Foreign Currency Translation Adjustment and Other	Interest Rate Swap	Total
			(Dollars in millions)		
Balance at December 31, 2021	$ (1,577)	(164)	(400)	(17)	(2,158)
Other comprehensive income (loss) before reclassifications	98	473	(134)	—	437
Amounts reclassified from accumulated other comprehensive loss	494	(1)	112	17	622
Net current-period other comprehensive income (loss)	592	472	(22)	17	1,059
Balance at December 31, 2022	$ (985)	308	(422)	—	(1,099)

The table below presents further information about our reclassifications out of accumulated other comprehensive loss by component for the year ended December 31, 2022:

Year Ended December 31, 2022	(Decrease) Increase in Net Loss	Affected Line Item in Consolidated Statement of Operations
	(Dollars in millions)	
Interest rate swap	$ 22	Interest expense
Income tax benefit	(5)	Income tax expense
Net of tax	$ 17	
Amortization of pension & post-retirement plans [1]		
Net actuarial loss	$ 121	Other (expense) income, net
Settlement charge	(2)	Other (expense) income, net
Reclassification of net actuarial loss and prior service credit to gain on the sale of business	539	Net loss (gain) on sale of businesses
Total before tax	658	
Income tax benefit	(165)	Income tax expense
Net of tax	$ 493	
Reclassification of realized loss on foreign currency translation to loss (gain) on sale of businesses	$ 112	Net loss (gain) on sale of businesses
Income tax benefit	—	Income tax expense
Net of tax	$ 112	

[1] See Note 11—Employee Benefits for additional information on our net periodic benefit (expense) income related to our pension and post-retirement plans.

(22) Labor Union Contracts

As of December 31, 2023, approximately 21% of our employees were represented by the Communications Workers of America ("CWA") or the International Brotherhood of Electrical Workers ("IBEW"). None of our collective bargaining agreements were in expired status as of December 31, 2023. Approximately 2% of our represented employees are subject to collective bargaining agreements that are scheduled to expire over the 12 month period ending December 31, 2024.

(23) Dividends

On November 2, 2022, we announced that our Board had terminated our quarterly cash dividend program; as a result no dividends were declared and paid in 2023.

Our Board of Directors declared the following dividends payable in 2022:

Date Declared	Record Date	Dividend Per Share	Total Amount	Payment Date
			(in millions)	
August 18, 2022	8/30/2022	$ 0.25	$ 253	9/9/2022
May 19, 2022	5/31/2022	0.25	253	6/10/2022
February 24, 2022	3/8/2022	0.25	253	3/18/2022

The declaration of dividends is solely at the discretion of our Board of Directors.

(24) Subsequent Events

Transaction Support Agreement

On January 22, 2024, the Company, Level 3, Qwest and a group of creditors holding a majority of our consolidated debt (the "TSA Parties") amended and restated the transaction support agreement that we originally entered into with a subset of the TSA Parties on October 31, 2023 (as amended and restated, the “Transaction Support Agreement”).

The Transaction Support Agreement defines the parties’ commitments to effect a series of transactions (the “TSA Transactions”) set forth in the term sheet attached thereto (the “Term Sheet”). Among other things and subject to the terms and conditions set forth therein, the Transaction Support Agreement, including the Term Sheet, contemplates:

- the incurrence by Level 3 of $ 1.325 billion in new money long term senior secured first lien indebtedness, which indebtedness will be backstopped by certain of the consenting lenders;
- a new revolving credit facility at Lumen in an amount expected to be approximately $ 1 billion;
- the extension of maturities, covenant modifications and rate increases of certain secured and unsecured indebtedness at the Company and Level 3 through a series of exchanges and other debt transactions with certain consenting lenders as set forth in the Term Sheet; and
- the repayment of certain indebtedness of the Company and Qwest.

The outside date for completion of the TSA Transactions under the Transaction Support Agreement is February 29, 2024, which the Company may unilaterally extend at its discretion to March 31, 2024. The Company expects to consummate the TSA Transactions in the first quarter of 2024, subject to the satisfaction of remaining closing conditions.

Following consummation of the TSA Transactions, the Company may assess potential follow-on transactions with respect to non-participating creditors.

Additional information about the Transaction Support Agreement and the TSA Transactions is available in our Current Report on Form 8-K filed with the Securities and Exchange Commission on January 25, 2024, and Exhibit 10.16 to this annual report.

Tax Refund

During the year ended December 31, 2023 we requested a U.S. Federal income tax refund of approximately $ 900 million. We applied approximately $200 million of that refund to pay our 2023 estimated taxes and, in January 2024, we received a cash refund of approximately $ 729 million, including interest.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Disclosure Controls and Procedures

We maintain disclosure controls and procedures (as defined in Rule 13a-15(e) promulgated under the Securities Exchange Act of 1934 (the "Exchange Act")) designed to provide reasonable assurance that the information required to be disclosed by us in the reports we file or furnish under the Exchange Act is recorded, processed, summarized, and reported within the time periods specified in the SEC's rules and forms. These include controls and procedures designed to ensure this information is accumulated and communicated to our senior management, including our Chief Executive Officer and Chief Financial Officer, as appropriate to allow timely decisions regarding required disclosure. Management, with the participation of our President and Chief Executive Officer, Kate Johnson, and our Executive Vice President and Chief Financial Officer, Chris Stansbury, evaluated the effectiveness of our disclosure controls and procedures as of December 31, 2023. Based on this evaluation, our Chief Executive Officer and Chief Financial Officer concluded our disclosure controls and procedures were effective, as of December 31, 2023, in providing reasonable assurance the information required to be disclosed by us in this report was accumulated and communicated in the manner provided above.

Changes in Internal Control Over Financial Reporting

Other than the implementation of controls over accounting and reporting for the completed divestiture of our EMEA business, there have been no changes in our internal control over financial reporting (as defined in Rule 13a-15(f) of the Exchange Act) that occurred during the fourth quarter of 2023 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Inherent Limitations of Internal Controls

The effectiveness of our or any system of disclosure controls and procedures is subject to certain limitations, including the exercise of judgment in designing, implementing and evaluating the controls and procedures, the assumptions used in identifying the likelihood of future events and the inability to eliminate misconduct completely. As a result, there can be no assurance that our disclosure controls and procedures will detect all errors or fraud. By their nature, our or any system of disclosure controls and procedures can provide only reasonable assurance regarding management's control objectives.

Internal Control Over Financial Reporting

Management's Report on Internal Control over Financial Reporting

Management of the Company is responsible for establishing and maintaining adequate internal control over financial reporting (as defined in Rule 13a-15(f) of the Exchange Act), a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles in the United States. Under the supervision and with the participation of management, including our principal executive officer and principal financial officer, we conducted an evaluation of the effectiveness of our internal control over financial reporting based on the framework in Internal Control-Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO"). Based on our evaluation under the framework of COSO, management concluded that our internal control over financial reporting was effective at December 31, 2023. The effectiveness of our internal control over financial reporting at December 31, 2023 has been audited by KPMG LLP, as stated in their report entitled "Opinion on Internal Control Over Financial Reporting" appearing in Item 8, which is incorporated into this item by reference.

Management's Report on the Consolidated Financial Statements

Management has prepared and is responsible for the integrity and objectivity of our consolidated financial statements for the year ended December 31, 2023. The consolidated financial statements included in this report have been prepared in accordance with accounting principles generally accepted in the United States and necessarily include amounts determined using our best judgments and estimates.

Our consolidated financial statements have been audited by KPMG LLP, an independent registered public accounting firm, who have expressed an unqualified opinion on the consolidated financial statements. Their audit was conducted in accordance with standards of the Public Company Accounting Oversight Board (United States).

ITEM 9B. OTHER INFORMATION

Not applicable.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

The information required by Item 10 is incorporated by reference to the Proxy Statement.

ITEM 11. EXECUTIVE COMPENSATION

The information required by Item 11 is incorporated by reference to the Proxy Statement.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The balance of the information required by Item 12 is incorporated by reference to the Proxy Statement.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS AND DIRECTOR INDEPENDENCE

The information required by Item 13 is incorporated by reference to the Proxy Statement.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by Item 14 is incorporated by reference to the Proxy Statement.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

Exhibits designated with an asterisk have been filed as part of this electronic submission. All others are on file with the SEC and are incorporated herein by reference. All references in this Item 15 to "Registrant" are to Lumen Technologies, Inc., which was formerly named CenturyLink, Inc. and Century Telephone Enterprises, Inc.

Exhibit Number	Description
2.1	Agreement, dated as of February 8, 2023, by and among certain affiliates of Registrant, and Colt Technology Services Group Limited (incorporated by reference to Exhibit 2.2 to Registrant's Annual Report on Form 10 K for the year ended December 31, 2022 (File No. 001-07784) filed with the Securities and Exchange Commission on February 23, 2023).
3.1	Composite Articles of Incorporation of Registrant, as amended through January 22, 2021 (incorporated by reference to Exhibit 3.1 to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
3.2	Bylaws of Registrant, as amended and restated through May 17, 2023 (incorporated by reference to Exhibit 3. 1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on May 17, 2023).
4.1*	Description of Registrant's, securities registered under Section 12 of the Securities Exchange Act of 1934, as amended.
4.2	Form of Registrant's common stock certificate (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 26, 2021).
4.3*	Second Amended and Restated Section 382 Rights Agreement by and between Registrant and Computershare Trust Company, N.A., dated as of November 15, 2023[1].
4.4	Instruments relating to Registrant's Senior Secured Credit Facilities.
	a. Restatement Agreement, dated as of January 31, 2020, by and among Registrant, as Borrower, Bank of America, N.A., as Administrative Agent and Collateral Agent, and the other lenders named therein (incorporated by reference to Exhibit 10.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 31, 2020).
	b. Amended and Restated Credit Agreement, dated as of January 31, 2020, by and among Registrant, as Borrower, Bank of America, N.A. as Administrative Agent and Collateral Agent, and the other lenders, agents, arrangers and bookrunners named therein (incorporated by reference to Exhibit 10.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 31, 2020).
	c. LIBOR Transition Amendment, dated as of March 17, 2023, by and among Registrant, the Guarantors party thereto, and Bank of America, N.A., as administrative agent and collateral agent, amending the parties' Amended and Restated Credit Agreement dated as of January 31, 2020 (incorporated by reference to Exhibit 10.1 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended March 31, 2023).
	d.* Amendment Agreement, dated as of February 15, 2024, by and among the Registrant, Bank of America, N.A., as a dministrative agent and collateral agent, and the lenders party thereto, amending the parties' Amended and Restated Credit Agreement dated as of January 31, 2020.
4.5	Instruments relating to Registrant's public senior debt. [2]
	a. Indenture, dated as of March 31, 1994, by and between Registrant and Regions Bank (successor-in-interest to First American Bank & Trust of Louisiana), as Trustee (incorporated by reference to Exhibit 4.4(a) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2014 filed with the Securities and Exchange Commission on February 24, 2015).
	(i). Form of 7.2% Senior Notes, Series D, due 2025 (incorporated by reference to Exhibit 4.27 to Registrant's Annual Report on Form 10-K for the year ended December 31, 1995 (File No. 001-07784) filed with the Securities and Exchange Commission on March 18, 1996).

Exhibit Number	Description
	(ii). Form of 6.875% Debentures, Series G, due 2028, (incorporated by reference to Exhibit 4.9 to Registrant's Annual Report on Form 10-K for the year ended December 31, 1997 (File No. 001-07784) filed with the Securities and Exchange Commission on March 16, 1998).
	(iii). Fifth Supplemental Indenture, dated as of September 21, 2009, by and between Registra nt and Regions Bank, as Trustee, designating and outlining the terms and conditions of Registrant's 7.60% Senior Notes, Series P, due 2039 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on September 22, 2009).
	(iv). Seventh Supplemental Indenture, dated as of March 12, 2012, by and between Registrant and Regions Bank, as Trustee, designating and outlining the terms and conditions of Registrant's 7.65% Senior Notes, Series U, due 2042 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on March 12, 2012).
	(v). Tenth Supplemental Indenture, dated as of March 19, 2015, by and between Registrant and Regions Bank, as Trustee, designating and outlining the terms and conditions of Registrant's 5.625% Senior Notes, Series X, due 2025 (incorporated by reference to Exhibit 4.2 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on March 19, 2015).
	b. Indenture, dated December 16, 2019, between Registrant and Regions Bank, as Trustee (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on December 16, 2019).
	(i). First Supplemental Indenture, dated December 16, 2019, between Registrant and Regions Bank, as Trustee, designating and outlining the terms and conditions of Registrant's 5.125% Senior Notes due 2026 (incorporated by reference to Exhibit 4.2 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on December 16, 2019).
	c. Indenture, dated January 24, 2020, between Registrant and Wells Fargo Bank, National Association, as Trustee and Notes Collateral Agent, designating and outlining the terms and conditions of Registrant's 4.000% Senior Secured Notes due 2027 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 24, 2020).
	d. Indenture, dated November 27, 2020, among Registrant's, as Issuer, and Regions Bank, as Trustee, designating and outlining the terms and conditions of Registrant, Inc. 4.500% Senior Notes due 2029 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) dated November 27, 2020).
	e. Indenture, dated June 15, 2021, among Registrant, as issuer, and Regions Bank, as trustee, relating to the issuance of Registrant's 5.375% Senior Notes due 2029 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on June 15, 2021).
4.6	Instruments relating to indebtedness of subsidiaries of Qwest Communications International, Inc. (2)
	a. Indenture, dated as of April 15, 1990, by and between The Mountain States Telephone and Telegraph Company (currently named Qwest Corporation) and The First National Bank of Chicago, under which Qwest Corporation's 7.375% Notes due 2030 were issued (incorporated by reference to Exhibit 4.2 to Qwest Corporation's Annual Report on Form 10-K for the year ended December 31, 2002 (File No. 001-03040) filed with the Securities and Exchange Commission on January 13, 2004).
	(i). First Supplemental Indenture, dated as of April 16, 1991, by and between U S WEST Communications, Inc. (currently named Qwest Corporation) and The First National Bank of Chicago (incorporated by reference to Exhibit 4.3 to Qwest Corporation's Annual Report on Form 10-K for the year ended December 31, 2002 (File No. 001-03040) filed with the Securities and Exchange Commission on January 13, 2004).

Exhibit Number	Description
	b. Indenture, dated as of April 15, 1990, by and between Northwestern Bell Telephone Company (predecessor to Qwest Corporation) and The First National Bank of Chicago, under which Qwest Corporation's 7.250% Notes due 2025 and 7.750% Notes due 2030 were issued (incorporated by reference to Exhibit 4.5(b) to Registrant's Quarterly Report on Form 10-Q for the period ended March 31, 2012 (File No. 001-07784) filed with the Securities and Exchange Commission on May 10, 2012).
	(i). First Supplemental Indenture, dated as of April 16, 1991, by and between U S WEST Communications, Inc. (currently named Qwest Corporation) and The First National Bank of Chicago (incorporated by reference to Exhibit 4.3 to Qwest Corporation's Annual Report on Form 10-K for the year ended December 31, 2002 (File No. 001-03040) filed with the Securities and Exchange Commission on January 13, 2004).
	c. Indenture, dated as of June 29, 1998, by and among U S WEST Capital Funding, Inc. (currently named Qwest Capital Funding, Inc.), U S WEST, Inc. (predecessor to Qwest Communications International Inc.) and The First National Bank of Chicago, as trustee (incorporated by reference to Exhibit 4(a) to U S WEST, Inc.'s Current Report on Form 8-K (File No. 001-14087) filed with the Securities and Exchange Commission on November 18, 1998).
	(i). First Supplemental Indenture, dated as of June 30, 2000, by and among U S WEST Capital Funding, Inc. (currently named Qwest Capital Funding, Inc.), U S WEST, Inc. (predecessor to Qwest Communications International Inc.) and Bank One Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.10 to Qwest Communications International Inc.'s Quarterly Report on Form 10-Q for the period ended June 30, 2000 (File No. 001-15577) filed with the Securities and Exchange Commission on August 11, 2000).
	d. Indenture, dated as of October 15, 1999, by and between US West Communications, Inc. (currently named Qwest Corporation) and Bank One Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4(b) to Qwest Corporation's Annual Report on Form 10-K for the year ended December 31, 1999 (File No. 001-03040) filed with the Securities and Exchange Commission on March 3, 2000).
	(i). Sixteenth Supplemental Indenture, dated as of August 22, 2016, by and between Qwest Corporation and U.S. Bank National Association, designating and outlining the terms and conditions of Qwest 6.500% Notes due 2056 (incorporated by reference to Exhibit 4.17 to Qwest Corporation's Form 8-A (File No. 001-03040) filed with Securities and Exchange Commission on August 22, 2016).
	(ii). Seventeenth Supplemental Indenture dated as of April 27, 2017, by and between Qwest Corporation and U.S. Bank National Association, designating and outlining the terms and conditions of Qwest Corporation's 6.750% Notes due 2057 (incorporated by reference to Exhibit 4.18 to Qwest Corporation's Form 8-A (File No. 03040) filed with the Securities and Exchange Commission on April 27, 2017).
	e. Amended and Restated Credit Agreement, dated as of October 23, 2020, by and among Qwest Corporation, the several lenders from time to time parties thereto, and CoBank, ACB, as administrative agent (incorporated by reference to Exhibit 4.6(e) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	(i). First Amendment to Amended and Restated Credit Agreement, dated as of March 27, 2023, by and between Qwest Corporation and CoBank, ACB, as administrative agent, amending the parties' Amended and Restated Credit Agreement dated as of October 23, 2020 (incorporated by reference to Exhibit 10.2 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended March 31, 2023).
4.7	Instruments relating to indebtedness of Level 3 Communications, Inc. and its subsidiaries. (2)
	a. Indenture, dated as of September 25, 2019, among Level 3 Parent, LLC, as Guarantor, Level 3 Financing, Inc., as Issuer, and the Bank of New York Mellon Trust Company, N.A., as Trustee, relating to the 4.625% Senior Notes due 2027 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 99.1 to Level 3 Parent, LLC's Current Report on Form 8-K (File No. 001-35134) filed with the Securities and Exchange Commission on September 26, 2019).

Exhibit Number	Description
	(i). First Supplemental Indenture, dated as of March 2, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, designating and outlining the terms and conditions of Level 3 Communications, LLC's unsecured guarantee of the 4.625% Senior Notes due 2027 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(d)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	(ii). Second Supplemental Indenture, dated as of March 2, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, relating to the subordination in any bankruptcy, liquidation or winding up proceeding of the guarantee by Level 3 Communications, LLC of the 4.625% Senior Notes due 2027 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(d)(ii) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	b. Indenture, dated as of November 29, 2019, among Level 3 Parent, LLC and the other guarantors party thereto, Level 3 Financing, Inc., as Issuer, and The Bank of New York Mellon Trust Company, N.A., as Trustee and Notes Collateral Agent, designating and outlining the terms and conditions of Level 3 Financing, Inc.'s 3.400% Senior Secured Notes due 2027 (incorporated by reference to Exhibit 10.2 to Level 3 Parent, LLC's Current Report on Form 8-K (File No. 001-35134) filed with the Securities and Exchange Commission on December 4, 2019).
	(i). Supplemental Indenture, dated as of April 15, 2020, among Level 3 Financing, Inc., as issuer, The Bank of New York Mellon Trust Company, N.A., as trustee, and Level 3 Parent, LLC and several of its subsidiaries, as guarantors, designating and outlining the terms and conditions of the secured guarantees of the 3.400% Senior Secured Notes due 2027 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(e)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	(ii). Supplement, dated as of October 26, 2023, to the Supplemental Indenture dated as of April 15, 2020, among Level 3 Financing, Inc., as Issuer, The Bank of New York Mellon Trust Company, N.A., as Trustee and Notes Collateral Agent, and Level 3 Parent, LLC and several of its subsidiaries, as guarantors, clarifying which subsidiaries are guarantors of the 3.400% Senior Secured Notes due 2027 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.1 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended September 30, 2023).
	(iii).* Second Supplemental Indenture, dated as of December 29, 2023, among Level 3 Financing, Inc., on behalf of itself as issuer and certain specified existing guarantors, The Bank of New York Mellon Trust Company, N.A., as trustee and collateral agent, Level 3 Parent, LLC, as guarantor, and several subsidiaries thereof, designating such subsidiaries as additional guarantors of Level 3 Financing, Inc.'s 3.400% Senior Secured Notes due 2027.
	c. Indenture, dated as of November 29, 2019, among Level 3 Parent, LLC and the other guarantors party thereto, Level 3 Financing, Inc., as Issuer, and The Bank of New York Mellon Trust Company, N.A., as Trustee and Notes Collateral Agent, designating and outlining the terms and conditions of Level 3 Financing, Inc.'s 3.875% Senior Secured Notes due 2029 (incorporated by reference to Exhibit 10.3 to Level 3 Parent, LLC's Current Report on Form 8-K (File No. 001-35134) filed with the Securities and Exchange Commission on December 4, 2019).
	(i). Supplemental Indenture, dated as of April 15, 2020, among Level 3 Financing, Inc., as issuer, The Bank of New York Mellon Trust Company, N.A., as trustee, and Level 3 Parent, LLC and several of its subsidiaries, as guarantors, designating and outlining the terms and conditions of the secured guarantees of the 3.875% Senior Secured Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(f)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).

Exhibit Number	Description
	(ii). Supplement, dated as of October 26, 2023, to the Supplemental Indenture dated as of April 15, 2020, among Level 3 Financing, Inc., as Issuer, The Bank of New York Mellon Trust Company, N.A., as Trustee and Notes Collateral Agent, and Level 3 Parent, LLC and several of its subsidiaries, as guarantors, clarifying which subsidiaries are guarantors of the 3.875% Senior Secured Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.2 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended September 30, 2023).
	(iii).* Second Supplemental Indenture, dated as of December 29, 2023, among Level 3 Financing, Inc., on behalf of itself as issuer and certain specified existing guarantors, The Bank of New York Mellon Trust Company, N.A., as trustee and collateral agent, Level 3 Parent, LLC, as guarantor, and several subsidiaries thereof, designating such subsidiaries as additional guarantors of Level 3 Financing, Inc.'s 3.875% Senior Secured Notes due 2029.
	d. Indenture, dated as of June 15, 2020, among Level 3 Parent, LLC, as Guarantor, Level 3 Financing, Inc., as Issuer, and The Bank of New York Mellon Trust Company, N.A., as Trustee, relating to the 4.250% Senior Notes due 2028 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on June 15, 2020).
	(i). First Supplemental Indenture, dated as of December 21, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, designating and outlining the terms and conditions of Level 3 Communications, LLC's unsecured guarantee of the 4.250% Senior Notes due 2028 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(g)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	(ii). Second Supplemental Indenture, dated as of December 21, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, relating to the subordination in any bankruptcy, liquidation or winding up proceeding of the guarantee by Level 3 Communications, LLC of the 4.250% Senior Notes due 2028 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(g)(ii) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	e. Indenture, dated August 12, 2020, among Level 3 Parent, LLC, as Guarantor, Level 3 Financing, Inc., as Issuer, and The Bank of New York Mellon Trust Company, N.A., as Trustee, relating to the 3.625% Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on August 12, 2020).
	(i). First Supplemental Indenture, dated as of December 21, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, designating and outlining the terms and conditions of Level 3 Communications, LLC's unsecured guarantee of the 3.625% Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(h)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).
	(ii). Second Supplemental Indenture, dated as of December 21, 2020, among Level 3 Parent LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, relating to the subordination in any bankruptcy, liquidation or winding up proceeding of the guarantee by Level 3 Communications, LLC of the 3.625% Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(h)(ii) to Registrant's Annual Report on Form 10-K (File No. 001-07784) for the year ended December 31, 2020 filed with the Securities and Exchange Commission on February 25, 2021).

Exhibit Number	Description
	f. Indenture, dated January 13, 2021, among Level 3 Parent, LLC, as Guarantor, Level 3 Financing, Inc., as Issuer, and The Bank of New York Mellon Trust Company, N.A., as Trustee, relating to the 3.750% Sustainability-Linked Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 13, 2021).
	(i). First Supplemental Indenture, dated as of May 7, 2021, among Level 3 Parent, LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, designating and outlining the terms and conditions of Level 3 Communications, LLC's unsecured guarantee of the 3.750% Sustainability-Linked Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(h)(i) to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
	(ii). Second Supplemental Indenture, dated as of May 7, 2021, among Level 3 Parent, LLC, as guarantor, Level 3 Communications, LLC, as guarantor, Level 3 Financing, Inc., as issuer, and The Bank of New York Mellon Trust Company, N.A., as trustee, relating to the subordination in any bankruptcy, liquidation or winding up proceeding of the guarantee by Level 3 Communications, LLC of the 3.750% Sustainability-Linked Senior Notes due 2029 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.8(h)(ii) to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
	g. Indenture dated March 31, 2023, among Level 3 Financing, Inc., as Issuer, Level 3 Parent, LLC, as Guarantor, the subsidiary guarantors party thereto, and The Bank of New York Mellon Trust Company, as Trustee and Note Collateral Agent, designating and outlining the terms and conditions of Level 3 Financing, Inc.'s 10.500% Senior Secured Notes due 2030 (incorporated by reference to Exhibit 4.1 to Registrant's Current Report on Form 8 K (File No. 001-07784) filed with the Securities and Exchange Commission on March 31, 2023).
	(i). Supplemental Indenture, dated as of October 23, 2023, among Level 3 Financing, Inc., as Issuer, The Bank of New York Mellon Trust Company, N.A., as Trustee and Notes Collateral Agent, and Level 3 Parent, LLC and several of its subsidiaries, as guarantors, designating and outlining the terms and conditions of certain specified secured guarantees of the 10.500% Senior Secured Notes due 2030 of Level 3 Financing, Inc. (incorporated by reference to Exhibit 4.3 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended September 30, 2023).
	(ii).* Second Supplemental Indenture, dated as of December 29, 2023, among Level 3 Financing, Inc., on behalf of itself as issuer and certain specified existing guarantors, The Bank of New York Mellon Trust Company, N.A., as trustee and collateral agent, Level 3 Parent, LLC, as guarantor, and several subsidiaries thereof, designating such subsidiaries as additional guarantors of Level 3 Financing, Inc.'s 10.500% Senior Secured Notes due 2030.
	h. Thirteenth Amendment Agreement to the Amended and Restated Credit Agreement, dated as of November 29, 2019, by and between Level 3 Parent, LLC, Level 3 Financing, Inc., the Lenders party thereto and Merrill Lynch Capital Corporation (incorporated by reference to Exhibit 10.1 to Level 3 Parent, LLC's Current Report on Form 8-K (File No. 001-35134) filed with the Securities and Exchange Commission on December 4, 2019).
	(i). LIBOR Transition Amendment, dated as of March 17, 2023, by and among Level 3 Parent, LLC, Level 3 Financing, Inc., the Subsidiary Loan Parties party thereto, and Merrill Lynch Capital Corporation, as administrative agent, amending the parties' Amended and Restated Credit Agreement dated as of November 29, 2019 (incorporated by reference to Exhibit 10.3 to Registrant's Quarterly Report on Form 10-Q (File No. 001-07784) for the quarterly period ended March 31, 2023).
10.1+	Registrant's Second Amended and Restated 2018 Equity Incentive Plan, as amended and re stated through May 17, 2023 (incorporated by reference to Exhibit 10.1 to Lumen Technologies, Inc.'s Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on May 17, 2023).

Exhibit Number	Description
	(i). Form of Restricted Stock Agreement for annual equity grants to non-management directors beginning in 2018 (incorporated by reference to Exhibit 10.1A to Registrant's Quarterly Report on Form 10-Q for the quarterly period ended June 30, 2018 (File No. 001-07784) filed with the Securities and Exchange Commission on August 9, 2018).
	(ii) Form of RSU Agreement for annual time-based equity grants to Jeffrey K. Storey (incorporated by reference to Exhibit 10.2(vi) to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
	(iii) Form of Restricted Stock Agreement for annual time-based equity grants to certain executive officers (incorporated by reference to Exhibit 10.2(viii) to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
	(iv) Form of Restricted Stock Agreement for annual performance-based equity grants to certain executive officers (incorporated by reference to Exhibit 10.2(ix) to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
	(v) Form of Restricted Stock Agreement for annual time-based equity grants to Kate Johnson (incorporated by reference to Exhibit 10.2(vi) to Registrant's Annual Report on Form 10-K for the year ended December 31, 2022 (File No. 001-00784) filed with the Securities and Exchange Commission on February 23, 2023).
	(vi) Restricted Stock Agreement for sign-on time-based award to Kate Johnson on November 7, 2022 (incorporated by reference to Exhibit 10.2(vii) to Registrant's Annual Report on Form 10 K for the year ended December 31, 2022 (File No. 001-00784) filed with the Securities and Exchange Commission on February 23, 2023).
10.2+	Registrant's Supplemental Dollars & Sense Plan, 2014 Restatement, effective January 1, 2014 (incorporated by reference to Exhibit 10.3 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2018 (File No. 001-07784) filed with the Securities and Exchange Commission on March 11, 2019).
10.3+	Registrant's Supplemental Defined Benefit Pension Plan, effective as of January 1, 2012 (incorporated by reference to Exhibit 10.5 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2011 (File No. 001-07784) filed with the Securities and Exchange Commission on February 28, 2012).
10.4+	Registrant's - Short-Term Incentive Plan - Annual (incorporated by reference to Exhibit 10.5 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2022 (File No. 001-07784) filed with the Securities and Exchange Commission on February 23, 2023).
10.5+	Form of Indemnification Agreement entered into between Registrant and each of its directors on or after February 24, 2016 (incorporated by reference to Exhibit 10.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 29, 2016).
10.6+	Form of Indemnification Agreement entered into between Registrant and each of its officers on or after February 24, 2016 (incorporated by reference to Exhibit 10.2 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 29, 2016).
10.7+	Change of Control Agreement, by and between Kate Johnson and Registrant (incorporated by reference to Exhibit 10.8 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2022 (File No. 001-07784) filed with the Securities and Exchange Commission on February 23, 2023.
10.8+	Form of Change of Control Agreement, on or after January 1, 2011 between Registrant and each of its other executive officers (incorporated by reference to Exhibit 10.12 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2010 (File No. 001-07784) filed with the Securities and Exchange Commission on March 1, 2011).
10.9+	Registrant's Amended and Restated Lumen Executive Severance Plan, effective October 10, 2017 (with updated exhibits and branding as of October 2020) (incorporated by reference to Exhibit 10.11 to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).

Exhibit Number	Description
10.10+	Amended and Restated CenturyLink, Inc. Bonus Life Insurance Plan for Executive Officers, dated as of April 3, 2008 (incorporated by reference to Exhibit 10.4 to Registrant's Quarterly Report on Form 10-Q for the period ended March 31, 2008 (File No. 001-07784) filed with the Securities and Exchange Commission on May 7, 2008) and First Amendment thereto (incorporated by reference to Exhibit 10.13 to Registrant's Quarterly Report on Form 10-Q for the period ended September 30, 2010 (File No. 001-07784) filed with the Securities and Exchange Commission on November 5, 2010).
10.11+	Registrant's Supplemental Savings Plan, as amended and restated (incorporated by reference to Exhibit 10.2 to Registrant's Quarterly Report on Form 10-Q for the period ended March 31, 2022 (File No. 001-07784) filed with the Securities and Exchange Commission on May 4, 2022).
10.12+	Registrant's Non-Employee Director Deferred Compensation Plan, effective April 18, 2019 (updated for branding as of October 2020) (incorporated by reference to Exhibit 10.15 to Registrant's Annual Report on Form 10-K (File No. 001-07784) filed with the Securities and Exchange Commission on February 24, 2022).
10.13+	Legacy Qwest Deferred Compensation Plan for Nonemployee Directors, as amended and restated, Amendment to Deferred Compensation Plan for Nonemployee Directors (incorporated by reference to Exhibit 10.2 to Qwest Communications International Inc.'s Current Report on Form 8-K (File No. 001-15577) filed with the Securities and Exchange Commission on December 16, 2005 and Exhibit 10.8 to Qwest Communication International Inc.'s Quarterly Report on Form 10-Q for the period ended September 30, 2008 (File No. 001-15577) filed with the Securities and Exchange Commission on October 29, 2008) and Amendment No. 2011-1 to Deferred Compensation Plan for Nonemployee Directors (incorporated by reference to Exhibit 10.15(c) to Registrant's Annual Report for the year ended December 31, 2011 (File No. 001-07784) filed with the Securities and Exchange Commission on February 28, 2012).
10.14+	Retention Letter between Lumen Technologies, Inc. and Stacey Goff, dated December 9, 2022 (incorporated by reference to Exhibit 10.20 to Registrant's Annual Report on Form 10-K for the year ended December 31, 2022 (File No. 001-00784) filed with the Securities and Exchange Commission on February 23, 2023).
10.15+	Offer Letters[3]
	a. Offer Letter dated September 12, 2022 between Registrant and Kate Johnson (incorporated by reference to Exhibit 10.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on September 13, 2022).
	b. Offer Letter dated March 24, 2022 between Registrant and Christopher D. Stansbury (incorporated by reference to Exhibit 10.1 to Registrant's Quarterly Report on Form 10-Q for the period ended March 31, 2022 (File No. 001-07784) filed with the Securities and Exchange Commission on May 4, 2022).
	c.* Offer Letter dated December 5, 2022 between Regist rant and Ashley Haynes-Gaspar.
10.16	Amended and Restated Transaction Support Agreement by and among Registrant, Level 3 Financing, Inc., Qwest Corporation, and the Consenting Parties identified therein, dated January 22, 2024 (incorporated by reference to Exhibit 10.1 to Registrant's Current Report on Form 8-K (File No. 001-07784) filed with the Securities and Exchange Commission on January 25, 2024).
21*	Subsidiaries of Registrant.
23*	Independent Registered Public Accounting Firm Consent.
31.1*	Certification of the Chief Executive Officer of Lumen Technologies, Inc. furnished pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
31.2*	Certification of the Chief Financial Officer of Lumen Technologies, Inc. furnished pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
32.1*	Certification of the Chief Executive Officer of Lumen Technologies, Inc. pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
32.2*	Certification of the Chief Financial Officer of Lumen Technologies, Inc. pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
97*	Registrant's Policy Relating to Recovery of Erroneously Awarded Compensation, adopted August 16, 2023.

Exhibit Number	Description
101*	Financial statements from the annual report on Form 10-K of Registrant for the period ended December 31, 2023, formatted in Inline XBRL: (i) the Consolidated Statements of Operations, (ii) the Consolidated Statements of Comprehensive (Loss) Income, (iii) the Consolidated Balance Sheets, (iv) the Consolidated Statements of Cash Flows, (v) the Consolidated Statements of Stockholders' Equity and (vi) the Notes to Consolidated Financial Statements.
104*	Cover page formatted as Inline XBRL and contained in Exhibit 101.

* Exhibit filed herewith.
\+ Indicates a management contract or compensatory plan or arrangement.

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(1) This corrected version of this instrument supersedes the prior version filed with the Securities and Exchange Commission on November 20, 2023.

(2) Certain of the items in Sections 4.5, 4.6 and 4.7 (i) omit supplemental indentures or other instruments governing debt that has been retired, or (ii) refer to trustees who may have been replaced, acquired or affected by similar changes. In accordance with applicable rules of the SEC, copies of certain instruments defining the rights of holders of certain of our long-term debt are not filed herewith. Additional documentation regarding the credit agreement of Level 3 Parent, LLC and its affiliates is available in reports filed by Level 3 Parent, LLC with the Securities and Exchange Commission.

(3) Offer letters present information regarding the executive's initial compensation only.

ITEM 16. SUMMARY OF BUSINESS AND FINANCIAL INFORMATION

Not applicable.

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this annual report to be signed on its behalf by the undersigned thereunto duly authorized.

Date: February 22, 2024

By: **Lumen Technologies, Inc.**

/s/ Andrea Genschaw

Andrea Genschaw

Senior Vice President, Controller *(Principal Accounting Officer)*

Pursuant to the requirements of the Securities Exchange Act of 1934, this annual report has been signed below by the following persons on behalf of the registrant and in the capacities and on the date indicated.

Signature	Title	Date
/s/ Kate Johnson Kate Johnson	President and Chief Executive Officer (Principal Executive Officer)	February 22, 2024
/s/ Chris Stansbury Chris Stansbury	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	February 22, 2024
/s/ Andrea Genschaw Andrea Genschaw	Senior Vice President, Controller (Principal Accounting Officer)	February 22, 2024
/s/ T. Michael Glenn T. Michael Glenn	Non-Executive Chairman of the Board	February 22, 2024
/s/ Jim Fowler Jim Fowler	Director	February 22, 2024
/s/ Quincy L. Allen Quincy L. Allen	Director	February 22, 2024
/s/ Martha Helena Bejar Martha Helena Bejar	Director	February 22, 2024
/s/ Peter C. Brown Peter C. Brown	Director	February 22, 2024
/s/ Kevin P. Chilton Kevin P. Chilton	Director	February 22, 2024
/s/ Steven T. "Terry" Clontz Steven T. "Terry" Clontz	Director	February 22, 2024
/s/ Hal Stanley Jones Hal Stanley Jones	Director	February 22, 2024
/s/ Michael Roberts Michael Roberts	Director	February 22, 2024
/s/ Laurie Siegel Laurie Siegel	Director	February 22, 2024